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| Tool Version : Vivado v.2023.1 (win64) Build 3865809 Sun May  7 15:05:29 MDT 2023
| Date        : Wed Sep  9 13:19:05 2026
| Host        : bedouinoureddine running 64-bit major release (build 9200)
| Command     : report_timing_summary -delay_type min_max -report_unconstrained
-check_timing_verbose -max_paths 10 -input_pins -routable_nets -name timing_2 -file
C:/Vivado_Projects/Harness_Test_System_3/Harness_Test_System/timing_report2.txt
| Design      : FPGA_Top
| Device      : 7a100t-csg324
| Speed File  : -1 PRODUCTION 1.23 2018-06-13
| Design State : Physopt postRoute
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Timing Summary Report

Timer Settings

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Enable Multi Corner Analysis      : Yes
Enable Pessimism Removal          : Yes
Pessimism Removal Resolution      : Nearest Common Node
Enable Input Delay Default Clock  : No
Enable Preset / Clear Arcs       : No
Disable Flight Delays             : No
Ignore I/O Paths                 : No
Timing Early Launch at Borrowing Latches : No
Borrow Time for Max Delay Exceptions : Yes
Merge Timing Exceptions           : Yes
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Corner Name	Analyze Max Paths	Analyze Min Paths
Slow	Yes	Yes
Fast	Yes	Yes

Report Methodology

Rule	Severity	Description	
Violations			
LUTAR-1	Warning	LUT drives async reset alert	1
XDCC-7	Warning	Scoped Clock constraint overwritten on the same source	1

Note: This report is based on the most recent report_methodology run and may not be up-to-date. Run report_methodology on the current design for the latest report.

check_timing report

Table of Contents

- 1. checking no_clock (0)
- 2. checking constant_clock (0)
- 3. checking pulse_width_clock (0)
- 4. checking unconstrained_internal_endpoints (0)
- 5. checking no_input_delay (17)
- 6. checking no_output_delay (2)
- 7. checking multiple_clock (0)
- 8. checking generated_clocks (0)
- 9. checking loops (0)
- 10. checking partial_input_delay (0)
- 11. checking partial_output_delay (0)
- 12. checking latch_loops (0)

1. checking no_clock (0)

There are 0 register/latch pins with no clock.

2. checking constant_clock (0)

There are 0 register/latch pins with constant_clock.

3. checking pulse_width_clock (0)

There are 0 register/latch pins which need pulse_width check

4. checking unconstrained_internal_endpoints (0)

There are 0 pins that are not constrained for maximum delay.

There are 0 pins that are not constrained for maximum delay due to constant clock.

5. checking no_input_delay (17)

There are 17 input ports with no input delay specified. (HIGH)

In1[0]
In1[10]
In1[11]
In1[12]
In1[13]
In1[14]
In1[15]
In1[1]
In1[2]
In1[3]
In1[4]
In1[5]
In1[6]
In1[7]
In1[8]
In1[9]
reset

There are 0 input ports with no input delay but user has a false path constraint.

6. checking no_output_delay (2)

There are 2 ports with no output delay specified. (HIGH)

Out3
ce_out

There are 0 ports with no output delay but user has a false path constraint

There are 0 ports with no output delay but with a timing clock defined on it or propagating through it

7. checking multiple_clock (0)

There are 0 register/latch pins with multiple clocks.

8. checking generated_clocks (0)

There are 0 generated clocks that are not connected to a clock source.

9. checking loops (0)

There are 0 combinational loops in the design.

10. checking partial_input_delay (0)

There are 0 input ports with partial input delay specified.

11. checking partial_output_delay (0)

There are 0 ports with partial output delay specified.

12. checking latch_loops (0)

There are 0 combinational latch loops in the design through latch input

Design Timing Summary

WNS(ns)		TNS(ns)		TNS Failing Endpoints		TNS Total Endpoints		WHS(ns)	
THS(ns)		THS Failing Endpoints		THS Total Endpoints		WPWS(ns)		TPWS(ns)	
TPWS Failing Endpoints		TPWS Total Endpoints							
-----		-----		-----		-----		-----	
0.093		0.000		0		2054		0.043	
0.000		0		1045		3.000		0.000	

All user specified timing constraints are met.

Clock Summary

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-----
Clock                Waveform(ns)      Period(ns)      Frequency(MHz)
-----
sys_clk_pin          {0.000 5.000}    10.000          100.000
  clk_out1_clk_wiz_0_1 {0.000 12.500}   25.000          40.000
  clkfbout_clk_wiz_0_1 {0.000 5.000}    10.000          100.000

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| Intra Clock Table
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Clock                WNS(ns)      TNS(ns)  TNS Failing Endpoints  TNS Total
Endpoints            WHS(ns)      THS(ns)  THS Failing Endpoints  THS Total Endpoints
WPWS(ns)            TPWS(ns)  TPWS  Failing Endpoints  TPWS Total Endpoints
-----
-----
-----
-----
sys_clk_pin

  3.000      0.000                0                1
  clk_out1_clk_wiz_0_1      0.093      0.000                0
    2054      0.043      0.000                0      2054
  12.000      0.000                0      1041
  clkfbout_clk_wiz_0_1

  7.845      0.000                0                3

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| Inter Clock Table
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From Clock    To Clock      WNS(ns)      TNS(ns)  TNS Failing Endpoints  TNS
Total Endpoints    WHS(ns)      THS(ns)  THS Failing Endpoints  THS Total
Endpoints
-----
-----
-----
-----

```


Other Path Groups Table

Path Group	From Clock	To Clock	WNS(ns)	TNS(ns)	TNS Failing
Endpoints	TNS Total	Endpoints	WHS(ns)	THS(ns)	THS Failing
THS Total	Endpoints				Endpoints
-----	-----	-----	-----	-----	
-----	-----	-----	-----	-----	
-----	-----	-----			

User Ignored Path Table

Path Group	From Clock	To Clock
-----	-----	-----

Unconstrained Path Table

Path Group	From Clock	To Clock
-----	-----	-----
(none)		
(none)	clk_out1_clk_wiz_0_1	
(none)	clkfbout_clk_wiz_0_1	
(none)		clk_out1_clk_wiz_0_1

Timing Details

From Clock: sys_clk_pin
To Clock: sys_clk_pin

Setup :	NA	Failing Endpoints,	Worst Slack	NA	, Total
Violation	NA				
Hold :	NA	Failing Endpoints,	Worst Slack	NA	, Total
Violation	NA				
PW :	0	Failing Endpoints,	Worst Slack	3.000ns	, Total
Violation	0.000ns				

Pulse Width Checks

Clock Name: sys_clk_pin
Waveform(ns): { 0.000 5.000 }
Period(ns): 10.000
Sources: { sys_clk }

Check Type	Corner	Lib Pin	Reference Pin	Required(ns)
Actual(ns)	Slack(ns)	Location	Pin	
Min Period	n/a	MMCME2_ADV/CLKIN1	n/a	1.249
8.751	MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKIN1		10.000
Max Period	n/a	MMCME2_ADV/CLKIN1	n/a	100.000
90.000	MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKIN1		10.000
Low Pulse Width	Slow	MMCME2_ADV/CLKIN1	n/a	2.000
3.000	MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKIN1		5.000
Low Pulse Width	Fast	MMCME2_ADV/CLKIN1	n/a	2.000
3.000	MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKIN1		5.000
High Pulse Width	Slow	MMCME2_ADV/CLKIN1	n/a	2.000
3.000	MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKIN1		5.000
High Pulse Width	Fast	MMCME2_ADV/CLKIN1	n/a	2.000
3.000	MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKIN1		5.000

From Clock: clk_out1_clk_wiz_0_1
To Clock: clk_out1_clk_wiz_0_1

Setup :	0	Failing Endpoints,	Worst Slack	0.093ns	, Total
Violation	0.000ns				
Hold :	0	Failing Endpoints,	Worst Slack	0.043ns	, Total
Violation	0.000ns				
PW :	0	Failing Endpoints,	Worst Slack	12.000ns	, Total
Violation	0.000ns				

MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.646	-0.894	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X12Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C				

SLICE_X12Y95	FDCE (Prop_fdce_C_Q)	0.478	-0.416	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/Q				
	net (fo=3, routed)	1.007	0.591	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[39][2]				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/I2				
SLICE_X13Y95	LUT3 (Prop_lut3_I2_0)	0.326	0.917	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/O				
	net (fo=2, routed)	0.690	1.607	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132_n_0				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/I3				
SLICE_X13Y95	LUT4 (Prop_lut4_I3_0)	0.327	1.934	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/O				
	net (fo=1, routed)	0.000	1.934	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135_n_0				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/S[3]				
SLICE_X13Y95	CARRY4 (Prop_carry4_S[3]_CO[3])	0.401	2.335	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/CO[3]				
	net (fo=1, routed)	0.000	2.335	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34_n_0				
SLICE_X13Y96				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/CI				
SLICE_X13Y96	CARRY4 (Prop_carry4_CI_CO[3])	0.114	2.449	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/CO[3]				
	net (fo=1, routed)	0.000	2.449	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168_n_0				
SLICE_X13Y97				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_168/CI				
SLICE_X13Y97	CARRY4 (Prop_carry4_CI_O[3])	0.313	2.762	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_168/O[3]				

	net (fo=2, routed)	1.013	3.774	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_168_n_4				
SLICE_X13Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_118/I2				
SLICE_X13Y91	LUT3 (Prop_lut3_I2_0)	0.336	4.110	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_118/O				
	net (fo=2, routed)	0.690	4.800	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_118_n_0				
SLICE_X13Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_122/I3				
SLICE_X13Y91	LUT4 (Prop_lut4_I3_0)	0.327	5.127	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_122/O				
	net (fo=1, routed)	0.000	5.127	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_122_n_0				
SLICE_X13Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_57/S[3]				
SLICE_X13Y91	CARRY4 (Prop_carry4_S[3]_CO[3])			
		0.401	5.528	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_57/CO[3]				
	net (fo=1, routed)	0.000	5.528	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_57_n_0				
SLICE_X13Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_79/CI				
SLICE_X13Y92	CARRY4 (Prop_carry4_CI_0[2])			
		0.239	5.767	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_79/O[2]				
	net (fo=2, routed)	0.967	6.734	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_79_n_5				
SLICE_X9Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_34/I1				
SLICE_X9Y92	LUT3 (Prop_lut3_I1_0)	0.332	7.066	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_34/O				
	net (fo=2, routed)	0.690	7.756	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_34_n_0				
SLICE_X9Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_38/I3				
SLICE_X9Y92	LUT4 (Prop_lut4_I3_0)	0.327	8.083	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_38/O				
	net (fo=1, routed)	0.000	8.083	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_38_n_0				
SLICE_X9Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15/S[3]				
SLICE_X9Y92	CARRY4 (Prop_carry4_S[3]_CO[3])			
		0.401	8.484	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15/CO[3]				
	net (fo=1, routed)	0.000	8.484	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15_n_0				
SLICE_X9Y93				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_11/CI				
SLICE_X9Y93	CARRY4 (Prop_carry4_CI_0[3])			

		0.313	8.797	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_11/0[3]				
	net (fo=3, routed)	0.730	9.527	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_11_n_4				
	SLICE_X7Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/I1				
	SLICE_X7Y92 LUT3 (Prop_lut3_I1_0)	0.306	9.833	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/0				
	net (fo=2, routed)	0.579	10.412	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13_n_0				
	SLICE_X5Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/I1				
	SLICE_X5Y92 LUT5 (Prop_lut5_I1_0)	0.124	10.536	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/0				
	net (fo=2, routed)	0.808	11.344	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3_n_0				
	SLICE_X7Y91			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/I0				
	SLICE_X7Y91 LUT6 (Prop_lut6_I0_0)	0.124	11.468	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/0				
	net (fo=1, routed)	0.000	11.468	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6_n_0				
	SLICE_X7Y91			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/S[0]				
	SLICE_X7Y91 CARRY4 (Prop_carry4_S[0]_O[2])			
		0.547	12.015	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/0[2]				
	net (fo=2, routed)	0.522	12.537	
u_FPGA_Core/u_Discrete_FIR_Filter/Discrete_FIR_Filter_out[21]				
	SLICE_X6Y91			r
u_FPGA_Core/u_Discrete_FIR_Filter/Sum_out1_carry__4_i_1/I1				
	SLICE_X6Y91 LUT2 (Prop_lut2_I1_0)	0.302	12.839	r
u_FPGA_Core/u_Discrete_FIR_Filter/Sum_out1_carry__4_i_1/0				
	net (fo=1, routed)	0.000	12.839	
u_FPGA_Core/u_Discrete_FIR_Filter_n_1				
	SLICE_X6Y91			r
u_FPGA_Core/Sum_out1_carry__4/S[1]				
	SLICE_X6Y91 CARRY4 (Prop_carry4_S[1]_CO[2])			
		0.574	13.413	r
u_FPGA_Core/Sum_out1_carry__4/CO[2]				
	net (fo=27, routed)	0.811	14.225	
u_FPGA_Core/u_Discrete_FIR_Filter/CO[0]				
	SLICE_X8Y91			r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_26/S[1]				
	SLICE_X8Y91 CARRY4 (Prop_carry4_S[1]_O[3])			
		0.953	15.178	r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_26/0[3]				
	net (fo=1, routed)	0.419	15.597	
u_FPGA_Core/u_Discrete_FIR_Filter/Abs_rsvd_y0[24]				
	SLICE_X8Y92			r

u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_1/I0			
SLICE_X8Y92	LUT2 (Prop_lut2_I0_0)	0.307	15.904 r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_1/O			
	net (fo=12, routed)	0.791	16.695
u_FPGA_Core/u_Discrete_FIR_Filter_n_45			
DSP48_X0Y36			r
u_FPGA_Core/Gain_mul_temp/A[28]			
DSP48_X0Y36	DSP48E1 (Prop_dsp48e1_A[28]_PCOUT[47])	4.036	20.731 r
u_FPGA_Core/Gain_mul_temp/PCOUT[47]			
	net (fo=1, routed)	0.002	20.733
u_FPGA_Core/Gain_mul_temp_n_106			
DSP48_X0Y37			r
u_FPGA_Core/Gain_mul_temp__0/PCIN[47]			
DSP48_X0Y37	DSP48E1 (Prop_dsp48e1_PCIN[47]_P[0])	1.518	22.251 f
u_FPGA_Core/Gain_mul_temp__0/P[0]			
	net (fo=2, routed)	0.760	23.011
u_FPGA_Core/Gain_out1[3]			
SLICE_X11Y91			f
u_FPGA_Core/Unit_Delay1_out10_carry_i_7/I1			
SLICE_X11Y91	LUT2 (Prop_lut2_I1_0)	0.124	23.135 r
u_FPGA_Core/Unit_Delay1_out10_carry_i_7/O			
	net (fo=1, routed)	0.000	23.135
u_FPGA_Core/Unit_Delay1_out10_carry_i_7_n_0			
SLICE_X11Y91			r
u_FPGA_Core/Unit_Delay1_out10_carry/S[1]			
SLICE_X11Y91	CARRY4 (Prop_carry4_S[1]_CO[3])	0.550	23.685 r
u_FPGA_Core/Unit_Delay1_out10_carry/CO[3]			
	net (fo=1, routed)	0.000	23.685
u_FPGA_Core/Unit_Delay1_out10_carry_n_0			
SLICE_X11Y92			r
u_FPGA_Core/Unit_Delay1_out10_carry__0/CI			
SLICE_X11Y92	CARRY4 (Prop_carry4_CI_CO[3])	0.114	23.799 r
u_FPGA_Core/Unit_Delay1_out10_carry__0/CO[3]			
	net (fo=1, routed)	0.000	23.799
u_FPGA_Core/Unit_Delay1_out10_carry__0_n_0			
SLICE_X11Y93			r
u_FPGA_Core/Unit_Delay1_out10_carry__1/CI			
SLICE_X11Y93	CARRY4 (Prop_carry4_CI_CO[3])	0.114	23.913 r
u_FPGA_Core/Unit_Delay1_out10_carry__1/CO[3]			
	net (fo=1, routed)	0.000	23.913
u_FPGA_Core/Unit_Delay1_out10_carry__1_n_0			
SLICE_X11Y94			r
u_FPGA_Core/Unit_Delay1_out10_carry__2/CI			
SLICE_X11Y94	CARRY4 (Prop_carry4_CI_CO[3])	0.114	24.027 r

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u_FPGA_Core/Unit_Delay1_out10_carry__2/C0[3]
    net (fo=2, routed)                0.000    24.027
u_FPGA_Core/Relational_Operator_out1
    SLICE_X11Y94      FDCE                                r
u_FPGA_Core/Unit_Delay1_out1_reg/D
-----
                                (clock clk_out1_clk_wiz_0_1 rise edge)
                                25.000    25.000 r
E3                                0.000    25.000 r  sys_clk
(IN)
    net (fo=0)                0.000    25.000
u_pll/inst/clk_in1
    E3                                r
u_pll/inst/clkin1_ibufg/I
    E3      IBUF (Prop_ibuf_I_0)    1.411    26.411 r
u_pll/inst/clkin1_ibufg/O
    net (fo=1, routed)        1.162    27.573
u_pll/inst/clk_in1_clk_wiz_0
    MMCME2_ADV_X1Y2                                r
u_pll/inst/mmcm_adv_inst/CLKIN1
    MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
                                -7.324    20.249 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
    net (fo=1, routed)        1.639    21.888
u_pll/inst/clk_out1_clk_wiz_0
    BUFGCTRL_X0Y16                                r
u_pll/inst/clkout1_buf/I
    BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)    0.091    21.979 r
u_pll/inst/clkout1_buf/O
    net (fo=1039, routed)    1.527    23.507
u_FPGA_Core/CLK
    SLICE_X11Y94      FDCE                                r
u_FPGA_Core/Unit_Delay1_out1_reg/C
    clock pessimism          0.559    24.066
    clock uncertainty        -0.085    23.982
    SLICE_X11Y94      FDCE (Setup_fdce_C_D)    0.138    24.120
u_FPGA_Core/Unit_Delay1_out1_reg
-----
                                required time          24.120
                                arrival time          -24.027
-----
                                slack                  0.093

```

Slack (MET) : 12.048ns (required time - arrival time)

Source:

```

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/C
    (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

```

Destination: u_FPGA_Core/Unit_Delay_out1_reg[20]/D
 (rising edge-triggered cell FDCE clocked by
 clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
 Path Group: clk_out1_clk_wiz_0_1
 Path Type: Setup (Max at Slow Process Corner)
 Requirement: 25.000ns (clk_out1_clk_wiz_0_1 rise@25.000ns -
 clk_out1_clk_wiz_0_1 rise@0.000ns)
 Data Path Delay: 12.811ns (logic 5.947ns (46.420%) route 6.864ns
 (53.580%))
 Logic Levels: 17 (CARRY4=8 LUT3=4 LUT4=3 LUT5=1 LUT6=1)
 Clock Path Skew: -0.107ns (DCD - SCD + CPR)
 Destination Clock Delay (DCD): -1.418ns = (23.582 - 25.000)
 Source Clock Delay (SCD): -0.831ns
 Clock Pessimism Removal (CPR): 0.480ns
 Clock Uncertainty: 0.085ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
 Total System Jitter (TSJ): 0.071ns
 Discrete Jitter (DJ): 0.154ns
 Phase Error (PE): 0.000ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.482	1.482	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.233	2.715	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.709	-0.831	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X4Y102	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/C				

```

-----
SLICE_X4Y102      FDCE (Prop_fdce_C_Q)      0.419   -0.412 r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/Q
      net (fo=3, routed)      0.848   0.436
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[53][0]
      SLICE_X2Y102      r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85/I1
      SLICE_X2Y102      LUT3 (Prop_lut3_I1_0)      0.325   0.761 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85/O
      net (fo=2, routed)      0.490   1.251
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85_n_0
      SLICE_X2Y102      r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88/I3
      SLICE_X2Y102      LUT4 (Prop_lut4_I3_0)      0.328   1.579 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88/O
      net (fo=1, routed)      0.000   1.579
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88_n_0
      SLICE_X2Y102      r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27/S[1]
      SLICE_X2Y102      CARRY4 (Prop_carry4_S[1]_CO[3])
      0.533   2.112 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27/CO[3]
      net (fo=1, routed)      0.000   2.112
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27_n_0
      SLICE_X2Y103      r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_161/CI
      SLICE_X2Y103      CARRY4 (Prop_carry4_CI_CO[3])
      0.117   2.229 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_161/CO[3]
      net (fo=1, routed)      0.000   2.229
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_161_n_0
      SLICE_X2Y104      r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_161/CI
      SLICE_X2Y104      CARRY4 (Prop_carry4_CI_O[0])
      0.219   2.448 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_161/O[0]
      net (fo=2, routed)      0.812   3.260
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_161_n_7
      SLICE_X3Y103      r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_105/I1
      SLICE_X3Y103      LUT3 (Prop_lut3_I1_0)      0.320   3.580 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_105/O
      net (fo=2, routed)      0.810   4.390
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_105_n_0
      SLICE_X3Y104      r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_109/I3
      SLICE_X3Y104      LUT4 (Prop_lut4_I3_0)      0.332   4.722 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_109/O
      net (fo=1, routed)      0.000   4.722

```

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_109_n_0			
SLICE_X3Y104			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_55/S[0]			
SLICE_X3Y104	CARRY4 (Prop_carry4_S[0]_C0[3])		
		0.532	5.254 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_55/C0[3]			
	net (fo=1, routed)	0.000	5.254
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_55_n_0			
SLICE_X3Y105			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_73/CI			
SLICE_X3Y105	CARRY4 (Prop_carry4_CI_0[3])		
		0.313	5.567 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_73/O[3]			
	net (fo=2, routed)	1.006	6.573
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_73_n_4			
SLICE_X5Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/I0			
SLICE_X5Y99	LUT3 (Prop_lut3_I0_0)	0.331	6.904 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/O			
	net (fo=2, routed)	0.817	7.722
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33_n_0			
SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/I3			
SLICE_X5Y100	LUT4 (Prop_lut4_I3_0)	0.332	8.054 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/O			
	net (fo=1, routed)	0.000	8.054
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37_n_0			
SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/S[0]			
SLICE_X5Y100	CARRY4 (Prop_carry4_S[0]_O[2])		
		0.547	8.601 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/O[2]			
	net (fo=3, routed)	0.779	9.380
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10_n_5			
SLICE_X6Y93			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10/I2			
SLICE_X6Y93	LUT3 (Prop_lut3_I2_0)	0.302	9.682 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10/O			
	net (fo=2, routed)	0.507	10.189
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10_n_0			
SLICE_X7Y93			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2/I1			
SLICE_X7Y93	LUT5 (Prop_lut5_I1_0)	0.124	10.313 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2/O			
	net (fo=2, routed)	0.795	11.107
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2_n_0			
SLICE_X7Y90			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6/I0			
SLICE_X7Y90	LUT6 (Prop_lut6_I0_0)	0.124	11.231 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6/O			

```

net (fo=1, routed) 0.000 11.231
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6_n_0
SLICE_X7Y90 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/S[3]
SLICE_X7Y90 CARRY4 (Prop_carry4_S[3]_CO[3])
0.401 11.632 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/CO[3]
net (fo=1, routed) 0.000 11.632
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1_n_0
SLICE_X7Y91 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/CI
SLICE_X7Y91 CARRY4 (Prop_carry4_CI_0[1])
0.348 11.980 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/O[1]
net (fo=3, routed) 0.000 11.980
u_FPGA_Core/Discrete_FIR_Filter_out1[36]
SLICE_X7Y91 FDCE r
u_FPGA_Core/Unit_Delay_out1_reg[20]/D
-----
-----
(clock clk_out1_clk_wiz_0_1 rise edge)
25.000 25.000 r
E3 0.000 25.000 r sys_clk
(IN)
net (fo=0) 0.000 25.000
u_pll/inst/clk_in1
E3 r
u_pll/inst/clkin1_ibufg/I
E3 IBUF (Prop_ibuf_I_0) 1.411 26.411 r
u_pll/inst/clkin1_ibufg/O
net (fo=1, routed) 1.162 27.573
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2 r
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
-7.324 20.249 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
net (fo=1, routed) 1.639 21.888
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16 r
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16 BUFG (Prop_bufg_I_0) 0.091 21.979 r
u_pll/inst/clkout1_buf/O
net (fo=1039, routed) 1.602 23.582
u_FPGA_Core/CLK
SLICE_X7Y91 FDCE r
u_FPGA_Core/Unit_Delay_out1_reg[20]/C
clock pessimism 0.480 24.062
clock uncertainty -0.085 23.978

```

SLICE_X7Y91	FDCE (Setup_fdce_C_D)	0.051	24.029
u_FPGA_Core/Unit_Delay_out1_reg[20]			

required time	24.029
arrival time	-11.980

slack	12.048
-------	--------

Slack (MET) : 12.093ns (required time - arrival time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C
(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination: u_FPGA_Core/Unit_Delay_out1_reg[23]/D

(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Setup (Max at Slow Process Corner)

Requirement: 25.000ns (clk_out1_clk_wiz_0_1 rise@25.000ns -

clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 12.909ns (logic 5.736ns (44.434%) route 7.173ns (55.566%))

Logic Levels: 17 (CARRY4=8 LUT3=4 LUT4=3 LUT5=1 LUT6=1)

Clock Path Skew: 0.035ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): -1.418ns = (23.582 - 25.000)

Source Clock Delay (SCD): -0.894ns

Clock Pessimism Removal (CPR): 0.559ns

Clock Uncertainty: 0.085ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE

Total System Jitter (TSJ): 0.071ns

Discrete Jitter (DJ): 0.154ns

Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist
Resource(s)				

(clock clk_out1_clk_wiz_0_1 rise edge)

0.000	0.000	r
-------	-------	---

E3	0.000	0.000	r	sys_clk
(IN)				

net (fo=0)	0.000	0.000
------------	-------	-------

u_pll/inst/clk_in1

E3			r
----	--	--	---

u_pll/inst/clkin1_ibufg/I

E3	IBUF (Prop_ibuf_I_0)	1.482	1.482	r
----	----------------------	-------	-------	---

u_pll/inst/clkin1_ibufg/O

net (fo=1, routed)	1.233	2.715
--------------------	-------	-------

u_pll/inst/clk_in1_clk_wiz_0

MMCME2_ADV_X1Y2			r
-----------------	--	--	---

u_pll/inst/mmcm_adv_inst/CLKIN1

MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.646	-0.894	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X12Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C				

SLICE_X12Y95	FDCE (Prop_fdce_C_Q)	0.478	-0.416	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/Q				
	net (fo=3, routed)	1.007	0.591	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[39][2]				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/I2				
SLICE_X13Y95	LUT3 (Prop_lut3_I2_0)	0.326	0.917	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/O				
	net (fo=2, routed)	0.690	1.607	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132_n_0				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/I3				
SLICE_X13Y95	LUT4 (Prop_lut4_I3_0)	0.327	1.934	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/O				
	net (fo=1, routed)	0.000	1.934	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135_n_0				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/S[3]				
SLICE_X13Y95	CARRY4 (Prop_carry4_S[3]_CO[3])	0.401	2.335	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/CO[3]				
	net (fo=1, routed)	0.000	2.335	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34_n_0				
SLICE_X13Y96				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/CI				
SLICE_X13Y96	CARRY4 (Prop_carry4_CI_CO[3])	0.114	2.449	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/CO[3]				
	net (fo=1, routed)	0.000	2.449	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168_n_0				
SLICE_X13Y97				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_168/CI				
SLICE_X13Y97	CARRY4 (Prop_carry4_CI_O[3])	0.313	2.762	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_168/O[3]				

	net (fo=2, routed)	1.013	3.774	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_168_n_4				
SLICE_X13Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_118/I2				
SLICE_X13Y91	LUT3 (Prop_lut3_I2_0)	0.336	4.110	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_118/O				
	net (fo=2, routed)	0.690	4.800	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_118_n_0				
SLICE_X13Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_122/I3				
SLICE_X13Y91	LUT4 (Prop_lut4_I3_0)	0.327	5.127	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_122/O				
	net (fo=1, routed)	0.000	5.127	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_122_n_0				
SLICE_X13Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_57/S[3]				
SLICE_X13Y91	CARRY4 (Prop_carry4_S[3]_CO[3])			
		0.401	5.528	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_57/CO[3]				
	net (fo=1, routed)	0.000	5.528	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_57_n_0				
SLICE_X13Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_79/CI				
SLICE_X13Y92	CARRY4 (Prop_carry4_CI_0[2])			
		0.239	5.767	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_79/O[2]				
	net (fo=2, routed)	0.967	6.734	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_79_n_5				
SLICE_X9Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_34/I1				
SLICE_X9Y92	LUT3 (Prop_lut3_I1_0)	0.332	7.066	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_34/O				
	net (fo=2, routed)	0.690	7.756	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_34_n_0				
SLICE_X9Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_38/I3				
SLICE_X9Y92	LUT4 (Prop_lut4_I3_0)	0.327	8.083	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_38/O				
	net (fo=1, routed)	0.000	8.083	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_38_n_0				
SLICE_X9Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15/S[3]				
SLICE_X9Y92	CARRY4 (Prop_carry4_S[3]_CO[3])			
		0.401	8.484	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15/CO[3]				
	net (fo=1, routed)	0.000	8.484	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15_n_0				
SLICE_X9Y93				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_11/CI				
SLICE_X9Y93	CARRY4 (Prop_carry4_CI_0[3])			

		0.313	8.797	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_11/0[3]				
	net (fo=3, routed)	0.730	9.527	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_11_n_4				
	SLICE_X7Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/I1				
	SLICE_X7Y92 LUT3 (Prop_lut3_I1_0)	0.306	9.833	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/0				
	net (fo=2, routed)	0.579	10.412	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13_n_0				
	SLICE_X5Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/I1				
	SLICE_X5Y92 LUT5 (Prop_lut5_I1_0)	0.124	10.536	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/0				
	net (fo=2, routed)	0.808	11.344	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3_n_0				
	SLICE_X7Y91			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/I0				
	SLICE_X7Y91 LUT6 (Prop_lut6_I0_0)	0.124	11.468	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/0				
	net (fo=1, routed)	0.000	11.468	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6_n_0				
	SLICE_X7Y91			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/S[0]				
	SLICE_X7Y91 CARRY4 (Prop_carry4_S[0]_0[2])			
		0.547	12.015	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/0[2]				
	net (fo=2, routed)	0.000	12.015	
u_FPGA_Core/Discrete_FIR_Filter_out1[37]				
	SLICE_X7Y91 FDCE			r
u_FPGA_Core/Unit_Delay_out1_reg[23]/D				

(clock clk_out1_clk_wiz_0_1 rise edge)				
		25.000	25.000	r
E3		0.000	25.000	r sys_clk
(IN)				
	net (fo=0)	0.000	25.000	
u_pll/inst/clk_in1				
	E3			r
u_pll/inst/clkin1_ibufg/I				
	E3 IBUF (Prop_ibuf_I_0)	1.411	26.411	r
u_pll/inst/clkin1_ibufg/0				
	net (fo=1, routed)	1.162	27.573	
u_pll/inst/clk_in1_clk_wiz_0				
	MMCME2_ADV_X1Y2			r
u_pll/inst/mmcm_adv_inst/CLKIN1				
	MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.324	20.249	r

u_pll/inst/mmcm_adv_inst/CLKOUT0			
net (fo=1, routed)	1.639	21.888	
u_pll/inst/clk_out1_clk_wiz_0			
BUFGCTRL_X0Y16			r
u_pll/inst/clkout1_buf/I			
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.091	21.979 r
u_pll/inst/clkout1_buf/O			
net (fo=1039, routed)	1.602	23.582	
u_FPGA_Core/CLK			
SLICE_X7Y91	FDCE		r
u_FPGA_Core/Unit_Delay_out1_reg[23]/C			
clock pessimism	0.559	24.141	
clock uncertainty	-0.085	24.057	
SLICE_X7Y91	FDCE (Setup_fdce_C_D)	0.051	24.108
u_FPGA_Core/Unit_Delay_out1_reg[23]			

	required time	24.108	
	arrival time	-12.015	

	slack	12.093	

Slack (MET) : 12.161ns (required time - arrival time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/C
 (rising edge-triggered cell FDCE clocked by
 clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination: u_FPGA_Core/Unit_Delay_out1_reg[19]/D
 (rising edge-triggered cell FDCE clocked by
 clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Setup (Max at Slow Process Corner)

Requirement: 25.000ns (clk_out1_clk_wiz_0_1 rise@25.000ns -
 clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 12.698ns (logic 5.834ns (45.943%) route 6.864ns
 (54.057%))

Logic Levels: 17 (CARRY4=8 LUT3=4 LUT4=3 LUT5=1 LUT6=1)

Clock Path Skew: -0.107ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): -1.418ns = (23.582 - 25.000)

Source Clock Delay (SCD): -0.831ns

Clock Pessimism Removal (CPR): 0.480ns

Clock Uncertainty: 0.085ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE

Total System Jitter (TSJ): 0.071ns

Discrete Jitter (DJ): 0.154ns

Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist
Resource(s)				

(clock clk_out1_clk_wiz_0_1 rise edge)

		0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.482	1.482	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.233	2.715	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.709	-0.831	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X4Y102	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/C				

SLICE_X4Y102	FDCE (Prop_fdce_C_Q)	0.419	-0.412	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/Q				
	net (fo=3, routed)	0.848	0.436	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[53][0]				
SLICE_X2Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85/I1				
SLICE_X2Y102	LUT3 (Prop_lut3_I1_0)	0.325	0.761	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85/O				
	net (fo=2, routed)	0.490	1.251	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85_n_0				
SLICE_X2Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88/I3				
SLICE_X2Y102	LUT4 (Prop_lut4_I3_0)	0.328	1.579	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88/O				
	net (fo=1, routed)	0.000	1.579	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88_n_0				
SLICE_X2Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27/S[1]				
SLICE_X2Y102	CARRY4 (Prop_carry4_S[1]_C0[3])	0.533	2.112	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27/C0[3]				
	net (fo=1, routed)	0.000	2.112	

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27_n_0			
SLICE_X2Y103			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_161/CI			
SLICE_X2Y103	CARRY4 (Prop_carry4_CI_CO[3])		
		0.117	2.229 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_161/CO[3]			
	net (fo=1, routed)	0.000	2.229
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_161_n_0			
SLICE_X2Y104			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_161/CI			
SLICE_X2Y104	CARRY4 (Prop_carry4_CI_O[0])		
		0.219	2.448 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_161/O[0]			
	net (fo=2, routed)	0.812	3.260
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_161_n_7			
SLICE_X3Y103			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_105/I1			
SLICE_X3Y103	LUT3 (Prop_lut3_I1_O)	0.320	3.580 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_105/O			
	net (fo=2, routed)	0.810	4.390
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_105_n_0			
SLICE_X3Y104			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_109/I3			
SLICE_X3Y104	LUT4 (Prop_lut4_I3_O)	0.332	4.722 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_109/O			
	net (fo=1, routed)	0.000	4.722
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_109_n_0			
SLICE_X3Y104			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_55/S[0]			
SLICE_X3Y104	CARRY4 (Prop_carry4_S[0]_CO[3])		
		0.532	5.254 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_55/CO[3]			
	net (fo=1, routed)	0.000	5.254
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_55_n_0			
SLICE_X3Y105			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_73/CI			
SLICE_X3Y105	CARRY4 (Prop_carry4_CI_O[3])		
		0.313	5.567 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_73/O[3]			
	net (fo=2, routed)	1.006	6.573
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_73_n_4			
SLICE_X5Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/I0			
SLICE_X5Y99	LUT3 (Prop_lut3_I0_O)	0.331	6.904 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/O			
	net (fo=2, routed)	0.817	7.722
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33_n_0			
SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/I3			
SLICE_X5Y100	LUT4 (Prop_lut4_I3_O)	0.332	8.054 r

```

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/O
      net (fo=1, routed) 0.000 8.054
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37_n_0
      SLICE_X5Y100 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/S[0]
      SLICE_X5Y100 CARRY4 (Prop_carry4_S[0]_O[2])
      0.547 8.601 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/O[2]
      net (fo=3, routed) 0.779 9.380
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10_n_5
      SLICE_X6Y93 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10/I2
      SLICE_X6Y93 LUT3 (Prop_lut3_I2_O) 0.302 9.682 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10/O
      net (fo=2, routed) 0.507 10.189
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10_n_0
      SLICE_X7Y93 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2/I1
      SLICE_X7Y93 LUT5 (Prop_lut5_I1_O) 0.124 10.313 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2/O
      net (fo=2, routed) 0.795 11.107
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2_n_0
      SLICE_X7Y90 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6/I0
      SLICE_X7Y90 LUT6 (Prop_lut6_I0_O) 0.124 11.231 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6/O
      net (fo=1, routed) 0.000 11.231
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6_n_0
      SLICE_X7Y90 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/S[3]
      SLICE_X7Y90 CARRY4 (Prop_carry4_S[3]_CO[3])
      0.401 11.632 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/CO[3]
      net (fo=1, routed) 0.000 11.632
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1_n_0
      SLICE_X7Y91 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/CI
      SLICE_X7Y91 CARRY4 (Prop_carry4_CI_O[0])
      0.235 11.867 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/O[0]
      net (fo=3, routed) 0.000 11.867
u_FPGA_Core/Discrete_FIR_Filter_out1[35]
      SLICE_X7Y91 FDCE r
u_FPGA_Core/Unit_Delay_out1_reg[19]/D
      -----
      -----
      (clock clk_out1_clk_wiz_0_1 rise edge)
      25.000 25.000 r
E3 0.000 25.000 r sys_clk

```

```

(IN)
net (fo=0) 0.000 25.000
u_pll/inst/clk_in1
E3 r
u_pll/inst/clkin1_ibufg/I
E3 IBUF (Prop_ibuf_I_0) 1.411 26.411 r
u_pll/inst/clkin1_ibufg/0
net (fo=1, routed) 1.162 27.573
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2 r
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
-7.324 20.249 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
net (fo=1, routed) 1.639 21.888
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16 r
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16 BUFG (Prop_bufg_I_0) 0.091 21.979 r
u_pll/inst/clkout1_buf/0
net (fo=1039, routed) 1.602 23.582
u_FPGA_Core/CLK
SLICE_X7Y91 FDCE r
u_FPGA_Core/Unit_Delay_out1_reg[19]/C
clock pessimism 0.480 24.062
clock uncertainty -0.085 23.978
SLICE_X7Y91 FDCE (Setup_fdce_C_D) 0.051 24.029
u_FPGA_Core/Unit_Delay_out1_reg[19]
-----
required time 24.029
arrival time -11.867
-----
slack 12.161

```

Slack (MET) : 12.486ns (required time - arrival time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/C

(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination: u_FPGA_Core/Unit_Delay_out1_reg[16]/D

(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Setup (Max at Slow Process Corner)

Requirement: 25.000ns (clk_out1_clk_wiz_0_1 rise@25.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 12.373ns (logic 5.795ns (46.835%) route 6.578ns
(53.165%))

Logic Levels: 15 (CARRY4=7 LUT3=4 LUT4=3 LUT5=1)

Clock Path Skew: -0.108ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): -1.419ns = (23.581 - 25.000)
 Source Clock Delay (SCD): -0.831ns
 Clock Pessimism Removal (CPR): 0.480ns
 Clock Uncertainty: 0.085ns $((TSJ^2 + DJ^2)^{1/2} / 2 + PE)$
 Total System Jitter (TSJ): 0.071ns
 Discrete Jitter (DJ): 0.154ns
 Phase Error (PE): 0.000ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.482	1.482	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.233	2.715	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.709	-0.831	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X4Y102	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/C				

SLICE_X4Y102	FDCE (Prop_fdce_C_Q)	0.419	-0.412	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/Q				
	net (fo=3, routed)	0.848	0.436	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[53][0]				
SLICE_X2Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85/I1				
SLICE_X2Y102	LUT3 (Prop_lut3_I1_0)	0.325	0.761	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85/O				
	net (fo=2, routed)	0.490	1.251	

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85_n_0				
SLICE_X2Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88/I3				
SLICE_X2Y102	LUT4 (Prop_lut4_I3_0)	0.328	1.579	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88/O				
	net (fo=1, routed)	0.000	1.579	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88_n_0				
SLICE_X2Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27/S[1]				
SLICE_X2Y102	CARRY4 (Prop_carry4_S[1]_O[3])			
		0.643	2.222	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27/O[3]				
	net (fo=2, routed)	0.808	3.030	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27_n_4				
SLICE_X3Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_102/I1				
SLICE_X3Y102	LUT3 (Prop_lut3_I1_0)	0.337	3.367	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_102/O				
	net (fo=2, routed)	0.690	4.057	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_102_n_0				
SLICE_X3Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_106/I3				
SLICE_X3Y102	LUT4 (Prop_lut4_I3_0)	0.327	4.384	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_106/O				
	net (fo=1, routed)	0.000	4.384	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_106_n_0				
SLICE_X3Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_55/S[3]				
SLICE_X3Y102	CARRY4 (Prop_carry4_S[3]_CO[3])			
		0.401	4.785	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_55/CO[3]				
	net (fo=1, routed)	0.000	4.785	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_55_n_0				
SLICE_X3Y103				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_55/CI				
SLICE_X3Y103	CARRY4 (Prop_carry4_CI_0[3])			
		0.313	5.098	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_55/O[3]				
	net (fo=2, routed)	0.980	6.077	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_55_n_4				
SLICE_X5Y97				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_29/I0				
SLICE_X5Y97	LUT3 (Prop_lut3_I0_0)	0.331	6.408	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_29/O				
	net (fo=2, routed)	0.834	7.242	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_29_n_0				
SLICE_X5Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_33/I3				
SLICE_X5Y98	LUT4 (Prop_lut4_I3_0)	0.332	7.574	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_33/O				

```

net (fo=1, routed) 0.000 7.574
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_33_n_0
SLICE_X5Y98 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/S[0]
SLICE_X5Y98 CARRY4 (Prop_carry4_S[0]_CO[3])
0.532 8.106 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/CO[3]
net (fo=1, routed) 0.000 8.106
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14_n_0
SLICE_X5Y99 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/CI
SLICE_X5Y99 CARRY4 (Prop_carry4_CI_0[1])
0.334 8.440 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/O[1]
net (fo=3, routed) 0.783 9.223
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14_n_6
SLICE_X6Y92 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11/I2
SLICE_X6Y92 LUT3 (Prop_lut3_I2_0) 0.303 9.526 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11/O
net (fo=2, routed) 0.515 10.041
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11_n_0
SLICE_X7Y92 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3/I1
SLICE_X7Y92 LUT5 (Prop_lut5_I1_0) 0.124 10.165 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3/O
net (fo=2, routed) 0.631 10.796
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3_n_0
SLICE_X7Y89 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/DI[2]
SLICE_X7Y89 CARRY4 (Prop_carry4_DI[2]_CO[3])
0.398 11.194 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/CO[3]
net (fo=1, routed) 0.000 11.194
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1_n_0
SLICE_X7Y90 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/CI
SLICE_X7Y90 CARRY4 (Prop_carry4_CI_0[1])
0.348 11.542 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/O[1]
net (fo=3, routed) 0.000 11.542
u_FPGA_Core/Discrete_FIR_Filter_out1[32]
SLICE_X7Y90 FDCE r
u_FPGA_Core/Unit_Delay_out1_reg[16]/D
-----
(clock clk_out1_clk_wiz_0_1 rise edge)
25.000 25.000 r
E3 0.000 25.000 r sys_clk

```

```

(IN)
net (fo=0) 0.000 25.000
u_pll/inst/clk_in1
E3 r
u_pll/inst/clkin1_ibufg/I
E3 IBUF (Prop_ibuf_I_0) 1.411 26.411 r
u_pll/inst/clkin1_ibufg/O
net (fo=1, routed) 1.162 27.573
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2 r
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
-7.324 20.249 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
net (fo=1, routed) 1.639 21.888
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16 r
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16 BUFG (Prop_bufg_I_0) 0.091 21.979 r
u_pll/inst/clkout1_buf/O
net (fo=1039, routed) 1.601 23.581
u_FPGA_Core/CLK
SLICE_X7Y90 FDCE r
u_FPGA_Core/Unit_Delay_out1_reg[16]/C
clock pessimism 0.480 24.061
clock uncertainty -0.085 23.977
SLICE_X7Y90 FDCE (Setup_fdce_C_D) 0.051 24.028
u_FPGA_Core/Unit_Delay_out1_reg[16]
-----
required time 24.028
arrival time -11.542
-----
slack 12.486

```

Slack (MET) : 12.487ns (required time - arrival time)

Source:

```

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Destination: u_FPGA_Core/Unit_Delay_out1_reg[18]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group: clk_out1_clk_wiz_0_1
Path Type: Setup (Max at Slow Process Corner)
Requirement: 25.000ns (clk_out1_clk_wiz_0_1 rise@25.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)
Data Path Delay: 12.514ns (logic 5.686ns (45.438%) route 6.828ns
(54.562%))
Logic Levels: 15 (CARRY4=6 LUT3=4 LUT4=3 LUT5=1 LUT6=1)
Clock Path Skew: 0.034ns (DCD - SCD + CPR)

```

Destination Clock Delay (DCD): -1.419ns = (23.581 - 25.000)
 Source Clock Delay (SCD): -0.894ns
 Clock Pessimism Removal (CPR): 0.559ns
 Clock Uncertainty: 0.085ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
 Total System Jitter (TSJ): 0.071ns
 Discrete Jitter (DJ): 0.154ns
 Phase Error (PE): 0.000ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
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(clock clk_out1_clk_wiz_0_1 rise edge)				
		0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.482	1.482	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.233	2.715	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.646	-0.894	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X12Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C				

SLICE_X12Y95	FDCE (Prop_fdce_C_Q)	0.478	-0.416	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/Q				
	net (fo=3, routed)	1.007	0.591	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[39][2]				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/I2				
SLICE_X13Y95	LUT3 (Prop_lut3_I2_0)	0.326	0.917	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/O				
	net (fo=2, routed)	0.690	1.607	

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132_n_0				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/I3				
SLICE_X13Y95	LUT4 (Prop_lut4_I3_0)	0.327	1.934	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/O				
	net (fo=1, routed)	0.000	1.934	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135_n_0				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/S[3]				
SLICE_X13Y95	CARRY4 (Prop_carry4_S[3]_CO[3])			
		0.401	2.335	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/CO[3]				
	net (fo=1, routed)	0.000	2.335	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34_n_0				
SLICE_X13Y96				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/CI				
SLICE_X13Y96	CARRY4 (Prop_carry4_CI_0[0])			
		0.223	2.558	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/O[0]				
	net (fo=2, routed)	1.005	3.562	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168_n_7				
SLICE_X13Y89				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121/I2				
SLICE_X13Y89	LUT3 (Prop_lut3_I2_0)	0.325	3.887	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121/O				
	net (fo=2, routed)	0.652	4.540	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121_n_0				
SLICE_X13Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125/I3				
SLICE_X13Y90	LUT4 (Prop_lut4_I3_0)	0.332	4.872	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125/O				
	net (fo=1, routed)	0.000	4.872	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125_n_0				
SLICE_X13Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57/S[0]				
SLICE_X13Y90	CARRY4 (Prop_carry4_S[0]_O[3])			
		0.606	5.478	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57/O[3]				
	net (fo=2, routed)	0.744	6.222	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57_n_4				
SLICE_X9Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37/I1				
SLICE_X9Y90	LUT3 (Prop_lut3_I1_0)	0.331	6.553	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37/O				
	net (fo=2, routed)	0.817	7.370	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37_n_0				
SLICE_X9Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41/I3				
SLICE_X9Y91	LUT4 (Prop_lut4_I3_0)	0.332	7.702	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41/O				

```

net (fo=1, routed) 0.000 7.702
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41_n_0
SLICE_X9Y91 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15/S[0]
SLICE_X9Y91 CARRY4 (Prop_carry4_S[0]_CO[3])
0.532 8.234 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15/CO[3]
net (fo=1, routed) 0.000 8.234
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15_n_0
SLICE_X9Y92 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15/CI
SLICE_X9Y92 CARRY4 (Prop_carry4_CI_0[3])
0.313 8.547 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15/O[3]
net (fo=3, routed) 0.819 9.366
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15_n_4
SLICE_X4Y91 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_17/I1
SLICE_X4Y91 LUT3 (Prop_lut3_I1_0) 0.306 9.672 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_17/O
net (fo=2, routed) 0.637 10.309
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_17_n_0
SLICE_X4Y90 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_5/I1
SLICE_X4Y90 LUT5 (Prop_lut5_I1_0) 0.124 10.433 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_5/O
net (fo=2, routed) 0.457 10.890
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_5_n_0
SLICE_X7Y90 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_9/I0
SLICE_X7Y90 LUT6 (Prop_lut6_I0_0) 0.124 11.014 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_9/O
net (fo=1, routed) 0.000 11.014
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_9_n_0
SLICE_X7Y90 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/S[0]
SLICE_X7Y90 CARRY4 (Prop_carry4_S[0]_O[3])
0.606 11.620 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/O[3]
net (fo=3, routed) 0.000 11.620
u_FPGA_Core/Discrete_FIR_Filter_out1[34]
SLICE_X7Y90 FDCE r
u_FPGA_Core/Unit_Delay_out1_reg[18]/D
-----
(clock clk_out1_clk_wiz_0_1 rise edge)
25.000 25.000 r
E3 0.000 25.000 r sys_clk
(IN)

```

u_pll/inst/clk_in1	net (fo=0)	0.000	25.000	
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.411	26.411	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.162	27.573	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.324	20.249	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.639	21.888	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.091	21.979	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.601	23.581	
u_FPGA_Core/CLK				
SLICE_X7Y90	FDCE			r
u_FPGA_Core/Unit_Delay_out1_reg[18]/C				
	clock pessimism	0.559	24.140	
	clock uncertainty	-0.085	24.056	
SLICE_X7Y90	FDCE (Setup_fdce_C_D)	0.051	24.107	
u_FPGA_Core/Unit_Delay_out1_reg[18]				

	required time		24.107	
	arrival time		-11.620	

	slack		12.487	

Slack (MET) : 12.546ns (required time - arrival time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination: u_FPGA_Core/Unit_Delay_out1_reg[17]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Setup (Max at Slow Process Corner)

Requirement: 25.000ns (clk_out1_clk_wiz_0_1 rise@25.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 12.455ns (logic 5.627ns (45.179%) route 6.828ns
(54.821%))

Logic Levels: 15 (CARRY4=6 LUT3=4 LUT4=3 LUT5=1 LUT6=1)

Clock Path Skew: 0.034ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): -1.419ns = (23.581 - 25.000)

Source Clock Delay (SCD): -0.894ns
 Clock Pessimism Removal (CPR): 0.559ns
 Clock Uncertainty: 0.085ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
 Total System Jitter (TSJ): 0.071ns
 Discrete Jitter (DJ): 0.154ns
 Phase Error (PE): 0.000ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
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(clock clk_out1_clk_wiz_0_1 rise edge)				
		0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.482	1.482	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.233	2.715	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.646	-0.894	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X12Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C				

SLICE_X12Y95	FDCE (Prop_fdce_C_Q)	0.478	-0.416	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/Q				
	net (fo=3, routed)	1.007	0.591	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[39][2]				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/I2				
SLICE_X13Y95	LUT3 (Prop_lut3_I2_0)	0.326	0.917	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/O				
	net (fo=2, routed)	0.690	1.607	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132_n_0				

SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/I3				
SLICE_X13Y95	LUT4 (Prop_lut4_I3_0)	0.327	1.934	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/0				
	net (fo=1, routed)	0.000	1.934	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135_n_0				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/S[3]				
SLICE_X13Y95	CARRY4 (Prop_carry4_S[3]_CO[3])			
		0.401	2.335	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/CO[3]				
	net (fo=1, routed)	0.000	2.335	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34_n_0				
SLICE_X13Y96				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/CI				
SLICE_X13Y96	CARRY4 (Prop_carry4_CI_0[0])			
		0.223	2.558	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/0[0]				
	net (fo=2, routed)	1.005	3.562	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168_n_7				
SLICE_X13Y89				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121/I2				
SLICE_X13Y89	LUT3 (Prop_lut3_I2_0)	0.325	3.887	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121/0				
	net (fo=2, routed)	0.652	4.540	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121_n_0				
SLICE_X13Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125/I3				
SLICE_X13Y90	LUT4 (Prop_lut4_I3_0)	0.332	4.872	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125/0				
	net (fo=1, routed)	0.000	4.872	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125_n_0				
SLICE_X13Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57/S[0]				
SLICE_X13Y90	CARRY4 (Prop_carry4_S[0]_O[3])			
		0.606	5.478	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57/0[3]				
	net (fo=2, routed)	0.744	6.222	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57_n_4				
SLICE_X9Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37/I1				
SLICE_X9Y90	LUT3 (Prop_lut3_I1_0)	0.331	6.553	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37/0				
	net (fo=2, routed)	0.817	7.370	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37_n_0				
SLICE_X9Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41/I3				
SLICE_X9Y91	LUT4 (Prop_lut4_I3_0)	0.332	7.702	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41/0				
	net (fo=1, routed)	0.000	7.702	

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41_n_0				
SLICE_X9Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15/S[0]				
SLICE_X9Y91	CARRY4 (Prop_carry4_S[0]_C0[3])	0.532	8.234	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15/C0[3]				
	net (fo=1, routed)	0.000	8.234	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15_n_0				
SLICE_X9Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15/CI				
SLICE_X9Y92	CARRY4 (Prop_carry4_CI_0[3])	0.313	8.547	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15/O[3]				
	net (fo=3, routed)	0.819	9.366	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15_n_4				
SLICE_X4Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_17/I1				
SLICE_X4Y91	LUT3 (Prop_lut3_I1_0)	0.306	9.672	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_17/O				
	net (fo=2, routed)	0.637	10.309	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_17_n_0				
SLICE_X4Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_5/I1				
SLICE_X4Y90	LUT5 (Prop_lut5_I1_0)	0.124	10.433	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_5/O				
	net (fo=2, routed)	0.457	10.890	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_5_n_0				
SLICE_X7Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_9/I0				
SLICE_X7Y90	LUT6 (Prop_lut6_I0_0)	0.124	11.014	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_9/O				
	net (fo=1, routed)	0.000	11.014	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_9_n_0				
SLICE_X7Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/S[0]				
SLICE_X7Y90	CARRY4 (Prop_carry4_S[0]_O[2])	0.547	11.561	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/O[2]				
	net (fo=3, routed)	0.000	11.561	
u_FPGA_Core/Discrete_FIR_Filter_out1[33]				
SLICE_X7Y90	FDCE			r
u_FPGA_Core/Unit_Delay_out1_reg[17]/D				

(clock clk_out1_clk_wiz_0_1 rise edge)				
		25.000	25.000	r
E3		0.000	25.000	r sys_clk
(IN)				
	net (fo=0)	0.000	25.000	

u_pll/inst/clk_in1			
E3			r
u_pll/inst/clkin1_ibufg/I			
E3	IBUF (Prop_ibuf_I_0)	1.411	26.411 r
u_pll/inst/clkin1_ibufg/O			
	net (fo=1, routed)	1.162	27.573
u_pll/inst/clk_in1_clk_wiz_0			
MMCME2_ADV_X1Y2			r
u_pll/inst/mmcm_adv_inst/CLKIN1			
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.324	20.249 r
u_pll/inst/mmcm_adv_inst/CLKOUT0			
	net (fo=1, routed)	1.639	21.888
u_pll/inst/clk_out1_clk_wiz_0			
BUFGCTRL_X0Y16			r
u_pll/inst/clkout1_buf/I			
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.091	21.979 r
u_pll/inst/clkout1_buf/O			
	net (fo=1039, routed)	1.601	23.581
u_FPGA_Core/CLK			
SLICE_X7Y90	FDCE		r
u_FPGA_Core/Unit_Delay_out1_reg[17]/C			
	clock pessimism	0.559	24.140
	clock uncertainty	-0.085	24.056
SLICE_X7Y90	FDCE (Setup_fdce_C_D)	0.051	24.107
u_FPGA_Core/Unit_Delay_out1_reg[17]			

	required time		24.107
	arrival time		-11.561

	slack		12.546

Slack (MET) : 12.599ns (required time - arrival time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/C
(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination: u_FPGA_Core/Unit_Delay_out1_reg[15]/D
(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Setup (Max at Slow Process Corner)

Requirement: 25.000ns (clk_out1_clk_wiz_0_1 rise@25.000ns -

clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 12.260ns (logic 5.682ns (46.345%) route 6.578ns

(53.655%))

Logic Levels: 15 (CARRY4=7 LUT3=4 LUT4=3 LUT5=1)

Clock Path Skew: -0.108ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): -1.419ns = (23.581 - 25.000)

Source Clock Delay (SCD): -0.831ns

Clock Pessimism Removal (CPR): 0.480ns
 Clock Uncertainty: 0.085ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
 Total System Jitter (TSJ): 0.071ns
 Discrete Jitter (DJ): 0.154ns
 Phase Error (PE): 0.000ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.482	1.482	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.233	2.715	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.709	-0.831	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X4Y102	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/C				

SLICE_X4Y102	FDCE (Prop_fdce_C_Q)	0.419	-0.412	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[53][0]/Q				
	net (fo=3, routed)	0.848	0.436	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[53][0]				
SLICE_X2Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85/I1				
SLICE_X2Y102	LUT3 (Prop_lut3_I1_0)	0.325	0.761	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85/O				
	net (fo=2, routed)	0.490	1.251	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_85_n_0				
SLICE_X2Y102				r

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88/I3	SLICE_X2Y102	LUT4 (Prop_lut4_I3_0)	0.328	1.579	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88/O		net (fo=1, routed)	0.000	1.579	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_88_n_0	SLICE_X2Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27/S[1]	SLICE_X2Y102	CARRY4 (Prop_carry4_S[1]_O[3])	0.643	2.222	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27/O[3]		net (fo=2, routed)	0.808	3.030	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_27_n_4	SLICE_X3Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_102/I1	SLICE_X3Y102	LUT3 (Prop_lut3_I1_0)	0.337	3.367	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_102/O		net (fo=2, routed)	0.690	4.057	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_102_n_0	SLICE_X3Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_106/I3	SLICE_X3Y102	LUT4 (Prop_lut4_I3_0)	0.327	4.384	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_106/O		net (fo=1, routed)	0.000	4.384	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_106_n_0	SLICE_X3Y102				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_55/S[3]	SLICE_X3Y102	CARRY4 (Prop_carry4_S[3]_CO[3])	0.401	4.785	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_55/CO[3]		net (fo=1, routed)	0.000	4.785	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_55_n_0	SLICE_X3Y103				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_55/CI	SLICE_X3Y103	CARRY4 (Prop_carry4_CI_0[3])	0.313	5.098	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_55/O[3]		net (fo=2, routed)	0.980	6.077	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_55_n_4	SLICE_X5Y97				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_29/I0	SLICE_X5Y97	LUT3 (Prop_lut3_I0_0)	0.331	6.408	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_29/O		net (fo=2, routed)	0.834	7.242	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_29_n_0	SLICE_X5Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_33/I3	SLICE_X5Y98	LUT4 (Prop_lut4_I3_0)	0.332	7.574	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_33/O		net (fo=1, routed)	0.000	7.574	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_33_n_0					

SLICE_X5Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/S[0]				
SLICE_X5Y98	CARRY4 (Prop_carry4_S[0]_CO[3])	0.532	8.106	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/CO[3]				
	net (fo=1, routed)	0.000	8.106	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14_n_0				
SLICE_X5Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/CI				
SLICE_X5Y99	CARRY4 (Prop_carry4_CI_0[1])	0.334	8.440	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/O[1]				
	net (fo=3, routed)	0.783	9.223	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14_n_6				
SLICE_X6Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11/I2				
SLICE_X6Y92	LUT3 (Prop_lut3_I2_0)	0.303	9.526	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11/O				
	net (fo=2, routed)	0.515	10.041	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11_n_0				
SLICE_X7Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3/I1				
SLICE_X7Y92	LUT5 (Prop_lut5_I1_0)	0.124	10.165	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3/O				
	net (fo=2, routed)	0.631	10.796	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3_n_0				
SLICE_X7Y89				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/DI[2]				
SLICE_X7Y89	CARRY4 (Prop_carry4_DI[2]_CO[3])	0.398	11.194	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/CO[3]				
	net (fo=1, routed)	0.000	11.194	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1_n_0				
SLICE_X7Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/CI				
SLICE_X7Y90	CARRY4 (Prop_carry4_CI_0[0])	0.235	11.429	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/O[0]				
	net (fo=3, routed)	0.000	11.429	
u_FPGA_Core/Discrete_FIR_Filter_out1[31]				
SLICE_X7Y90	FDCE			r
u_FPGA_Core/Unit_Delay_out1_reg[15]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)	25.000	25.000	r
E3		0.000	25.000	r sys_clk
(IN)				
	net (fo=0)	0.000	25.000	

u_pll/inst/clk_in1			
E3			r
u_pll/inst/clkin1_ibufg/I			
E3	IBUF (Prop_ibuf_I_0)	1.411	26.411 r
u_pll/inst/clkin1_ibufg/O			
	net (fo=1, routed)	1.162	27.573
u_pll/inst/clk_in1_clk_wiz_0			
MMCME2_ADV_X1Y2			r
u_pll/inst/mmcm_adv_inst/CLKIN1			
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.324	20.249 r
u_pll/inst/mmcm_adv_inst/CLKOUT0			
	net (fo=1, routed)	1.639	21.888
u_pll/inst/clk_out1_clk_wiz_0			
BUFGCTRL_X0Y16			r
u_pll/inst/clkout1_buf/I			
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.091	21.979 r
u_pll/inst/clkout1_buf/O			
	net (fo=1039, routed)	1.601	23.581
u_FPGA_Core/CLK			
SLICE_X7Y90	FDCE		r
u_FPGA_Core/Unit_Delay_out1_reg[15]/C			
	clock pessimism	0.480	24.061
	clock uncertainty	-0.085	23.977
SLICE_X7Y90	FDCE (Setup_fdce_C_D)	0.051	24.028
u_FPGA_Core/Unit_Delay_out1_reg[15]			

	required time		24.028
	arrival time		-11.429

	slack		12.599

Slack (MET) : 12.708ns (required time - arrival time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C
(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination: u_FPGA_Core/Unit_Delay_out1_reg[12]/D
(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Setup (Max at Slow Process Corner)

Requirement: 25.000ns (clk_out1_clk_wiz_0_1 rise@25.000ns -

clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 12.292ns (logic 5.527ns (44.964%) route 6.765ns (55.036%))

Logic Levels: 15 (CARRY4=6 LUT3=4 LUT4=3 LUT5=1 LUT6=1)

Clock Path Skew: 0.034ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): -1.419ns = (23.581 - 25.000)

Source Clock Delay (SCD): -0.894ns

Clock Pessimism Removal (CPR): 0.559ns
 Clock Uncertainty: 0.085ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
 Total System Jitter (TSJ): 0.071ns
 Discrete Jitter (DJ): 0.154ns
 Phase Error (PE): 0.000ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.482	1.482	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.233	2.715	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.646	-0.894	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X12Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C				

SLICE_X12Y95	FDCE (Prop_fdce_C_Q)	0.478	-0.416	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/Q				
	net (fo=3, routed)	1.007	0.591	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[39][2]				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/I2				
SLICE_X13Y95	LUT3 (Prop_lut3_I2_0)	0.326	0.917	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/O				
	net (fo=2, routed)	0.690	1.607	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132_n_0				
SLICE_X13Y95				r

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/I3	SLICE_X13Y95	LUT4 (Prop_lut4_I3_0)	0.327	1.934	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/O		net (fo=1, routed)	0.000	1.934	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135_n_0	SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/S[3]	SLICE_X13Y95	CARRY4 (Prop_carry4_S[3]_CO[3])	0.401	2.335	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/CO[3]		net (fo=1, routed)	0.000	2.335	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34_n_0	SLICE_X13Y96				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/CI	SLICE_X13Y96	CARRY4 (Prop_carry4_CI_0[0])	0.223	2.558	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/O[0]		net (fo=2, routed)	1.005	3.562	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168_n_7	SLICE_X13Y89				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121/I2	SLICE_X13Y89	LUT3 (Prop_lut3_I2_0)	0.325	3.887	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121/O		net (fo=2, routed)	0.652	4.540	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121_n_0	SLICE_X13Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125/I3	SLICE_X13Y90	LUT4 (Prop_lut4_I3_0)	0.332	4.872	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125/O		net (fo=1, routed)	0.000	4.872	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125_n_0	SLICE_X13Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57/S[0]	SLICE_X13Y90	CARRY4 (Prop_carry4_S[0]_O[3])	0.606	5.478	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57/O[3]		net (fo=2, routed)	0.744	6.222	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57_n_4	SLICE_X9Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37/I1	SLICE_X9Y90	LUT3 (Prop_lut3_I1_0)	0.331	6.553	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37/O		net (fo=2, routed)	0.817	7.370	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37_n_0	SLICE_X9Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41/I3	SLICE_X9Y91	LUT4 (Prop_lut4_I3_0)	0.332	7.702	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41/O		net (fo=1, routed)	0.000	7.702	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41_n_0					

SLICE_X9Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15/S[0]				
SLICE_X9Y91	CARRY4 (Prop_carry4_S[0]_O[2])	0.547	8.249	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15/O[2]				
	net (fo=3, routed)	0.858	9.106	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15_n_5				
SLICE_X5Y88				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_10/I1				
SLICE_X5Y88	LUT3 (Prop_lut3_I1_0)	0.302	9.408	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_10/O				
	net (fo=2, routed)	0.694	10.103	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_10_n_0				
SLICE_X5Y88				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_2/I1				
SLICE_X5Y88	LUT5 (Prop_lut5_I1_0)	0.124	10.227	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_2/O				
	net (fo=2, routed)	0.299	10.525	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_2_n_0				
SLICE_X7Y88				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_6/I0				
SLICE_X7Y88	LUT6 (Prop_lut6_I0_0)	0.124	10.649	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_6/O				
	net (fo=1, routed)	0.000	10.649	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_6_n_0				
SLICE_X7Y88				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_1/S[3]				
SLICE_X7Y88	CARRY4 (Prop_carry4_S[3]_CO[3])	0.401	11.050	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_1/CO[3]				
	net (fo=1, routed)	0.000	11.050	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_1_n_0				
SLICE_X7Y89				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/CI				
SLICE_X7Y89	CARRY4 (Prop_carry4_CI_0[1])	0.348	11.398	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/O[1]				
	net (fo=3, routed)	0.000	11.398	
u_FPGA_Core/Discrete_FIR_Filter_out1[28]				
SLICE_X7Y89	FDCE			r
u_FPGA_Core/Unit_Delay_out1_reg[12]/D				

(clock clk_out1_clk_wiz_0_1 rise edge)				
		25.000	25.000	r
E3		0.000	25.000	r sys_clk
(IN)				
	net (fo=0)	0.000	25.000	
u_pll/inst/clk_in1				

E3			r
u_pll/inst/clkin1_ibufg/I			
E3	IBUF (Prop_ibuf_I_0)	1.411	26.411 r
u_pll/inst/clkin1_ibufg/O			
	net (fo=1, routed)	1.162	27.573
u_pll/inst/clk_in1_clk_wiz_0			
MMCME2_ADV_X1Y2			r
u_pll/inst/mmcm_adv_inst/CLKIN1			
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.324	20.249 r
u_pll/inst/mmcm_adv_inst/CLKOUT0			
	net (fo=1, routed)	1.639	21.888
u_pll/inst/clk_out1_clk_wiz_0			
BUFGCTRL_X0Y16			r
u_pll/inst/clkout1_buf/I			
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.091	21.979 r
u_pll/inst/clkout1_buf/O			
	net (fo=1039, routed)	1.601	23.581
u_FPGA_Core/CLK			
SLICE_X7Y89	FDCE		r
u_FPGA_Core/Unit_Delay_out1_reg[12]/C			
	clock pessimism	0.559	24.140
	clock uncertainty	-0.085	24.056
SLICE_X7Y89	FDCE (Setup_fdce_C_D)	0.051	24.107
u_FPGA_Core/Unit_Delay_out1_reg[12]			

	required time		24.107
	arrival time		-11.398

	slack		12.708

Slack (MET) : 12.727ns (required time - arrival time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination: u_FPGA_Core/Unit_Delay_out1_reg[14]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Setup (Max at Slow Process Corner)

Requirement: 25.000ns (clk_out1_clk_wiz_0_1 rise@25.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 12.273ns (logic 5.508ns (44.879%) route 6.765ns
(55.121%))

Logic Levels: 15 (CARRY4=6 LUT3=4 LUT4=3 LUT5=1 LUT6=1)

Clock Path Skew: 0.034ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): -1.419ns = (23.581 - 25.000)

Source Clock Delay (SCD): -0.894ns

Clock Pessimism Removal (CPR): 0.559ns

Clock Uncertainty: 0.085ns $((TSJ^2 + DJ^2)^{1/2}) / 2 + PE$
 Total System Jitter (TSJ): 0.071ns
 Discrete Jitter (DJ): 0.154ns
 Phase Error (PE): 0.000ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.482	1.482	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.233	2.715	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.646	-0.894	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X12Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C				

SLICE_X12Y95	FDCE (Prop_fdce_C_Q)	0.478	-0.416	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/Q				
	net (fo=3, routed)	1.007	0.591	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[39][2]				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/I2				
SLICE_X13Y95	LUT3 (Prop_lut3_I2_0)	0.326	0.917	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/O				
	net (fo=2, routed)	0.690	1.607	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132_n_0				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/I3				

SLICE_X13Y95	LUT4 (Prop_lut4_I3_0)	0.327	1.934	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/0	net (fo=1, routed)	0.000	1.934	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135_n_0				
SLICE_X13Y95				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/S[3]				
SLICE_X13Y95	CARRY4 (Prop_carry4_S[3]_CO[3])	0.401	2.335	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/CO[3]	net (fo=1, routed)	0.000	2.335	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34_n_0				
SLICE_X13Y96				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/CI				
SLICE_X13Y96	CARRY4 (Prop_carry4_CI_0[0])	0.223	2.558	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/0[0]	net (fo=2, routed)	1.005	3.562	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168_n_7				
SLICE_X13Y89				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121/I2				
SLICE_X13Y89	LUT3 (Prop_lut3_I2_0)	0.325	3.887	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121/0	net (fo=2, routed)	0.652	4.540	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_121_n_0				
SLICE_X13Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125/I3				
SLICE_X13Y90	LUT4 (Prop_lut4_I3_0)	0.332	4.872	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125/0	net (fo=1, routed)	0.000	4.872	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_125_n_0				
SLICE_X13Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57/S[0]				
SLICE_X13Y90	CARRY4 (Prop_carry4_S[0]_O[3])	0.606	5.478	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57/0[3]	net (fo=2, routed)	0.744	6.222	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_57_n_4				
SLICE_X9Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37/I1				
SLICE_X9Y90	LUT3 (Prop_lut3_I1_0)	0.331	6.553	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37/0	net (fo=2, routed)	0.817	7.370	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_37_n_0				
SLICE_X9Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41/I3				
SLICE_X9Y91	LUT4 (Prop_lut4_I3_0)	0.332	7.702	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41/0	net (fo=1, routed)	0.000	7.702	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_41_n_0				
SLICE_X9Y91				r

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u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15/S[0]
  SLICE_X9Y91      CARRY4 (Prop_carry4_S[0]_O[2])
                                0.547      8.249 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15/O[2]
  net (fo=3, routed)      0.858      9.106
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_15_n_5
  SLICE_X5Y88
                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_10/I1
  SLICE_X5Y88      LUT3 (Prop_lut3_I1_0)      0.302      9.408 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_10/O
  net (fo=2, routed)      0.694      10.103
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_10_n_0
  SLICE_X5Y88
                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_2/I1
  SLICE_X5Y88      LUT5 (Prop_lut5_I1_0)      0.124      10.227 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_2/O
  net (fo=2, routed)      0.299      10.525
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_2_n_0
  SLICE_X7Y88
                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_6/I0
  SLICE_X7Y88      LUT6 (Prop_lut6_I0_0)      0.124      10.649 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_6/O
  net (fo=1, routed)      0.000      10.649
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[10]_i_6_n_0
  SLICE_X7Y88
                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_1/S[3]
  SLICE_X7Y88      CARRY4 (Prop_carry4_S[3]_CO[3])
                                0.401      11.050 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_1/CO[3]
  net (fo=1, routed)      0.000      11.050
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[10]_i_1_n_0
  SLICE_X7Y89
                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/CI
  SLICE_X7Y89      CARRY4 (Prop_carry4_CI_0[3])
                                0.329      11.379 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/O[3]
  net (fo=3, routed)      0.000      11.379
u_FPGA_Core/Discrete_FIR_Filter_out1[30]
  SLICE_X7Y89      FDCE
                                r
u_FPGA_Core/Unit_Delay_out1_reg[14]/D
  -----
  -----

                                (clock clk_out1_clk_wiz_0_1 rise edge)
                                25.000      25.000 r
E3                                0.000      25.000 r sys_clk
(IN)
                                net (fo=0)      0.000      25.000
u_pll/inst/clk_in1
E3
                                r

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u_pll/inst/clkin1_ibufg/I			
E3	IBUF (Prop_ibuf_I_0)	1.411	26.411 r
u_pll/inst/clkin1_ibufg/O			
	net (fo=1, routed)	1.162	27.573
u_pll/inst/clk_in1_clk_wiz_0			
MMCME2_ADV_X1Y2			r
u_pll/inst/mmcm_adv_inst/CLKIN1			
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.324	20.249 r
u_pll/inst/mmcm_adv_inst/CLKOUT0			
	net (fo=1, routed)	1.639	21.888
u_pll/inst/clk_out1_clk_wiz_0			
BUFGCTRL_X0Y16			r
u_pll/inst/clkout1_buf/I			
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.091	21.979 r
u_pll/inst/clkout1_buf/O			
	net (fo=1039, routed)	1.601	23.581
u_FPGA_Core/CLK			
SLICE_X7Y89	FDCE		r
u_FPGA_Core/Unit_Delay_out1_reg[14]/C			
	clock pessimism	0.559	24.140
	clock uncertainty	-0.085	24.056
SLICE_X7Y89	FDCE (Setup_fdce_C_D)	0.051	24.107
u_FPGA_Core/Unit_Delay_out1_reg[14]			

	required time		24.107
	arrival time		-11.379

	slack		12.727

Min Delay Paths

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Slack (MET) :          0.043ns (arrival time - required time)
Source:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[41][7]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Destination:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][7]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group:          clk_out1_clk_wiz_0_1
Path Type:           Hold (Min at Fast Process Corner)
Requirement:         0.000ns (clk_out1_clk_wiz_0_1 rise@0.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

```

Data Path Delay: 0.339ns (logic 0.148ns (43.647%) route 0.191ns (56.353%))

Logic Levels: 0

Clock Path Skew: 0.277ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): -0.826ns

Source Clock Delay (SCD): -0.594ns

Clock Pessimism Removal (CPR): -0.509ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

(clock clk_out1_clk_wiz_0_1 rise edge)				
		0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.570	-0.594	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X10Y100	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[41][7]/C				

SLICE_X10Y100	FDCE (Prop_fdce_C_Q)	0.148	-0.446	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[41][7]/Q				
	net (fo=3, routed)	0.191	-0.255	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[41][7]				
SLICE_X11Y97	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][7]/D				


```

(clock clk_out1_clk_wiz_0_1 rise edge)
0.000 0.000 r
E3 0.000 0.000 r sys_clk
(IN)
net (fo=0) 0.000 0.000
u_pll/inst/clk_in1
E3 r
u_pll/inst/clkin1_ibufg/I
E3 IBUF (Prop_ibuf_I_0) 0.438 0.438 r
u_pll/inst/clkin1_ibufg/O
net (fo=1, routed) 0.480 0.918
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2 r
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
-3.163 -2.245 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
net (fo=1, routed) 0.544 -1.702
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16 r
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16 BUFG (Prop_bufg_I_0) 0.029 -1.673 r
u_pll/inst/clkout1_buf/O
net (fo=1039, routed) 0.847 -0.826
u_FPGA_Core/u_Discrete_FIR_Filter/CLK
SLICE_X11Y97 FDCE r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][7]/C
clock pessimism 0.509 -0.317
SLICE_X11Y97 FDCE (Hold_fdce_C_D) 0.019 -0.298
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][7]
-----
required time 0.298
arrival time -0.255
-----
slack 0.043

```

Slack (MET) : 0.045ns (arrival time - required time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[41][6]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][6]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (clk_out1_clk_wiz_0_1 rise@0.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 0.332ns (logic 0.148ns (44.624%) route 0.184ns)

(55.376%))

Logic Levels: 0
Clock Path Skew: 0.265ns (DCD - SCD - CPR)
Destination Clock Delay (DCD): -0.832ns
Source Clock Delay (SCD): -0.588ns
Clock Pessimism Removal (CPR): -0.509ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
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(clock clk_out1_clk_wiz_0_1 rise edge)				
		0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.576	-0.588	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X10Y99	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[41][6]/C				

SLICE_X10Y99	FDCE (Prop_fdce_C_Q)	0.148	-0.440	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[41][6]/Q				
	net (fo=3, routed)	0.184	-0.257	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[41][6]				
SLICE_X10Y102	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][6]/D				

(clock clk_out1_clk_wiz_0_1 rise edge)

E3		0.000	0.000	r
(IN)		0.000	0.000	r sys_clk
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.438	0.438	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.480	0.918	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.544	-1.702	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.841	-0.832	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X10Y102	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][6]/C				
	clock pessimism	0.509	-0.323	
SLICE_X10Y102	FDCE (Hold_fdce_C_D)	0.022	-0.301	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][6]				

	required time		0.301	
	arrival time		-0.257	

	slack		0.045	

Slack (MET) : 0.070ns (arrival time - required time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][3]/C

(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][3]/D

(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (clk_out1_clk_wiz_0_1 rise@0.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 0.407ns (logic 0.164ns (40.260%) route 0.243ns
(59.740%))

Logic Levels: 0
 Clock Path Skew: 0.265ns (DCD - SCD - CPR)
 Destination Clock Delay (DCD): -0.832ns
 Source Clock Delay (SCD): -0.588ns
 Clock Pessimism Removal (CPR): -0.509ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.576	-0.588	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X10Y99	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][3]/C				

SLICE_X10Y99	FDCE (Prop_fdce_C_Q)	0.164	-0.424	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][3]/Q				
	net (fo=3, routed)	0.243	-0.181	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[42][3]				
SLICE_X13Y100	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][3]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r

E3		0.000	0.000	r	sys_clk
(IN)					
	net (fo=0)	0.000	0.000		
u_pll/inst/clk_in1					
E3				r	
u_pll/inst/clkin1_ibufg/I					
E3	IBUF (Prop_ibuf_I_0)	0.438	0.438	r	
u_pll/inst/clkin1_ibufg/O					
	net (fo=1, routed)	0.480	0.918		
u_pll/inst/clk_in1_clk_wiz_0					
MMCME2_ADV_X1Y2				r	
u_pll/inst/mmcm_adv_inst/CLKIN1					
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245	r	
u_pll/inst/mmcm_adv_inst/CLKOUT0					
	net (fo=1, routed)	0.544	-1.702		
u_pll/inst/clk_out1_clk_wiz_0					
BUFGCTRL_X0Y16				r	
u_pll/inst/clkout1_buf/I					
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673	r	
u_pll/inst/clkout1_buf/O					
	net (fo=1039, routed)	0.841	-0.832		
u_FPGA_Core/u_Discrete_FIR_Filter/CLK					
SLICE_X13Y100	FDCE			r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][3]/C					
	clock pessimism	0.509	-0.323		
SLICE_X13Y100	FDCE (Hold_fdce_C_D)	0.072	-0.251		
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][3]					

	required time		0.251		
	arrival time		-0.181		

	slack		0.070		

Slack (MET) : 0.102ns (arrival time - required time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][1]/C

(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[44][1]/D

(rising edge-triggered cell FDCE clocked by

clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (clk_out1_clk_wiz_0_1 rise@0.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 0.449ns (logic 0.164ns (36.537%) route 0.285ns
(63.463%))

Logic Levels: 0

Clock Path Skew: 0.277ns (DCD - SCD - CPR)
 Destination Clock Delay (DCD): -0.826ns
 Source Clock Delay (SCD): -0.594ns
 Clock Pessimism Removal (CPR): -0.509ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.570	-0.594	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X12Y100	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][1]/C				

SLICE_X12Y100	FDCE (Prop_fdce_C_Q)	0.164	-0.430	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][1]/Q				
	net (fo=3, routed)	0.285	-0.145	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[43][1]				
SLICE_X9Y98	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[44][1]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk

```

(IN)
net (fo=0) 0.000 0.000
u_pll/inst/clk_in1
E3 r
u_pll/inst/clkin1_ibufg/I
E3 IBUF (Prop_ibuf_I_0) 0.438 0.438 r
u_pll/inst/clkin1_ibufg/O
net (fo=1, routed) 0.480 0.918
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2 r
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
-3.163 -2.245 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
net (fo=1, routed) 0.544 -1.702
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16 r
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16 BUFG (Prop_bufg_I_0) 0.029 -1.673 r
u_pll/inst/clkout1_buf/O
net (fo=1039, routed) 0.847 -0.826
u_FPGA_Core/u_Discrete_FIR_Filter/CLK
SLICE_X9Y98 FDCE r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[44][1]/C
clock pessimism 0.509 -0.317
SLICE_X9Y98 FDCE (Hold_fdce_C_D) 0.070 -0.247
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[44][1]
-----
required time 0.247
arrival time -0.145
-----
slack 0.102

```

```

Slack (MET) : 0.107ns (arrival time - required time)
Source:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][6]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Destination:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][6]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group: clk_out1_clk_wiz_0_1
Path Type: Hold (Min at Fast Process Corner)
Requirement: 0.000ns (clk_out1_clk_wiz_0_1 rise@0.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)
Data Path Delay: 0.456ns (logic 0.164ns (35.985%) route 0.292ns
(64.015%))
Logic Levels: 0
Clock Path Skew: 0.277ns (DCD - SCD - CPR)

```

Destination Clock Delay (DCD): -0.826ns
Source Clock Delay (SCD): -0.594ns
Clock Pessimism Removal (CPR): -0.509ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.570	-0.594	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X10Y102	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][6]/C				

	SLICE_X10Y102	FDCE (Prop_fdce_C_Q)	0.164	-0.430 r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[42][6]/Q				
	net (fo=3, routed)	0.292	-0.138	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[42][6]				
SLICE_X9Y99	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][6]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				

u_pll/inst/clk_in1	net (fo=0)	0.000	0.000	
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.438	0.438	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.480	0.918	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.544	-1.702	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.847	-0.826	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X9Y99	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][6]/C				
	clock pessimism	0.509	-0.317	
SLICE_X9Y99	FDCE (Hold_fdce_C_D)	0.072	-0.245	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][6]				

	required time		0.245	
	arrival time		-0.138	

	slack		0.107	

Slack (MET) : 0.108ns (arrival time - required time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][8]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[44][8]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (clk_out1_clk_wiz_0_1 rise@0.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 0.455ns (logic 0.164ns (36.073%) route 0.291ns
(63.927%))

Logic Levels: 0

Clock Path Skew: 0.264ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): -0.833ns

Source Clock Delay (SCD): -0.588ns
Clock Pessimism Removal (CPR): -0.509ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)			
E3		0.000	0.000 r	
(IN)		0.000	0.000 r	sys_clk
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3			r	
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250 r	
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2			r	
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689 r	
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16			r	
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164 r	
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.576	-0.588	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X8Y99	FDCE		r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][8]/C				

SLICE_X8Y99	FDCE (Prop_fdce_C_Q)	0.164	-0.424 r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[43][8]/Q				
	net (fo=3, routed)	0.291	-0.134	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[43][8]				
SLICE_X8Y104	FDCE		r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[44][8]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)			
E3		0.000	0.000 r	
(IN)		0.000	0.000 r	sys_clk
	net (fo=0)	0.000	0.000	

u_pll/inst/clk_in1			
E3			r
u_pll/inst/clkin1_ibufg/I			
E3	IBUF (Prop_ibuf_I_0)	0.438	0.438 r
u_pll/inst/clkin1_ibufg/O			
	net (fo=1, routed)	0.480	0.918
u_pll/inst/clk_in1_clk_wiz_0			
MMCME2_ADV_X1Y2			r
u_pll/inst/mmcm_adv_inst/CLKIN1			
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245 r
u_pll/inst/mmcm_adv_inst/CLKOUT0			
	net (fo=1, routed)	0.544	-1.702
u_pll/inst/clk_out1_clk_wiz_0			
BUFGCTRL_X0Y16			r
u_pll/inst/clkout1_buf/I			
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673 r
u_pll/inst/clkout1_buf/O			
	net (fo=1039, routed)	0.840	-0.833
u_FPGA_Core/u_Discrete_FIR_Filter/CLK			
SLICE_X8Y104	FDCE		r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[44][8]/C			
	clock pessimism	0.509	-0.324
SLICE_X8Y104	FDCE (Hold_fdce_C_D)	0.083	-0.241
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[44][8]			

	required time		0.241
	arrival time		-0.134

	slack		0.108

Slack (MET) : 0.109ns (arrival time - required time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[57][5]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][5]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (clk_out1_clk_wiz_0_1 rise@0.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 0.382ns (logic 0.128ns (33.503%) route 0.254ns
(66.497%))

Logic Levels: 0

Clock Path Skew: 0.279ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): -0.795ns

Source Clock Delay (SCD): -0.565ns

Clock Pessimism Removal (CPR): -0.509ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)			
E3		0.000	0.000	r
(IN)		0.000	0.000	r sys_clk
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.599	-0.565	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X0Y102	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[57][5]/C				

SLICE_X0Y102	FDCE (Prop_fdce_C_Q)	0.128	-0.437	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[57][5]/Q				
	net (fo=3, routed)	0.254	-0.183	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[57][5]				
SLICE_X0Y97	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][5]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)			
E3		0.000	0.000	r
(IN)		0.000	0.000	r sys_clk
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				

E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.438	0.438	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.480	0.918	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.544	-1.702	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_bufg/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673	r
u_pll/inst/clkout1_bufg/O				
	net (fo=1039, routed)	0.878	-0.795	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X0Y97	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][5]/C				
	clock pessimism	0.509	-0.286	
SLICE_X0Y97	FDCE (Hold_fdce_C_D)	-0.006	-0.292	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][5]				

	required time		0.292	
	arrival time		-0.183	

	slack		0.109	

Slack (MET) : 0.109ns (arrival time - required time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[57][1]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][1]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (clk_out1_clk_wiz_0_1 rise@0.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 0.409ns (logic 0.128ns (31.290%) route 0.281ns
(68.710%))

Logic Levels: 0

Clock Path Skew: 0.278ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): -0.796ns

Source Clock Delay (SCD): -0.565ns

Clock Pessimism Removal (CPR): -0.509ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)			
E3		0.000	0.000 r	sys_clk
(IN)		0.000	0.000 r	
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3			r	
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250 r	
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2			r	
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689 r	
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16			r	
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164 r	
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.599	-0.565	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X0Y102	FDCE		r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[57][1]/C				

SLICE_X0Y102	FDCE (Prop_fdce_C_Q)	0.128	-0.437 r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[57][1]/Q				
	net (fo=3, routed)	0.281	-0.156	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[57][1]				
SLICE_X2Y96	FDCE		r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][1]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)			
E3		0.000	0.000 r	sys_clk
(IN)		0.000	0.000 r	
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3			r	

u_pll/inst/clkin1_ibufg/I			
E3	IBUF (Prop_ibuf_I_0)	0.438	0.438 r
u_pll/inst/clkin1_ibufg/O			
	net (fo=1, routed)	0.480	0.918
u_pll/inst/clk_in1_clk_wiz_0			
MMCME2_ADV_X1Y2			r
u_pll/inst/mmcm_adv_inst/CLKIN1			
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245 r
u_pll/inst/mmcm_adv_inst/CLKOUT0			
	net (fo=1, routed)	0.544	-1.702
u_pll/inst/clk_out1_clk_wiz_0			
BUFGCTRL_X0Y16			r
u_pll/inst/clkout1_buf/I			
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673 r
u_pll/inst/clkout1_buf/O			
	net (fo=1039, routed)	0.877	-0.796
u_FPGA_Core/u_Discrete_FIR_Filter/CLK			
SLICE_X2Y96	FDCE		r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][1]/C			
	clock pessimism	0.509	-0.287
SLICE_X2Y96	FDCE (Hold_fdce_C_D)	0.022	-0.265
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][1]			

	required time		0.265
	arrival time		-0.156

	slack		0.109

Slack (MET) : 0.110ns (arrival time - required time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[57][6]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][6]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (clk_out1_clk_wiz_0_1 rise@0.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 0.383ns (logic 0.128ns (33.390%) route 0.255ns
(66.610%))

Logic Levels: 0

Clock Path Skew: 0.279ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): -0.795ns

Source Clock Delay (SCD): -0.565ns

Clock Pessimism Removal (CPR): -0.509ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.599	-0.565	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X0Y102	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[57][6]/C				

SLICE_X0Y102	FDCE (Prop_fdce_C_Q)	0.128	-0.437	r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[57][6]/Q				
	net (fo=3, routed)	0.255	-0.182	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[57][6]				
SLICE_X0Y97	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][6]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				

E3	IBUF (Prop_ibuf_I_0)	0.438	0.438	r
u_pll/inst/clkin1_ibufg/0				
	net (fo=1, routed)	0.480	0.918	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.544	-1.702	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673	r
u_pll/inst/clkout1_buf/0				
	net (fo=1039, routed)	0.878	-0.795	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X0Y97	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][6]/C				
	clock pessimism	0.509	-0.286	
SLICE_X0Y97	FDCE (Hold_fdce_C_D)	-0.006	-0.292	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[58][6]				

	required time		0.292	
	arrival time		-0.182	

	slack		0.110	

Slack (MET) : 0.115ns (arrival time - required time)

Source:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][8]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Destination:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[40][8]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: clk_out1_clk_wiz_0_1

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (clk_out1_clk_wiz_0_1 rise@0.000ns -
clk_out1_clk_wiz_0_1 rise@0.000ns)

Data Path Delay: 0.379ns (logic 0.148ns (39.096%) route 0.231ns
(60.904%))

Logic Levels: 0

Clock Path Skew: 0.265ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): -0.832ns

Source Clock Delay (SCD): -0.588ns

Clock Pessimism Removal (CPR): -0.509ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist
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Resource(s)

```

-----
                                (clock clk_out1_clk_wiz_0_1 rise edge)
                                0.000    0.000 r
E3                                0.000    0.000 r sys_clk
(IN)
                                net (fo=0)          0.000    0.000
u_pll/inst/clk_in1
E3                                r
u_pll/inst/clkin1_ibufg/I
E3                                IBUF (Prop_ibuf_I_0)    0.250    0.250 r
u_pll/inst/clkin1_ibufg/O
                                net (fo=1, routed)    0.440    0.690
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2                                r
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2    MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
                                -2.379    -1.689 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
                                net (fo=1, routed)    0.499    -1.190
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16                                r
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16    BUFG (Prop_bufg_I_0)    0.026    -1.164 r
u_pll/inst/clkout1_buf/O
                                net (fo=1039, routed)    0.576    -0.588
u_FPGA_Core/u_Discrete_FIR_Filter/CLK
SLICE_X12Y97    FDCE                                r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][8]/C
-----
SLICE_X12Y97    FDCE (Prop_fdce_C_Q)    0.148    -0.440 r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][8]/Q
                                net (fo=3, routed)    0.231    -0.210
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[39][8]
SLICE_X10Y100    FDCE                                r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[40][8]/D
-----

```

```

                                (clock clk_out1_clk_wiz_0_1 rise edge)
                                0.000    0.000 r
E3                                0.000    0.000 r sys_clk
(IN)
                                net (fo=0)          0.000    0.000
u_pll/inst/clk_in1
E3                                r
u_pll/inst/clkin1_ibufg/I
E3                                IBUF (Prop_ibuf_I_0)    0.438    0.438 r

```

```

u_pll/inst/clkin1_ibufg/0
    net (fo=1, routed)          0.480      0.918
u_pll/inst/clk_in1_clk_wiz_0
    MMCME2_ADV_X1Y2                                r
u_pll/inst/mmcm_adv_inst/CLKIN1
    MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
                                -3.163      -2.245 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
    net (fo=1, routed)          0.544      -1.702
u_pll/inst/clk_out1_clk_wiz_0
    BUFGCTRL_X0Y16                                r
u_pll/inst/clkout1_buf/I
    BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)          0.029      -1.673 r
u_pll/inst/clkout1_buf/O
    net (fo=1039, routed)       0.841      -0.832
u_FPGA_Core/u_Discrete_FIR_Filter/CLK
    SLICE_X10Y100      FDCE                                r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[40][8]/C
    clock pessimism          0.509      -0.323
    SLICE_X10Y100      FDCE (Hold_fdce_C_D)          -0.001      -0.324
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[40][8]
-----
                                required time          0.324
                                arrival time          -0.210
-----
                                slack          0.115

```

Pulse Width Checks

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```

Clock Name:      clk_out1_clk_wiz_0_1
Waveform(ns):    { 0.000 12.500 }
Period(ns):      25.000
Sources:         { u_pll/inst/mmcm_adv_inst/CLKOUT0 }
```

Check Type	Corner	Lib Pin	Reference Pin	Required(ns)
Actual(ns)	Slack(ns)	Location	Pin	
Min Period	n/a	BUFG/I	n/a	2.155 25.000
22.845		BUFGCTRL_X0Y16	u_pll/inst/clkout1_buf/I	
Min Period	n/a	MMCME2_ADV/CLKOUT0	n/a	1.249 25.000
23.751		MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKOUT0	
Min Period	n/a	FDCE/C	n/a	1.000 25.000
24.000		SLICE_X1Y109	enable_counter_reg[0]/C	
Min Period	n/a	FDCE/C	n/a	1.000 25.000
24.000		SLICE_X1Y109	enable_counter_reg[1]/C	
Min Period	n/a	FDCE/C	n/a	1.000 25.000

24.000	SLICE_X1Y110	enable_counter_reg[2]/C		
Min Period	n/a	FDCE/C	n/a	1.000 25.000
24.000	SLICE_X1Y110	enable_counter_reg[3]/C		
Min Period	n/a	FDCE/C	n/a	1.000 25.000
24.000	SLICE_X1Y110	enable_counter_reg[4]/C		
Min Period	n/a	FDCE/C	n/a	1.000 25.000
24.000	SLICE_X1Y110	enable_counter_reg[5]/C		
Min Period	n/a	FDCE/C	n/a	1.000 25.000
24.000	SLICE_X1Y109	enable_pulse_reg/C		
Min Period	n/a	FDCE/C	n/a	1.000 25.000
24.000	SLICE_X1Y109	enable_pulse_reg_lopt_replica/C		
Max Period	n/a	MMCME2_ADV/CLKOUT0	n/a	213.360 25.000
188.360	MMCME2_ADV_X1Y2	u_pll/inst/mmc Adv_inst/CLKOUT0		
Low Pulse Width	Slow	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y109	enable_counter_reg[0]/C		
Low Pulse Width	Fast	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y109	enable_counter_reg[0]/C		
Low Pulse Width	Slow	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y109	enable_counter_reg[1]/C		
Low Pulse Width	Fast	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y109	enable_counter_reg[1]/C		
Low Pulse Width	Slow	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y110	enable_counter_reg[2]/C		
Low Pulse Width	Fast	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y110	enable_counter_reg[2]/C		
Low Pulse Width	Slow	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y110	enable_counter_reg[3]/C		
Low Pulse Width	Fast	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y110	enable_counter_reg[3]/C		
Low Pulse Width	Slow	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y110	enable_counter_reg[4]/C		
Low Pulse Width	Fast	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y110	enable_counter_reg[4]/C		
High Pulse Width	Slow	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y109	enable_counter_reg[0]/C		
High Pulse Width	Fast	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y109	enable_counter_reg[0]/C		
High Pulse Width	Slow	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y109	enable_counter_reg[1]/C		
High Pulse Width	Fast	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y109	enable_counter_reg[1]/C		
High Pulse Width	Slow	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y110	enable_counter_reg[2]/C		
High Pulse Width	Fast	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y110	enable_counter_reg[2]/C		
High Pulse Width	Slow	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y110	enable_counter_reg[3]/C		
High Pulse Width	Fast	FDCE/C	n/a	0.500 12.500
12.000	SLICE_X1Y110	enable_counter_reg[3]/C		
High Pulse Width	Slow	FDCE/C	n/a	0.500 12.500

12.000	SLICE_X1Y110	enable_counter_reg[4]/C		
High Pulse Width Fast	FDCE/C	n/a	0.500	12.500
12.000	SLICE_X1Y110	enable_counter_reg[4]/C		

 From Clock: clkfbout_clk_wiz_0_1
 To Clock: clkfbout_clk_wiz_0_1

Setup :	NA	Failing Endpoints, Worst Slack	NA	, Total
Violation	NA			
Hold :	NA	Failing Endpoints, Worst Slack	NA	, Total
Violation	NA			
PW :	0	Failing Endpoints, Worst Slack	7.845ns	, Total
Violation	0.000ns			

Pulse Width Checks

 Clock Name: clkfbout_clk_wiz_0_1
 Waveform(ns): { 0.000 5.000 }
 Period(ns): 10.000
 Sources: { u_pll/inst/mmcm_adv_inst/CLKFBOUT }

Check Type	Corner	Lib Pin	Reference Pin	Required(ns)	Actual(ns)
Slack(ns)	Location	Pin			
Min Period	n/a	BUFG/I	n/a	2.155	10.000
7.845	BUFGCTRL_X0Y17	u_pll/inst/clkf_buf/I			
Min Period	n/a	MMCME2_ADV/CLKFBOUT	n/a	1.249	10.000
8.751	MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKFBOUT			
Min Period	n/a	MMCME2_ADV/CLKFBIN	n/a	1.249	10.000
8.751	MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKFBIN			
Max Period	n/a	MMCME2_ADV/CLKFBIN	n/a	100.000	10.000
90.000	MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKFBIN			
Max Period	n/a	MMCME2_ADV/CLKFBOUT	n/a	213.360	10.000
203.360	MMCME2_ADV_X1Y2	u_pll/inst/mmcm_adv_inst/CLKFBOUT			

Path Group: (none)
 From Clock:
 To Clock:

Max Delay 1 Endpoint
Min Delay 1 Endpoint

Max Delay Paths

Slack: inf
Source: In1[8]
(input port)
Destination: Out3
(output port)
Path Group: (none)
Path Type: Max at Slow Process Corner
Data Path Delay: 36.764ns (logic 18.853ns (51.282%) route 17.911ns
(48.718%))
Logic Levels: 29 (CARRY4=12 DSP48E1=2 IBUF=1 LUT2=4 LUT3=4 LUT4=3
LUT5=1 LUT6=1 OBUF=1)

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
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T8 (IN)		0.000	0.000 r	In1[8]
	net (fo=0)	0.000	0.000	In1[8]
T8			r	
In1_IBUF[8]_inst/I				
T8	IBUF (Prop_ibuf_I_0)	1.497	1.497 r	
In1_IBUF[8]_inst/O				
	net (fo=3, routed)	5.467	6.964	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8]				
SLICE_X1Y98			r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2				
SLICE_X1Y98	LUT3 (Prop_lut3_I2_0)	0.152	7.116 r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/O				
	net (fo=2, routed)	0.469	7.585	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0				
SLICE_X1Y98			r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3				
SLICE_X1Y98	LUT4 (Prop_lut4_I3_0)	0.326	7.911 r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/O				
	net (fo=1, routed)	0.000	7.911	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0				
SLICE_X1Y98			r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]				
SLICE_X1Y98	CARRY4 (Prop_carry4_S[1]_CO[3])	0.550	8.461 r	

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/CO[3]			
net (fo=1, routed)	0.000	8.461	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_0			r
SLICE_X1Y99			
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/CI			
SLICE_X1Y99 CARRY4 (Prop_carry4_CI_0[1])	0.334	8.795	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/O[1]			
net (fo=2, routed)	0.582	9.377	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183_n_6			r
SLICE_X4Y99			
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/I1			
SLICE_X4Y99 LUT3 (Prop_lut3_I1_0)	0.329	9.706	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/O			
net (fo=2, routed)	0.872	10.578	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87_n_0			r
SLICE_X4Y99			
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/I0			
SLICE_X4Y99 LUT4 (Prop_lut4_I0_0)	0.332	10.910	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/O			
net (fo=1, routed)	0.000	10.910	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91_n_0			r
SLICE_X4Y99			
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/S[2]			
SLICE_X4Y99 CARRY4 (Prop_carry4_S[2]_CO[3])	0.398	11.308	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/CO[3]			
net (fo=1, routed)	0.001	11.309	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53_n_0			r
SLICE_X4Y100			
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/CI			
SLICE_X4Y100 CARRY4 (Prop_carry4_CI_0[0])	0.222	11.531	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/O[0]			
net (fo=2, routed)	0.720	12.251	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71_n_7			r
SLICE_X5Y99			
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/I1			
SLICE_X5Y99 LUT3 (Prop_lut3_I1_0)	0.324	12.575	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/O			
net (fo=2, routed)	0.817	13.392	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33_n_0			r
SLICE_X5Y100			
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/I3			
SLICE_X5Y100 LUT4 (Prop_lut4_I3_0)	0.332	13.724	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/O			
net (fo=1, routed)	0.000	13.724	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37_n_0			r
SLICE_X5Y100			
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/S[0]			

SLICE_X5Y100	CARRY4 (Prop_carry4_S[0]_O[3])	0.606	14.330	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/O[3]				
	net (fo=3, routed)	0.771	15.102	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10_n_4				
SLICE_X7Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/I2				
SLICE_X7Y92	LUT3 (Prop_lut3_I2_0)	0.306	15.408	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/O				
	net (fo=2, routed)	0.579	15.987	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13_n_0				
SLICE_X5Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/I1				
SLICE_X5Y92	LUT5 (Prop_lut5_I1_0)	0.124	16.111	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/O				
	net (fo=2, routed)	0.808	16.918	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3_n_0				
SLICE_X7Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/I0				
SLICE_X7Y91	LUT6 (Prop_lut6_I0_0)	0.124	17.042	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/O				
	net (fo=1, routed)	0.000	17.042	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6_n_0				
SLICE_X7Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/S[0]				
SLICE_X7Y91	CARRY4 (Prop_carry4_S[0]_O[2])	0.547	17.589	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/O[2]				
	net (fo=2, routed)	0.522	18.112	
u_FPGA_Core/u_Discrete_FIR_Filter/Discrete_FIR_Filter_out[21]				
SLICE_X6Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Sum_out1_carry__4_i_1/I1				
SLICE_X6Y91	LUT2 (Prop_lut2_I1_0)	0.302	18.414	r
u_FPGA_Core/u_Discrete_FIR_Filter/Sum_out1_carry__4_i_1/O				
	net (fo=1, routed)	0.000	18.414	
u_FPGA_Core/u_Discrete_FIR_Filter_n_1				
SLICE_X6Y91				r
u_FPGA_Core/Sum_out1_carry__4/S[1]				
SLICE_X6Y91	CARRY4 (Prop_carry4_S[1]_CO[2])	0.574	18.988	r
u_FPGA_Core/Sum_out1_carry__4/CO[2]				
	net (fo=27, routed)	0.811	19.799	
u_FPGA_Core/u_Discrete_FIR_Filter/CO[0]				
SLICE_X8Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_26/S[1]				
SLICE_X8Y91	CARRY4 (Prop_carry4_S[1]_O[3])	0.953	20.752	r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_26/O[3]				
	net (fo=1, routed)	0.419	21.171	
u_FPGA_Core/u_Discrete_FIR_Filter/Abs_rsvd_y0[24]				

SLICE_X8Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_1/I0			
SLICE_X8Y92	LUT2 (Prop_lut2_I0_0)	0.307	21.478 r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_1/O			
	net (fo=12, routed)	0.791	22.269
u_FPGA_Core/u_Discrete_FIR_Filter_n_45			
DSP48_X0Y36			r
u_FPGA_Core/Gain_mul_temp/A[28]			
DSP48_X0Y36	DSP48E1 (Prop_dsp48e1_A[28]_PCOUT[47])	4.036	26.305 r
u_FPGA_Core/Gain_mul_temp/PCOUT[47]			
	net (fo=1, routed)	0.002	26.307
u_FPGA_Core/Gain_mul_temp_n_106			
DSP48_X0Y37			r
u_FPGA_Core/Gain_mul_temp__0/PCIN[47]			
DSP48_X0Y37	DSP48E1 (Prop_dsp48e1_PCIN[47]_P[0])	1.518	27.825 f
u_FPGA_Core/Gain_mul_temp__0/P[0]			
	net (fo=2, routed)	0.760	28.585
u_FPGA_Core/Gain_out1[3]			
SLICE_X11Y91			f
u_FPGA_Core/Unit_Delay1_out10_carry_i_7/I1			
SLICE_X11Y91	LUT2 (Prop_lut2_I1_0)	0.124	28.709 r
u_FPGA_Core/Unit_Delay1_out10_carry_i_7/O			
	net (fo=1, routed)	0.000	28.709
u_FPGA_Core/Unit_Delay1_out10_carry_i_7_n_0			
SLICE_X11Y91			r
u_FPGA_Core/Unit_Delay1_out10_carry/S[1]			
SLICE_X11Y91	CARRY4 (Prop_carry4_S[1]_CO[3])	0.550	29.259 r
u_FPGA_Core/Unit_Delay1_out10_carry/CO[3]			
	net (fo=1, routed)	0.000	29.259
u_FPGA_Core/Unit_Delay1_out10_carry_n_0			
SLICE_X11Y92			r
u_FPGA_Core/Unit_Delay1_out10_carry__0/CI			
SLICE_X11Y92	CARRY4 (Prop_carry4_CI_CO[3])	0.114	29.373 r
u_FPGA_Core/Unit_Delay1_out10_carry__0/CO[3]			
	net (fo=1, routed)	0.000	29.373
u_FPGA_Core/Unit_Delay1_out10_carry__0_n_0			
SLICE_X11Y93			r
u_FPGA_Core/Unit_Delay1_out10_carry__1/CI			
SLICE_X11Y93	CARRY4 (Prop_carry4_CI_CO[3])	0.114	29.487 r
u_FPGA_Core/Unit_Delay1_out10_carry__1/CO[3]			
	net (fo=1, routed)	0.000	29.487
u_FPGA_Core/Unit_Delay1_out10_carry__1_n_0			
SLICE_X11Y94			r
u_FPGA_Core/Unit_Delay1_out10_carry__2/CI			
SLICE_X11Y94	CARRY4 (Prop_carry4_CI_CO[3])		

u_FPGA_Core/Unit_Delay1_out10_carry__2/C0[3]		0.114	29.601	r	
net (fo=2, routed)		0.978	30.579		
u_FPGA_Core/Relational_Operator_out1					
SLICE_X8Y94				r	
u_FPGA_Core/Out3_OBUF_inst_i_1/I0					
SLICE_X8Y94 LUT2 (Prop_lut2_I0_0)		0.124	30.703	r	
u_FPGA_Core/Out3_OBUF_inst_i_1/0					
net (fo=1, routed)		2.541	33.244		Out3_OBUF
H17				r	
Out3_OBUF_inst/I					
H17 OBUF (Prop_obuf_I_0)		3.520	36.764	r	
Out3_OBUF_inst/0					
net (fo=0)		0.000	36.764		Out3
H17				r	Out3 (OUT)

Min Delay Paths

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Slack:                inf
  Source:              In1[0]
                      (input port)
  Destination:        Out3
                      (output port)
  Path Group:          (none)
  Path Type:           Min at Fast Process Corner
  Data Path Delay:     6.611ns  (logic 3.064ns (46.346%)  route 3.547ns
(53.654%))
  Logic Levels:        15  (CARRY4=6 DSP48E1=1 IBUF=1 LUT2=2 LUT3=2 LUT5=1
LUT6=1 OBUF=1)

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

J15 (IN)		0.000	0.000 r	In1[0]
	net (fo=0)	0.000	0.000	In1[0]
J15 In1_IBUF[0]_inst/I			r	
J15 In1_IBUF[0]_inst/O	IBUF (Prop_ibuf_I_0)	0.245	0.245 r	
	net (fo=3, routed)	0.482	0.727	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[0]				

SLICE_X1Y96			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_196/I2			
SLICE_X1Y96	LUT3 (Prop_lut3_I2_0)	0.045	0.772 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_196/O			
	net (fo=1, routed)	0.000	0.772
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_196_n_0			
SLICE_X1Y96			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_182/S[0]			
SLICE_X1Y96	CARRY4 (Prop_carry4_S[0]_O[0])		
		0.070	0.842 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_182/O[0]			
	net (fo=2, routed)	0.222	1.064
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_182_n_7			
SLICE_X4Y96			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_96/I1			
SLICE_X4Y96	LUT3 (Prop_lut3_I1_0)	0.107	1.171 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_96/O			
	net (fo=1, routed)	0.000	1.171
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_96_n_0			
SLICE_X4Y96			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_28/S[0]			
SLICE_X4Y96	CARRY4 (Prop_carry4_S[0]_O[0])		
		0.070	1.241 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_28/O[0]			
	net (fo=2, routed)	0.200	1.441
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_28_n_7			
SLICE_X7Y97			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_14/I4			
SLICE_X7Y97	LUT5 (Prop_lut5_I4_0)	0.107	1.548 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_14/O			
	net (fo=3, routed)	0.693	2.241
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_14_n_0			
SLICE_X7Y86			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_9/I4			
SLICE_X7Y86	LUT6 (Prop_lut6_I4_0)	0.045	2.286 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_9/O			
	net (fo=1, routed)	0.000	2.286
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_9_n_0			
SLICE_X7Y86			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_1/S[0]			
SLICE_X7Y86	CARRY4 (Prop_carry4_S[0]_O[1])		
		0.106	2.392 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_1/O[1]			
	net (fo=3, routed)	0.216	2.608
u_FPGA_Core/u_Discrete_FIR_Filter/Discrete_FIR_Filter_out[0]			
SLICE_X6Y86			r
u_FPGA_Core/u_Discrete_FIR_Filter/Sum_out1_carry_i_4/I0			
SLICE_X6Y86	LUT2 (Prop_lut2_I0_0)	0.107	2.715 r
u_FPGA_Core/u_Discrete_FIR_Filter/Sum_out1_carry_i_4/O			
	net (fo=1, routed)	0.000	2.715

u_FPGA_Core/u_Discrete_FIR_Filter_n_28				
SLICE_X6Y86				r
u_FPGA_Core/Sum_out1_carry/S[0]				
SLICE_X6Y86	CARRY4 (Prop_carry4_S[0]_O[0])	0.070	2.785	r
u_FPGA_Core/Sum_out1_carry/O[0]				
	net (fo=3, routed)	0.475	3.260	
u_FPGA_Core/RESIZE[0]				
DSP48_X0Y37				r
u_FPGA_Core/Gain_mul_temp__0/A[0]				
DSP48_X0Y37	DSP48E1 (Prop_dsp48e1_A[0]_P[18])	0.669	3.929	r
u_FPGA_Core/Gain_mul_temp__0/P[18]				
	net (fo=2, routed)	0.176	4.105	
u_FPGA_Core/Gain_out1[21]				
SLICE_X11Y93				r
u_FPGA_Core/Unit_Delay1_out10_carry__1/DI[2]				
SLICE_X11Y93	CARRY4 (Prop_carry4_DI[2]_CO[3])	0.117	4.222	r
u_FPGA_Core/Unit_Delay1_out10_carry__1/CO[3]				
	net (fo=1, routed)	0.000	4.222	
u_FPGA_Core/Unit_Delay1_out10_carry__1_n_0				
SLICE_X11Y94				r
u_FPGA_Core/Unit_Delay1_out10_carry__2/CI				
SLICE_X11Y94	CARRY4 (Prop_carry4_CI_CO[3])	0.039	4.261	r
u_FPGA_Core/Unit_Delay1_out10_carry__2/CO[3]				
	net (fo=2, routed)	0.331	4.592	
u_FPGA_Core/Relational_Operator_out1				
SLICE_X8Y94				r
u_FPGA_Core/Out3_OBUF_inst_i_1/I0				
SLICE_X8Y94	LUT2 (Prop_lut2_I0_0)	0.045	4.637	r
u_FPGA_Core/Out3_OBUF_inst_i_1/O				
	net (fo=1, routed)	0.753	5.389	Out3_OBUF
H17				r
Out3_OBUF_inst/I				
H17	OBUF (Prop_obuf_I_0)	1.221	6.611	r
Out3_OBUF_inst/O				
	net (fo=0)	0.000	6.611	Out3
H17				r Out3 (OUT)

Path Group: (none)

From Clock: clk_out1_clk_wiz_0_1
To Clock:

Max Delay 2 Endpoints
Min Delay 2 Endpoints

Max Delay Paths

Slack: inf
Source:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Destination: Out3
(output port)
Path Group: (none)
Path Type: Max at Slow Process Corner
Data Path Delay: 32.084ns (logic 18.086ns (56.372%) route 13.998ns
(43.628%))
Logic Levels: 30 (CARRY4=14 DSP48E1=2 LUT2=4 LUT3=4 LUT4=3 LUT5=1
LUT6=1 OBUF=1)
Clock Uncertainty: 0.179ns $((TSJ^2 + DJ^2)^{1/2} / 2 + PE$
Total System Jitter (TSJ): 0.050ns
Discrete Jitter (DJ): 0.154ns
Phase Error (PE): 0.099ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
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	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.482	1.482	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.233	2.715	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.070	-4.355	r

u_pll/inst/mmcm_adv_inst/CLKOUT0			
net (fo=1, routed)	1.719	-2.636	
u_pll/inst/clk_out1_clk_wiz_0			
BUFGCTRL_X0Y16			r
u_pll/inst/clkout1_buf/I			
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540 r
u_pll/inst/clkout1_buf/O			
net (fo=1039, routed)	1.646	-0.894	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK			
SLICE_X12Y95	FDCE		r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/C			

SLICE_X12Y95	FDCE (Prop_fdce_C_Q)	0.478	-0.416 r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[39][2]/Q			
net (fo=3, routed)	1.007	0.591	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg_n_0_[39][2]			
SLICE_X13Y95			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/I2			
SLICE_X13Y95	LUT3 (Prop_lut3_I2_0)	0.326	0.917 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132/O			
net (fo=2, routed)	0.690	1.607	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_132_n_0			
SLICE_X13Y95			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/I3			
SLICE_X13Y95	LUT4 (Prop_lut4_I3_0)	0.327	1.934 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135/O			
net (fo=1, routed)	0.000	1.934	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[2]_i_135_n_0			
SLICE_X13Y95			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/S[3]			
SLICE_X13Y95	CARRY4 (Prop_carry4_S[3]_CO[3])		
		0.401	2.335 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34/CO[3]			
net (fo=1, routed)	0.000	2.335	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[2]_i_34_n_0			
SLICE_X13Y96			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/CI			
SLICE_X13Y96	CARRY4 (Prop_carry4_CI_CO[3])		
		0.114	2.449 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168/CO[3]			
net (fo=1, routed)	0.000	2.449	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_168_n_0			
SLICE_X13Y97			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_168/CI			
SLICE_X13Y97	CARRY4 (Prop_carry4_CI_O[3])		
		0.313	2.762 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_168/O[3]			
net (fo=2, routed)	1.013	3.774	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_168_n_4			

SLICE_X13Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_118/I2				
SLICE_X13Y91	LUT3 (Prop_lut3_I2_0)	0.336	4.110	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_118/O				
	net (fo=2, routed)	0.690	4.800	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_118_n_0				
SLICE_X13Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_122/I3				
SLICE_X13Y91	LUT4 (Prop_lut4_I3_0)	0.327	5.127	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_122/O				
	net (fo=1, routed)	0.000	5.127	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_122_n_0				
SLICE_X13Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_57/S[3]				
SLICE_X13Y91	CARRY4 (Prop_carry4_S[3]_CO[3])			
		0.401	5.528	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_57/CO[3]				
	net (fo=1, routed)	0.000	5.528	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_57_n_0				
SLICE_X13Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_79/CI				
SLICE_X13Y92	CARRY4 (Prop_carry4_CI_0[2])			
		0.239	5.767	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_79/O[2]				
	net (fo=2, routed)	0.967	6.734	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_79_n_5				
SLICE_X9Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_34/I1				
SLICE_X9Y92	LUT3 (Prop_lut3_I1_0)	0.332	7.066	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_34/O				
	net (fo=2, routed)	0.690	7.756	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_34_n_0				
SLICE_X9Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_38/I3				
SLICE_X9Y92	LUT4 (Prop_lut4_I3_0)	0.327	8.083	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_38/O				
	net (fo=1, routed)	0.000	8.083	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_38_n_0				
SLICE_X9Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15/S[3]				
SLICE_X9Y92	CARRY4 (Prop_carry4_S[3]_CO[3])			
		0.401	8.484	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15/CO[3]				
	net (fo=1, routed)	0.000	8.484	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_15_n_0				
SLICE_X9Y93				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_11/CI				
SLICE_X9Y93	CARRY4 (Prop_carry4_CI_0[3])			
		0.313	8.797	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_11/O[3]				

	net (fo=3, routed)	0.730	9.527	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_11_n_4				
SLICE_X7Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/I1				
SLICE_X7Y92	LUT3 (Prop_lut3_I1_0)	0.306	9.833	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/O				
	net (fo=2, routed)	0.579	10.412	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13_n_0				
SLICE_X5Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/I1				
SLICE_X5Y92	LUT5 (Prop_lut5_I1_0)	0.124	10.536	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/O				
	net (fo=2, routed)	0.808	11.344	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3_n_0				
SLICE_X7Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/I0				
SLICE_X7Y91	LUT6 (Prop_lut6_I0_0)	0.124	11.468	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/O				
	net (fo=1, routed)	0.000	11.468	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6_n_0				
SLICE_X7Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/S[0]				
SLICE_X7Y91	CARRY4 (Prop_carry4_S[0]_O[2])	0.547	12.015	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/O[2]				
	net (fo=2, routed)	0.522	12.537	
u_FPGA_Core/u_Discrete_FIR_Filter/Discrete_FIR_Filter_out[21]				
SLICE_X6Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Sum_out1_carry__4_i_1/I1				
SLICE_X6Y91	LUT2 (Prop_lut2_I1_0)	0.302	12.839	r
u_FPGA_Core/u_Discrete_FIR_Filter/Sum_out1_carry__4_i_1/O				
	net (fo=1, routed)	0.000	12.839	
u_FPGA_Core/u_Discrete_FIR_Filter_n_1				
SLICE_X6Y91				r
u_FPGA_Core/Sum_out1_carry__4/S[1]				
SLICE_X6Y91	CARRY4 (Prop_carry4_S[1]_CO[2])	0.574	13.413	r
u_FPGA_Core/Sum_out1_carry__4/CO[2]				
	net (fo=27, routed)	0.811	14.225	
u_FPGA_Core/u_Discrete_FIR_Filter/CO[0]				
SLICE_X8Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_26/S[1]				
SLICE_X8Y91	CARRY4 (Prop_carry4_S[1]_O[3])	0.953	15.178	r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_26/O[3]				
	net (fo=1, routed)	0.419	15.597	
u_FPGA_Core/u_Discrete_FIR_Filter/Abs_rsvd_y0[24]				
SLICE_X8Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_1/I0				
SLICE_X8Y92	LUT2 (Prop_lut2_I0_0)	0.307	15.904	r

u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_1/0			
net (fo=12, routed)	0.791	16.695	
u_FPGA_Core/u_Discrete_FIR_Filter_n_45			
DSP48_X0Y36			r
u_FPGA_Core/Gain_mul_temp/A[28]			
DSP48_X0Y36 DSP48E1 (Prop_dsp48e1_A[28]_PCOUT[47])	4.036	20.731	r
u_FPGA_Core/Gain_mul_temp/PCOUT[47]			
net (fo=1, routed)	0.002	20.733	
u_FPGA_Core/Gain_mul_temp_n_106			
DSP48_X0Y37			r
u_FPGA_Core/Gain_mul_temp__0/PCIN[47]			
DSP48_X0Y37 DSP48E1 (Prop_dsp48e1_PCIN[47]_P[0])	1.518	22.251	f
u_FPGA_Core/Gain_mul_temp__0/P[0]			
net (fo=2, routed)	0.760	23.011	
u_FPGA_Core/Gain_out1[3]			
SLICE_X11Y91			f
u_FPGA_Core/Unit_Delay1_out10_carry_i_7/I1			
SLICE_X11Y91 LUT2 (Prop_lut2_I1_0)	0.124	23.135	r
u_FPGA_Core/Unit_Delay1_out10_carry_i_7/0			
net (fo=1, routed)	0.000	23.135	
u_FPGA_Core/Unit_Delay1_out10_carry_i_7_n_0			
SLICE_X11Y91			r
u_FPGA_Core/Unit_Delay1_out10_carry/S[1]			
SLICE_X11Y91 CARRY4 (Prop_carry4_S[1]_CO[3])	0.550	23.685	r
u_FPGA_Core/Unit_Delay1_out10_carry/CO[3]			
net (fo=1, routed)	0.000	23.685	
u_FPGA_Core/Unit_Delay1_out10_carry_n_0			
SLICE_X11Y92			r
u_FPGA_Core/Unit_Delay1_out10_carry__0/CI			
SLICE_X11Y92 CARRY4 (Prop_carry4_CI_CO[3])	0.114	23.799	r
u_FPGA_Core/Unit_Delay1_out10_carry__0/CO[3]			
net (fo=1, routed)	0.000	23.799	
u_FPGA_Core/Unit_Delay1_out10_carry__0_n_0			
SLICE_X11Y93			r
u_FPGA_Core/Unit_Delay1_out10_carry__1/CI			
SLICE_X11Y93 CARRY4 (Prop_carry4_CI_CO[3])	0.114	23.913	r
u_FPGA_Core/Unit_Delay1_out10_carry__1/CO[3]			
net (fo=1, routed)	0.000	23.913	
u_FPGA_Core/Unit_Delay1_out10_carry__1_n_0			
SLICE_X11Y94			r
u_FPGA_Core/Unit_Delay1_out10_carry__2/CI			
SLICE_X11Y94 CARRY4 (Prop_carry4_CI_CO[3])	0.114	24.027	r
u_FPGA_Core/Unit_Delay1_out10_carry__2/CO[3]			
net (fo=2, routed)	0.978	25.005	

```

u_FPGA_Core/Relational_Operator_out1
  SLICE_X8Y94
u_FPGA_Core/Out3_OBUF_inst_i_1/I0
  SLICE_X8Y94      LUT2 (Prop_lut2_I0_0)      0.124      25.129 r
u_FPGA_Core/Out3_OBUF_inst_i_1/O
  net (fo=1, routed)      2.541      27.670      Out3_OBUF
  H17
Out3_OBUF_inst/I
  H17      OBUF (Prop_obuf_I_0)      3.520      31.190 r
Out3_OBUF_inst/O
  net (fo=0)      0.000      31.190      Out3
  H17
  r      Out3 (OUT)
-----
-----

```

```

Slack:      inf
Source:      enable_pulse_reg_lopt_replica/C
              (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Destination: ce_out
              (output port)
Path Group:  (none)
Path Type:   Max at Slow Process Corner
Data Path Delay: 5.865ns (logic 3.991ns (68.053%) route 1.874ns
(31.947%))
Logic Levels: 1 (OBUF=1)
Clock Uncertainty: 0.179ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
  Total System Jitter (TSJ): 0.050ns
  Discrete Jitter (DJ): 0.154ns
  Phase Error (PE): 0.099ns

```

Location	Delay type	Incr(ns)	Path(ns)	Netlist
Resource(s)				

```

-----
(clock clk_out1_clk_wiz_0_1 rise edge)
E3      0.000      0.000 r
(IN)      0.000      0.000 r sys_clk
net (fo=0)      0.000      0.000
u_pll/inst/clk_in1
  E3
u_pll/inst/clkin1_ibufg/I
  E3      IBUF (Prop_ibuf_I_0)      1.482      1.482 r
u_pll/inst/clkin1_ibufg/O
  net (fo=1, routed)      1.233      2.715
u_pll/inst/clk_in1_clk_wiz_0
  MMCME2_ADV_X1Y2
u_pll/inst/mmcm_adv_inst/CLKIN1
  MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)

```

u_pll/inst/mmcm_adv_inst/CLKOUT0		-7.070	-4.355	r	
net (fo=1, routed)		1.719	-2.636		
u_pll/inst/clk_out1_clk_wiz_0					
BUFGCTRL_X0Y16				r	
u_pll/inst/clkout1_buf/I					
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.096	-2.540	r	
u_pll/inst/clkout1_buf/O					
	net (fo=1039, routed)	1.707	-0.833		clk_40mhz
SLICE_X1Y109	FDCE			r	
enable_pulse_reg_lopt_replica/C					

SLICE_X1Y109	FDCE (Prop_fdce_C_Q)	0.456	-0.377	r	
enable_pulse_reg_lopt_replica/Q					
	net (fo=1, routed)	1.874	1.497		
enable_pulse_reg_lopt_replica_1					
K15				r	
ce_out_OBUF_inst/I					
K15	OBUF (Prop_obuf_I_0)	3.535	5.032	r	
ce_out_OBUF_inst/O					
	net (fo=0)	0.000	5.032		ce_out
K15				r	ce_out
(OUT)					

Min Delay Paths

```

-----
---
Slack:                inf
Source:               enable_pulse_reg_lopt_replica/C
                     (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Destination:         ce_out
                     (output port)
Path Group:           (none)
Path Type:            Min at Fast Process Corner
Data Path Delay:      1.789ns  (logic 1.377ns (76.985%)  route 0.412ns
(23.015%))
Logic Levels:        1  (OBUF=1)
Clock Uncertainty:    0.179ns  ((TSJ^2 + DJ^2)^1/2) / 2 + PE
  Total System Jitter    (TSJ):    0.050ns
  Discrete Jitter        (DJ):     0.154ns
  Phase Error            (PE):     0.099ns

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.597	-0.567	clk_40mhz
SLICE_X1Y109	FDCE			r
enable_pulse_reg_lopt_replica/C				

SLICE_X1Y109	FDCE (Prop_fdce_C_Q)	0.141	-0.426	r
enable_pulse_reg_lopt_replica/Q				
	net (fo=1, routed)	0.412	-0.014	
enable_pulse_reg_lopt_replica_1				
K15				r
ce_out_OBUF_inst/I				
K15	OBUF (Prop_obuf_I_0)	1.236	1.222	r
ce_out_OBUF_inst/O				
	net (fo=0)	0.000	1.222	ce_out
K15				r ce_out
(OUT)				

Slack:	inf			
Source:	u_FPGA_Core/Unit_Delay1_out1_reg/C			
	(rising edge-triggered cell FDCE clocked by			
clk_out1_clk_wiz_0_1	{rise@0.000ns fall@12.500ns period=25.000ns})			

Destination: Out3
 (output port)
 Path Group: (none)
 Path Type: Min at Fast Process Corner
 Data Path Delay: 2.468ns (logic 1.407ns (57.035%) route 1.060ns (42.965%))
 Logic Levels: 2 (LUT2=1 OBUF=1)
 Clock Uncertainty: 0.179ns $((TSJ^2 + DJ^2)^{1/2} / 2 + PE)$
 Total System Jitter (TSJ): 0.050ns
 Discrete Jitter (DJ): 0.154ns
 Phase Error (PE): 0.099ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3 (IN)		0.000	0.000	r sys_clk
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.250	0.250	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.440	0.690	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-2.379	-1.689	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.499	-1.190	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.026	-1.164	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.575	-0.589	
u_FPGA_Core/CLK				
SLICE_X11Y94	FDCE			r
u_FPGA_Core/Unit_Delay1_out1_reg/C				

SLICE_X11Y94	FDCE (Prop_fdce_C_Q)	0.141	-0.448	r
u_FPGA_Core/Unit_Delay1_out1_reg/Q				
	net (fo=1, routed)	0.308	-0.141	
u_FPGA_Core/Unit_Delay1_out1				
SLICE_X8Y94				r

u_FPGA_Core/Out3_OBUF_inst_i_1/I1				
SLICE_X8Y94	LUT2 (Prop_lut2_I1_0)	0.045	-0.096	r
u_FPGA_Core/Out3_OBUF_inst_i_1/0				
	net (fo=1, routed)	0.753	0.657	Out3_OBUF
H17				r
Out3_OBUF_inst/I				
H17	OBUF (Prop_obuf_I_0)	1.221	1.878	r
Out3_OBUF_inst/0				
	net (fo=0)	0.000	1.878	Out3
H17				r Out3 (OUT)

Path Group: (none)
From Clock: clkfbout_clk_wiz_0_1
To Clock:

Max Delay	1 Endpoint
Min Delay	1 Endpoint

Max Delay Paths

Slack: inf
Source: u_pll/inst/mmcm_adv_inst/CLKFBOUT
(clock source 'clkfbout_clk_wiz_0_1' {rise@0.000ns
fall@5.000ns period=10.000ns})
Destination: u_pll/inst/mmcm_adv_inst/CLKFBIN
Path Group: (none)
Path Type: Max at Fast Process Corner
Data Path Delay: 1.396ns (logic 0.029ns (2.077%) route 1.367ns
(97.923%))
Logic Levels: 1 (BUFG=1)
Clock Uncertainty: 0.147ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
Total System Jitter (TSJ): 0.050ns
Discrete Jitter (DJ): 0.083ns
Phase Error (PE): 0.099ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist
Resource(s)				

```

-----
                                (clock clkfbout_clk_wiz_0_1 fall edge)
                                5.000      5.000 f
E3                                0.000      5.000 f  sys_clk
(IN)
                                net (fo=0)      0.000      5.000
u_pll/inst/clk_in1
E3                                f
u_pll/inst/clkin1_ibufg/I
E3                                IBUF (Prop_ibuf_I_0)      0.438      5.438 f
u_pll/inst/clkin1_ibufg/O
                                net (fo=1, routed)      0.480      5.918
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2                                f
u_pll/inst/mmcm_adv_inst/CLKIN1
-----

```

```

-----
MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKFBOUT)
                                -3.163      2.755 f
u_pll/inst/mmcm_adv_inst/CLKFBOUT
                                net (fo=1, routed)      0.544      3.298
u_pll/inst/clkfbout_clk_wiz_0
BUFGCTRL_X0Y17                                f
u_pll/inst/clkf_buf/I
BUFGCTRL_X0Y17      BUFG (Prop_bufg_I_0)      0.029      3.327 f
u_pll/inst/clkf_buf/O
                                net (fo=1, routed)      0.824      4.151
u_pll/inst/clkfbout_buf_clk_wiz_0
MMCME2_ADV_X1Y2      MMCME2_ADV                                f
u_pll/inst/mmcm_adv_inst/CLKFBIN
-----

```

Min Delay Paths

```

-----
---
Slack:      inf
Source:      u_pll/inst/mmcm_adv_inst/CLKFBOUT
              (clock source 'clkfbout_clk_wiz_0_1' {rise@0.000ns
fall@5.000ns period=10.000ns})
Destination: u_pll/inst/mmcm_adv_inst/CLKFBIN
Path Group:  (none)
Path Type:   Min at Slow Process Corner
Data Path Delay: 3.236ns (logic 0.091ns (2.812%) route 3.145ns
(97.188%))
Logic Levels: 1 (BUFG=1)

```

Clock Uncertainty: 0.147ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
 Total System Jitter (TSJ): 0.050ns
 Discrete Jitter (DJ): 0.083ns
 Phase Error (PE): 0.099ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
-------------------------	------------	----------	----------	---------

(clock clkfbout_clk_wiz_0_1 rise edge)				
E3		0.000	0.000	r
(IN)		0.000	0.000	r sys_clk
u_pll/inst/clk_in1	net (fo=0)	0.000	0.000	
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.411	1.411	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.162	2.573	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				

MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKFBOUT)	-7.324	-4.751	r
u_pll/inst/mmcm_adv_inst/CLKFBOUT				
	net (fo=1, routed)	1.639	-3.112	
u_pll/inst/clkfbout_clk_wiz_0				
BUFGCTRL_X0Y17				r
u_pll/inst/clkf_buf/I				
BUFGCTRL_X0Y17	BUFG (Prop_bufg_I_0)	0.091	-3.021	r
u_pll/inst/clkf_buf/O				
	net (fo=1, routed)	1.506	-1.515	
u_pll/inst/clkfbout_buf_clk_wiz_0				
MMCME2_ADV_X1Y2	MMCME2_ADV			r
u_pll/inst/mmcm_adv_inst/CLKFBIN				

Path Group: (none)
 From Clock:
 To Clock: clk_out1_clk_wiz_0_1

Max Delay 1078 Endpoints
Min Delay 1078 Endpoints

Max Delay Paths

Slack: inf
Source: In1[8]
(input port)
Destination: u_FPGA_Core/Unit_Delay1_out1_reg/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group: (none)
Path Type: Setup (Max at Slow Process Corner)
Data Path Delay: 29.601ns (logic 15.209ns (51.380%) route 14.392ns
(48.620%))
Logic Levels: 27 (CARRY4=12 DSP48E1=2 IBUF=1 LUT2=3 LUT3=4 LUT4=3
LUT5=1 LUT6=1)
Clock Path Skew: -1.493ns (DCD - SCD + CPR)
Destination Clock Delay (DCD): -1.493ns
Source Clock Delay (SCD): 0.000ns
Clock Pessimism Removal (CPR): 0.000ns
Clock Uncertainty: 0.179ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
Total System Jitter (TSJ): 0.050ns
Discrete Jitter (DJ): 0.154ns
Phase Error (PE): 0.099ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist
Resource(s)				

T8		0.000	0.000	r	In1[8]
(IN)					
	net (fo=0)	0.000	0.000		In1[8]
T8				r	
In1_IBUF[8]_inst/I					
T8	IBUF (Prop_ibuf_I_0)	1.497	1.497	r	
In1_IBUF[8]_inst/O					
	net (fo=3, routed)	5.467	6.964		
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8]					
SLICE_X1Y98				r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2					
SLICE_X1Y98	LUT3 (Prop_lut3_I2_0)	0.152	7.116	r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/O					
	net (fo=2, routed)	0.469	7.585		
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0					

SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3				
SLICE_X1Y98	LUT4 (Prop_lut4_I3_0)	0.326	7.911	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/O				
	net (fo=1, routed)	0.000	7.911	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]				
SLICE_X1Y98	CARRY4 (Prop_carry4_S[1]_CO[3])			
		0.550	8.461	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/CO[3]				
	net (fo=1, routed)	0.000	8.461	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_0				
SLICE_X1Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/CI				
SLICE_X1Y99	CARRY4 (Prop_carry4_CI_0[1])			
		0.334	8.795	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/O[1]				
	net (fo=2, routed)	0.582	9.377	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183_n_6				
SLICE_X4Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/I1				
SLICE_X4Y99	LUT3 (Prop_lut3_I1_0)	0.329	9.706	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/O				
	net (fo=2, routed)	0.872	10.578	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87_n_0				
SLICE_X4Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/I0				
SLICE_X4Y99	LUT4 (Prop_lut4_I0_0)	0.332	10.910	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/O				
	net (fo=1, routed)	0.000	10.910	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91_n_0				
SLICE_X4Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/S[2]				
SLICE_X4Y99	CARRY4 (Prop_carry4_S[2]_CO[3])			
		0.398	11.308	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/CO[3]				
	net (fo=1, routed)	0.001	11.309	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53_n_0				
SLICE_X4Y100				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/CI				
SLICE_X4Y100	CARRY4 (Prop_carry4_CI_0[0])			
		0.222	11.531	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/O[0]				
	net (fo=2, routed)	0.720	12.251	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71_n_7				
SLICE_X5Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/I1				
SLICE_X5Y99	LUT3 (Prop_lut3_I1_0)	0.324	12.575	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/O				

	net (fo=2, routed)	0.817	13.392	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33_n_0	SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/I3	SLICE_X5Y100 LUT4 (Prop_lut4_I3_0)	0.332	13.724	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/O	net (fo=1, routed)	0.000	13.724	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37_n_0	SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/S[0]	SLICE_X5Y100 CARRY4 (Prop_carry4_S[0]_O[3])	0.606	14.330	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/O[3]	net (fo=3, routed)	0.771	15.102	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10_n_4	SLICE_X7Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/I2	SLICE_X7Y92 LUT3 (Prop_lut3_I2_0)	0.306	15.408	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/O	net (fo=2, routed)	0.579	15.987	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13_n_0	SLICE_X5Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/I1	SLICE_X5Y92 LUT5 (Prop_lut5_I1_0)	0.124	16.111	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/O	net (fo=2, routed)	0.808	16.918	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3_n_0	SLICE_X7Y91			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/I0	SLICE_X7Y91 LUT6 (Prop_lut6_I0_0)	0.124	17.042	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/O	net (fo=1, routed)	0.000	17.042	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6_n_0	SLICE_X7Y91			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/S[0]	SLICE_X7Y91 CARRY4 (Prop_carry4_S[0]_O[2])	0.547	17.589	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/O[2]	net (fo=2, routed)	0.522	18.112	
u_FPGA_Core/u_Discrete_FIR_Filter/Discrete_FIR_Filter_out[21]	SLICE_X6Y91			r
u_FPGA_Core/u_Discrete_FIR_Filter/Sum_out1_carry__4_i_1/I1	SLICE_X6Y91 LUT2 (Prop_lut2_I1_0)	0.302	18.414	r
u_FPGA_Core/u_Discrete_FIR_Filter/Sum_out1_carry__4_i_1/O	net (fo=1, routed)	0.000	18.414	
u_FPGA_Core/u_Discrete_FIR_Filter_n_1	SLICE_X6Y91			r
u_FPGA_Core/Sum_out1_carry__4/S[1]	SLICE_X6Y91 CARRY4 (Prop_carry4_S[1]_CO[2])	0.574	18.988	r

u_FPGA_Core/Sum_out1_carry__4/CO[2]			
net (fo=27, routed)	0.811	19.799	
u_FPGA_Core/u_Discrete_FIR_Filter/CO[0]			
SLICE_X8Y91			r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_26/S[1]			
SLICE_X8Y91 CARRY4 (Prop_carry4_S[1]_O[3])	0.953	20.752	r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_26/O[3]			
net (fo=1, routed)	0.419	21.171	
u_FPGA_Core/u_Discrete_FIR_Filter/Abs_rsvd_y0[24]			
SLICE_X8Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_1/I0			
SLICE_X8Y92 LUT2 (Prop_lut2_I0_0)	0.307	21.478	r
u_FPGA_Core/u_Discrete_FIR_Filter/Gain_mul_temp_i_1/O			
net (fo=12, routed)	0.791	22.269	
u_FPGA_Core/u_Discrete_FIR_Filter_n_45			
DSP48_X0Y36			r
u_FPGA_Core/Gain_mul_temp/A[28]			
DSP48_X0Y36 DSP48E1 (Prop_dsp48e1_A[28]_PCOUT[47])	4.036	26.305	r
u_FPGA_Core/Gain_mul_temp/PCOUT[47]			
net (fo=1, routed)	0.002	26.307	
u_FPGA_Core/Gain_mul_temp_n_106			
DSP48_X0Y37			r
u_FPGA_Core/Gain_mul_temp__0/PCIN[47]			
DSP48_X0Y37 DSP48E1 (Prop_dsp48e1_PCIN[47]_P[0])	1.518	27.825	f
u_FPGA_Core/Gain_mul_temp__0/P[0]			
net (fo=2, routed)	0.760	28.585	
u_FPGA_Core/Gain_out1[3]			
SLICE_X11Y91			f
u_FPGA_Core/Unit_Delay1_out10_carry_i_7/I1			
SLICE_X11Y91 LUT2 (Prop_lut2_I1_0)	0.124	28.709	r
u_FPGA_Core/Unit_Delay1_out10_carry_i_7/O			
net (fo=1, routed)	0.000	28.709	
u_FPGA_Core/Unit_Delay1_out10_carry_i_7_n_0			
SLICE_X11Y91			r
u_FPGA_Core/Unit_Delay1_out10_carry/S[1]			
SLICE_X11Y91 CARRY4 (Prop_carry4_S[1]_CO[3])	0.550	29.259	r
u_FPGA_Core/Unit_Delay1_out10_carry/CO[3]			
net (fo=1, routed)	0.000	29.259	
u_FPGA_Core/Unit_Delay1_out10_carry_n_0			
SLICE_X11Y92			r
u_FPGA_Core/Unit_Delay1_out10_carry__0/CI			
SLICE_X11Y92 CARRY4 (Prop_carry4_CI_CO[3])	0.114	29.373	r
u_FPGA_Core/Unit_Delay1_out10_carry__0/CO[3]			
net (fo=1, routed)	0.000	29.373	
u_FPGA_Core/Unit_Delay1_out10_carry__0_n_0			

SLICE_X11Y93				r
u_FPGA_Core/Unit_Delay1_out10_carry__1/CI				
SLICE_X11Y93	CARRY4 (Prop_carry4_CI_CO[3])	0.114	29.487	r
u_FPGA_Core/Unit_Delay1_out10_carry__1/CO[3]				
	net (fo=1, routed)	0.000	29.487	
u_FPGA_Core/Unit_Delay1_out10_carry__1_n_0				
SLICE_X11Y94				r
u_FPGA_Core/Unit_Delay1_out10_carry__2/CI				
SLICE_X11Y94	CARRY4 (Prop_carry4_CI_CO[3])	0.114	29.601	r
u_FPGA_Core/Unit_Delay1_out10_carry__2/CO[3]				
	net (fo=2, routed)	0.000	29.601	
u_FPGA_Core/Relational_Operator_out1				
SLICE_X11Y94	FDCE			r
u_FPGA_Core/Unit_Delay1_out1_reg/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)			
		0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.411	1.411	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.162	2.573	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.324	-4.751	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.639	-3.112	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.091	-3.021	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	1.527	-1.493	
u_FPGA_Core/CLK				
SLICE_X11Y94	FDCE			r
u_FPGA_Core/Unit_Delay1_out1_reg/C				
Slack:	inf			
Source:	In1[8]			
	(input port)			

Destination: u_FPGA_Core/Unit_Delay_out1_reg[20]/D
 (rising edge-triggered cell FDCE clocked by
 clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
 Path Group: (none)
 Path Type: Setup (Max at Slow Process Corner)
 Data Path Delay: 17.651ns (logic 6.642ns (37.630%) route 11.009ns
 (62.370%))
 Logic Levels: 17 (CARRY4=7 IBUF=1 LUT3=4 LUT4=3 LUT5=1 LUT6=1)
 Clock Path Skew: -1.418ns (DCD - SCD + CPR)
 Destination Clock Delay (DCD): -1.418ns
 Source Clock Delay (SCD): 0.000ns
 Clock Pessimism Removal (CPR): 0.000ns
 Clock Uncertainty: 0.179ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
 Total System Jitter (TSJ): 0.050ns
 Discrete Jitter (DJ): 0.154ns
 Phase Error (PE): 0.099ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
T8 (IN)		0.000	0.000 r	In1[8]
T8	net (fo=0)	0.000	0.000 r	In1[8]
In1_IBUF[8]_inst/I				
T8	IBUF (Prop_ibuf_I_0)	1.497	1.497 r	
In1_IBUF[8]_inst/O				
	net (fo=3, routed)	5.467	6.964	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8]				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2				
SLICE_X1Y98	LUT3 (Prop_lut3_I2_0)	0.152	7.116 r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/O				
	net (fo=2, routed)	0.469	7.585	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3				
SLICE_X1Y98	LUT4 (Prop_lut4_I3_0)	0.326	7.911 r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/O				
	net (fo=1, routed)	0.000	7.911	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]				
SLICE_X1Y98	CARRY4 (Prop_carry4_S[1]_CO[3])	0.550	8.461 r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/CO[3]				
	net (fo=1, routed)	0.000	8.461	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_0				
SLICE_X1Y99				r

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/CI			
SLICE_X1Y99	CARRY4 (Prop_carry4_CI_0[1])		
		0.334	8.795 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/O[1]			
	net (fo=2, routed)	0.582	9.377
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183_n_6			
SLICE_X4Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/I1			
SLICE_X4Y99	LUT3 (Prop_lut3_I1_0)	0.329	9.706 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/O			
	net (fo=2, routed)	0.872	10.578
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87_n_0			
SLICE_X4Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/I0			
SLICE_X4Y99	LUT4 (Prop_lut4_I0_0)	0.332	10.910 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/O			
	net (fo=1, routed)	0.000	10.910
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91_n_0			
SLICE_X4Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/S[2]			
SLICE_X4Y99	CARRY4 (Prop_carry4_S[2]_CO[3])		
		0.398	11.308 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/CO[3]			
	net (fo=1, routed)	0.001	11.309
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53_n_0			
SLICE_X4Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/CI			
SLICE_X4Y100	CARRY4 (Prop_carry4_CI_0[0])		
		0.222	11.531 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/O[0]			
	net (fo=2, routed)	0.720	12.251
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71_n_7			
SLICE_X5Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/I1			
SLICE_X5Y99	LUT3 (Prop_lut3_I1_0)	0.324	12.575 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/O			
	net (fo=2, routed)	0.817	13.392
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33_n_0			
SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/I3			
SLICE_X5Y100	LUT4 (Prop_lut4_I3_0)	0.332	13.724 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/O			
	net (fo=1, routed)	0.000	13.724
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37_n_0			
SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/S[0]			
SLICE_X5Y100	CARRY4 (Prop_carry4_S[0]_O[2])		
		0.547	14.271 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/O[2]			
	net (fo=3, routed)	0.779	15.050

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10_n_5				
SLICE_X6Y93				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10/I2				
SLICE_X6Y93	LUT3 (Prop_lut3_I2_0)	0.302	15.352	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10/O				
	net (fo=2, routed)	0.507	15.860	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10_n_0				
SLICE_X7Y93				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2/I1				
SLICE_X7Y93	LUT5 (Prop_lut5_I1_0)	0.124	15.984	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2/O				
	net (fo=2, routed)	0.795	16.778	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2_n_0				
SLICE_X7Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6/I0				
SLICE_X7Y90	LUT6 (Prop_lut6_I0_0)	0.124	16.902	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6/O				
	net (fo=1, routed)	0.000	16.902	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6_n_0				
SLICE_X7Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/S[3]				
SLICE_X7Y90	CARRY4 (Prop_carry4_S[3]_C0[3])	0.401	17.303	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/C0[3]				
	net (fo=1, routed)	0.000	17.303	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1_n_0				
SLICE_X7Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/CI				
SLICE_X7Y91	CARRY4 (Prop_carry4_CI_0[1])	0.348	17.651	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/O[1]				
	net (fo=3, routed)	0.000	17.651	
u_FPGA_Core/Discrete_FIR_Filter_out1[36]				
SLICE_X7Y91	FDCE			r
u_FPGA_Core/Unit_Delay_out1_reg[20]/D				

(clock clk_out1_clk_wiz_0_1 rise edge)				
		0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.411	1.411	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	1.162	2.573	
u_pll/inst/clk_in1_clk_wiz_0				

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    MMCME2_ADV_X1Y2
u_pll/inst/mmcm_adv_inst/CLKIN1
    MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
                                -7.324      -4.751 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
                                net (fo=1, routed)      1.639      -3.112
u_pll/inst/clk_out1_clk_wiz_0
    BUFGCTRL_X0Y16
u_pll/inst/clkout1_buf/I
    BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)      0.091      -3.021 r
u_pll/inst/clkout1_buf/O
                                net (fo=1039, routed)      1.602      -1.418
u_FPGA_Core/CLK
    SLICE_X7Y91      FDCE
u_FPGA_Core/Unit_Delay_out1_reg[20]/C

Slack:      inf
Source:      In1[8]
              (input port)
Destination: u_FPGA_Core/Unit_Delay_out1_reg[23]/D
              (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group:  (none)
Path Type:   Setup (Max at Slow Process Corner)
Data Path Delay: 17.589ns (logic 6.503ns (36.971%) route 11.086ns
(63.029%))
Logic Levels: 16 (CARRY4=6 IBUF=1 LUT3=4 LUT4=3 LUT5=1 LUT6=1)
Clock Path Skew: -1.418ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD): -1.418ns
  Source Clock Delay (SCD): 0.000ns
  Clock Pessimism Removal (CPR): 0.000ns
Clock Uncertainty: 0.179ns (((TSJ^2 + DJ^2)^1/2) / 2 + PE
  Total System Jitter (TSJ): 0.050ns
  Discrete Jitter (DJ): 0.154ns
  Phase Error (PE): 0.099ns

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Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

T8 (IN)		0.000	0.000 r	In1[8]
	net (fo=0)	0.000	0.000	In1[8]
T8			r	
In1_IBUF[8]_inst/I				
T8	IBUF (Prop_ibuf_I_0)	1.497	1.497 r	
In1_IBUF[8]_inst/O				
	net (fo=3, routed)	5.467	6.964	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8]				
SLICE_X1Y98			r	

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2	SLICE_X1Y98	LUT3 (Prop_lut3_I2_0)	0.152	7.116	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/O		net (fo=2, routed)	0.469	7.585	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0	SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3	SLICE_X1Y98	LUT4 (Prop_lut4_I3_0)	0.326	7.911	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/O		net (fo=1, routed)	0.000	7.911	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0	SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]	SLICE_X1Y98	CARRY4 (Prop_carry4_S[1]_CO[3])	0.550	8.461	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/CO[3]		net (fo=1, routed)	0.000	8.461	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_0	SLICE_X1Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/CI	SLICE_X1Y99	CARRY4 (Prop_carry4_CI_0[1])	0.334	8.795	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/O[1]		net (fo=2, routed)	0.582	9.377	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183_n_6	SLICE_X4Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/I1	SLICE_X4Y99	LUT3 (Prop_lut3_I1_0)	0.329	9.706	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/O		net (fo=2, routed)	0.872	10.578	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87_n_0	SLICE_X4Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/I0	SLICE_X4Y99	LUT4 (Prop_lut4_I0_0)	0.332	10.910	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/O		net (fo=1, routed)	0.000	10.910	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91_n_0	SLICE_X4Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/S[2]	SLICE_X4Y99	CARRY4 (Prop_carry4_S[2]_CO[3])	0.398	11.308	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/CO[3]		net (fo=1, routed)	0.001	11.309	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53_n_0	SLICE_X4Y100				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/CI	SLICE_X4Y100	CARRY4 (Prop_carry4_CI_0[0])	0.222	11.531	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/O[0]		net (fo=2, routed)	0.720	12.251	

```

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71_n_7
    SLICE_X5Y99
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/I1
    SLICE_X5Y99          LUT3 (Prop_lut3_I1_0)          0.324    12.575 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/O
    net (fo=2, routed)          0.817    13.392
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33_n_0
    SLICE_X5Y100
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/I3
    SLICE_X5Y100          LUT4 (Prop_lut4_I3_0)          0.332    13.724 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/O
    net (fo=1, routed)          0.000    13.724
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37_n_0
    SLICE_X5Y100
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/S[0]
    SLICE_X5Y100          CARRY4 (Prop_carry4_S[0]_O[3])
                                0.606    14.330 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/O[3]
    net (fo=3, routed)          0.771    15.102
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10_n_4
    SLICE_X7Y92
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/I2
    SLICE_X7Y92          LUT3 (Prop_lut3_I2_0)          0.306    15.408 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13/O
    net (fo=2, routed)          0.579    15.987
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_13_n_0
    SLICE_X5Y92
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/I1
    SLICE_X5Y92          LUT5 (Prop_lut5_I1_0)          0.124    16.111 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3/O
    net (fo=2, routed)          0.808    16.918
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_3_n_0
    SLICE_X7Y91
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/I0
    SLICE_X7Y91          LUT6 (Prop_lut6_I0_0)          0.124    17.042 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6/O
    net (fo=1, routed)          0.000    17.042
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_6_n_0
    SLICE_X7Y91
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/S[0]
    SLICE_X7Y91          CARRY4 (Prop_carry4_S[0]_O[2])
                                0.547    17.589 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/O[2]
    net (fo=2, routed)          0.000    17.589
u_FPGA_Core/Discrete_FIR_Filter_out1[37]
    SLICE_X7Y91          FDCE
u_FPGA_Core/Unit_Delay_out1_reg[23]/D
    -----
    -----

```

```

(clock clk_out1_clk_wiz_0_1 rise edge)
0.000 0.000 r
E3 0.000 0.000 r sys_clk
(IN)
net (fo=0) 0.000 0.000
u_pll/inst/clk_in1
E3 r
u_pll/inst/clkin1_ibufg/I
E3 IBUF (Prop_ibuf_I_0) 1.411 1.411 r
u_pll/inst/clkin1_ibufg/O
net (fo=1, routed) 1.162 2.573
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2 r
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
-7.324 -4.751 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
net (fo=1, routed) 1.639 -3.112
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16 r
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16 BUFG (Prop_bufg_I_0) 0.091 -3.021 r
u_pll/inst/clkout1_buf/O
net (fo=1039, routed) 1.602 -1.418
u_FPGA_Core/CLK
SLICE_X7Y91 FDCE r
u_FPGA_Core/Unit_Delay_out1_reg[23]/C

```

```

Slack: inf
Source: In1[8]
(input port)
Destination: u_FPGA_Core/Unit_Delay_out1_reg[19]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group: (none)
Path Type: Setup (Max at Slow Process Corner)
Data Path Delay: 17.538ns (logic 6.529ns (37.228%) route 11.009ns
(62.772%))
Logic Levels: 17 (CARRY4=7 IBUF=1 LUT3=4 LUT4=3 LUT5=1 LUT6=1)
Clock Path Skew: -1.418ns (DCD - SCD + CPR)
Destination Clock Delay (DCD): -1.418ns
Source Clock Delay (SCD): 0.000ns
Clock Pessimism Removal (CPR): 0.000ns
Clock Uncertainty: 0.179ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
Total System Jitter (TSJ): 0.050ns
Discrete Jitter (DJ): 0.154ns
Phase Error (PE): 0.099ns

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
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-----
T8                                0.000    0.000 r  In1[8]
(IN)
      net (fo=0)                  0.000    0.000    In1[8]
      T8                                r
In1_IBUF[8]_inst/I
      T8      IBUF (Prop_ibuf_I_0)    1.497    1.497 r
In1_IBUF[8]_inst/O
      net (fo=3, routed)            5.467    6.964
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8]
      SLICE_X1Y98                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2
      SLICE_X1Y98      LUT3 (Prop_lut3_I2_0)    0.152    7.116 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/O
      net (fo=2, routed)            0.469    7.585
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0
      SLICE_X1Y98                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3
      SLICE_X1Y98      LUT4 (Prop_lut4_I3_0)    0.326    7.911 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/O
      net (fo=1, routed)            0.000    7.911
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0
      SLICE_X1Y98                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]
      SLICE_X1Y98      CARRY4 (Prop_carry4_S[1]_C0[3])
                                0.550    8.461 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/C0[3]
      net (fo=1, routed)            0.000    8.461
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_0
      SLICE_X1Y99                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/CI
      SLICE_X1Y99      CARRY4 (Prop_carry4_CI_0[1])
                                0.334    8.795 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/O[1]
      net (fo=2, routed)            0.582    9.377
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183_n_6
      SLICE_X4Y99                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/I1
      SLICE_X4Y99      LUT3 (Prop_lut3_I1_0)    0.329    9.706 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/O
      net (fo=2, routed)            0.872    10.578
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87_n_0
      SLICE_X4Y99                                r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/I0
      SLICE_X4Y99      LUT4 (Prop_lut4_I0_0)    0.332    10.910 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/O
      net (fo=1, routed)            0.000    10.910
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91_n_0
      SLICE_X4Y99                                r

```

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/S[2]			
SLICE_X4Y99	CARRY4 (Prop_carry4_S[2]_CO[3])	0.398	11.308 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/CO[3]	net (fo=1, routed)	0.001	11.309
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53_n_0			
SLICE_X4Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/CI			
SLICE_X4Y100	CARRY4 (Prop_carry4_CI_0[0])	0.222	11.531 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/O[0]	net (fo=2, routed)	0.720	12.251
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71_n_7			
SLICE_X5Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/I1			
SLICE_X5Y99	LUT3 (Prop_lut3_I1_0)	0.324	12.575 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/O	net (fo=2, routed)	0.817	13.392
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33_n_0			
SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/I3			
SLICE_X5Y100	LUT4 (Prop_lut4_I3_0)	0.332	13.724 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/O	net (fo=1, routed)	0.000	13.724
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37_n_0			
SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/S[0]			
SLICE_X5Y100	CARRY4 (Prop_carry4_S[0]_O[2])	0.547	14.271 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/O[2]	net (fo=3, routed)	0.779	15.050
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10_n_5			
SLICE_X6Y93			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10/I2			
SLICE_X6Y93	LUT3 (Prop_lut3_I2_0)	0.302	15.352 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10/O	net (fo=2, routed)	0.507	15.860
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10_n_0			
SLICE_X7Y93			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2/I1			
SLICE_X7Y93	LUT5 (Prop_lut5_I1_0)	0.124	15.984 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2/O	net (fo=2, routed)	0.795	16.778
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2_n_0			
SLICE_X7Y90			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6/I0			
SLICE_X7Y90	LUT6 (Prop_lut6_I0_0)	0.124	16.902 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6/O	net (fo=1, routed)	0.000	16.902
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6_n_0			

```

SLICE_X7Y90
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/S[3]
SLICE_X7Y90 CARRY4 (Prop_carry4_S[3]_CO[3])
0.401 17.303 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/CO[3]
net (fo=1, routed) 0.000 17.303
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1_n_0
SLICE_X7Y91
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/CI
SLICE_X7Y91 CARRY4 (Prop_carry4_CI_0[0])
0.235 17.538 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_1/O[0]
net (fo=3, routed) 0.000 17.538
u_FPGA_Core/Discrete_FIR_Filter_out1[35]
SLICE_X7Y91 FDCE
u_FPGA_Core/Unit_Delay_out1_reg[19]/D
-----
(clock clk_out1_clk_wiz_0_1 rise edge)
0.000 0.000 r
E3 0.000 0.000 r sys_clk
(IN)
net (fo=0) 0.000 0.000
u_pll/inst/clk_in1
E3
u_pll/inst/clkin1_ibufg/I
E3 IBUF (Prop_ibuf_I_0) 1.411 1.411 r
u_pll/inst/clkin1_ibufg/O
net (fo=1, routed) 1.162 2.573
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
-7.324 -4.751 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
net (fo=1, routed) 1.639 -3.112
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16 BUFG (Prop_bufg_I_0) 0.091 -3.021 r
u_pll/inst/clkout1_buf/O
net (fo=1039, routed) 1.602 -1.418
u_FPGA_Core/CLK
SLICE_X7Y91 FDCE
u_FPGA_Core/Unit_Delay_out1_reg[19]/C

Slack: inf
Source: In1[8]
(input port)

```

Destination: u_FPGA_Core/Unit_Delay_out1_reg[18]/D
 (rising edge-triggered cell FDCE clocked by
 clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
 Path Group: (none)
 Path Type: Setup (Max at Slow Process Corner)
 Data Path Delay: 17.150ns (logic 6.141ns (35.808%) route 11.009ns
 (64.192%))
 Logic Levels: 16 (CARRY4=6 IBUF=1 LUT3=4 LUT4=3 LUT5=1 LUT6=1)
 Clock Path Skew: -1.419ns (DCD - SCD + CPR)
 Destination Clock Delay (DCD): -1.419ns
 Source Clock Delay (SCD): 0.000ns
 Clock Pessimism Removal (CPR): 0.000ns
 Clock Uncertainty: 0.179ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
 Total System Jitter (TSJ): 0.050ns
 Discrete Jitter (DJ): 0.154ns
 Phase Error (PE): 0.099ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
T8 (IN)		0.000	0.000 r	In1[8]
T8	net (fo=0)	0.000	0.000 r	In1[8]
In1_IBUF[8]_inst/I				
T8	IBUF (Prop_ibuf_I_0)	1.497	1.497 r	
In1_IBUF[8]_inst/O				
	net (fo=3, routed)	5.467	6.964	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8]				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2				
SLICE_X1Y98	LUT3 (Prop_lut3_I2_0)	0.152	7.116 r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/O				
	net (fo=2, routed)	0.469	7.585	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3				
SLICE_X1Y98	LUT4 (Prop_lut4_I3_0)	0.326	7.911 r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/O				
	net (fo=1, routed)	0.000	7.911	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]				
SLICE_X1Y98	CARRY4 (Prop_carry4_S[1]_CO[3])	0.550	8.461 r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/CO[3]				
	net (fo=1, routed)	0.000	8.461	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_0				
SLICE_X1Y99				r

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/CI			
SLICE_X1Y99	CARRY4 (Prop_carry4_CI_0[1])		
		0.334	8.795 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/O[1]			
	net (fo=2, routed)	0.582	9.377
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183_n_6			
SLICE_X4Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/I1			
SLICE_X4Y99	LUT3 (Prop_lut3_I1_0)	0.329	9.706 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/O			
	net (fo=2, routed)	0.872	10.578
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87_n_0			
SLICE_X4Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/I0			
SLICE_X4Y99	LUT4 (Prop_lut4_I0_0)	0.332	10.910 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/O			
	net (fo=1, routed)	0.000	10.910
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91_n_0			
SLICE_X4Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/S[2]			
SLICE_X4Y99	CARRY4 (Prop_carry4_S[2]_CO[3])		
		0.398	11.308 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/CO[3]			
	net (fo=1, routed)	0.001	11.309
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53_n_0			
SLICE_X4Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/CI			
SLICE_X4Y100	CARRY4 (Prop_carry4_CI_0[0])		
		0.222	11.531 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/O[0]			
	net (fo=2, routed)	0.720	12.251
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71_n_7			
SLICE_X5Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/I1			
SLICE_X5Y99	LUT3 (Prop_lut3_I1_0)	0.324	12.575 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/O			
	net (fo=2, routed)	0.817	13.392
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33_n_0			
SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/I3			
SLICE_X5Y100	LUT4 (Prop_lut4_I3_0)	0.332	13.724 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/O			
	net (fo=1, routed)	0.000	13.724
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37_n_0			
SLICE_X5Y100			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/S[0]			
SLICE_X5Y100	CARRY4 (Prop_carry4_S[0]_O[2])		
		0.547	14.271 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/O[2]			
	net (fo=3, routed)	0.779	15.050

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u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10_n_5
  SLICE_X6Y93
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10/I2
  SLICE_X6Y93      LUT3 (Prop_lut3_I2_0)      0.302      15.352 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10/O
  net (fo=2, routed)      0.507      15.860
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_10_n_0
  SLICE_X7Y93
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2/I1
  SLICE_X7Y93      LUT5 (Prop_lut5_I1_0)      0.124      15.984 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2/O
  net (fo=2, routed)      0.795      16.778
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_2_n_0
  SLICE_X7Y90
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6/I0
  SLICE_X7Y90      LUT6 (Prop_lut6_I0_0)      0.124      16.902 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6/O
  net (fo=1, routed)      0.000      16.902
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_6_n_0
  SLICE_X7Y90
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/S[3]
  SLICE_X7Y90      CARRY4 (Prop_carry4_S[3]_O[3])      0.248      17.150 r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/O[3]
  net (fo=3, routed)      0.000      17.150
u_FPGA_Core/Discrete_FIR_Filter_out1[34]
  SLICE_X7Y90      FDCE
u_FPGA_Core/Unit_Delay_out1_reg[18]/D
  -----
  (clock clk_out1_clk_wiz_0_1 rise edge)
  E3      0.000      0.000 r
(IN)      0.000      0.000 r sys_clk
  net (fo=0)      0.000      0.000
u_pll/inst/clk_in1
  E3
u_pll/inst/clkin1_ibufg/I
  E3      IBUF (Prop_ibuf_I_0)      1.411      1.411 r
u_pll/inst/clkin1_ibufg/O
  net (fo=1, routed)      1.162      2.573
u_pll/inst/clk_in1_clk_wiz_0
  MMCME2_ADV_X1Y2
u_pll/inst/mmcm_adv_inst/CLKIN1
  MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)      -7.324      -4.751 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
  net (fo=1, routed)      1.639      -3.112
u_pll/inst/clk_out1_clk_wiz_0

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    BUFGCTRL_X0Y16
u_pll/inst/clkout1_buf/I
    BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)      0.091    -3.021 r
u_pll/inst/clkout1_buf/O
    net (fo=1039, routed)      1.601    -1.419
u_FPGA_Core/CLK
    SLICE_X7Y90      FDCE
u_FPGA_Core/Unit_Delay_out1_reg[18]/C

Slack:      inf
Source:      In1[8]
              (input port)
Destination: u_FPGA_Core/Unit_Delay_out1_reg[17]/D
              (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group:  (none)
Path Type:   Setup (Max at Slow Process Corner)
Data Path Delay: 16.818ns (logic 6.170ns (36.687%) route 10.648ns
(63.313%))
Logic Levels: 16 (CARRY4=6 IBUF=1 LUT3=4 LUT4=3 LUT5=1 LUT6=1)
Clock Path Skew: -1.419ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD): -1.419ns
  Source Clock Delay (SCD): 0.000ns
  Clock Pessimism Removal (CPR): 0.000ns
Clock Uncertainty: 0.179ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
  Total System Jitter (TSJ): 0.050ns
  Discrete Jitter (DJ): 0.154ns
  Phase Error (PE): 0.099ns

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Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
T8 (IN)		0.000	0.000 r	In1[8]
	net (fo=0)	0.000	0.000	In1[8]
T8 In1_IBUF[8]_inst/I			r	
T8 In1_IBUF[8]_inst/O	IBUF (Prop_ibuf_I_0)	1.497	1.497 r	
	net (fo=3, routed)	5.467	6.964	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8] SLICE_X1Y98			r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2 SLICE_X1Y98	LUT3 (Prop_lut3_I2_0)	0.152	7.116 r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/O	net (fo=2, routed)	0.469	7.585	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0 SLICE_X1Y98			r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3				

SLICE_X1Y98	LUT4 (Prop_lut4_I3_0)	0.326	7.911	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/0	net (fo=1, routed)	0.000	7.911	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]				
SLICE_X1Y98	CARRY4 (Prop_carry4_S[1]_CO[3])	0.550	8.461	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/CO[3]	net (fo=1, routed)	0.000	8.461	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_0				
SLICE_X1Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/CI				
SLICE_X1Y99	CARRY4 (Prop_carry4_CI_0[1])	0.334	8.795	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183/O[1]	net (fo=2, routed)	0.582	9.377	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_183_n_6				
SLICE_X4Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/I1				
SLICE_X4Y99	LUT3 (Prop_lut3_I1_0)	0.329	9.706	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87/0	net (fo=2, routed)	0.872	10.578	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_87_n_0				
SLICE_X4Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/I0				
SLICE_X4Y99	LUT4 (Prop_lut4_I0_0)	0.332	10.910	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91/0	net (fo=1, routed)	0.000	10.910	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_91_n_0				
SLICE_X4Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/S[2]				
SLICE_X4Y99	CARRY4 (Prop_carry4_S[2]_CO[3])	0.398	11.308	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53/CO[3]	net (fo=1, routed)	0.001	11.309	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_53_n_0				
SLICE_X4Y100				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/CI				
SLICE_X4Y100	CARRY4 (Prop_carry4_CI_0[0])	0.222	11.531	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71/O[0]	net (fo=2, routed)	0.720	12.251	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_71_n_7				
SLICE_X5Y99				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/I1				
SLICE_X5Y99	LUT3 (Prop_lut3_I1_0)	0.324	12.575	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33/0	net (fo=2, routed)	0.817	13.392	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_33_n_0				

SLICE_X5Y100				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/I3				
SLICE_X5Y100	LUT4 (Prop_lut4_I3_0)	0.332	13.724	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37/O				
	net (fo=1, routed)	0.000	13.724	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[23]_i_37_n_0				
SLICE_X5Y100				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/S[0]				
SLICE_X5Y100	CARRY4 (Prop_carry4_S[0]_O[0])	0.247	13.971	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10/O[0]				
	net (fo=3, routed)	0.720	14.692	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[23]_i_10_n_7				
SLICE_X6Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_12/I2				
SLICE_X6Y92	LUT3 (Prop_lut3_I2_0)	0.299	14.991	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_12/O				
	net (fo=2, routed)	0.421	15.412	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_12_n_0				
SLICE_X5Y91				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_4/I1				
SLICE_X5Y91	LUT5 (Prop_lut5_I1_0)	0.124	15.536	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_4/O				
	net (fo=2, routed)	0.578	16.114	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_4_n_0				
SLICE_X7Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_8/I0				
SLICE_X7Y90	LUT6 (Prop_lut6_I0_0)	0.124	16.238	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_8/O				
	net (fo=1, routed)	0.000	16.238	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_8_n_0				
SLICE_X7Y90				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/S[1]				
SLICE_X7Y90	CARRY4 (Prop_carry4_S[1]_O[2])	0.580	16.818	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/O[2]				
	net (fo=3, routed)	0.000	16.818	
u_FPGA_Core/Discrete_FIR_Filter_out1[33]				
SLICE_X7Y90	FDCE			r
u_FPGA_Core/Unit_Delay_out1_reg[17]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)			
		0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r

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u_pll/inst/clkin1_ibufg/I
  E3          IBUF (Prop_ibuf_I_0)          1.411      1.411 r
u_pll/inst/clkin1_ibufg/O
  net (fo=1, routed)          1.162      2.573
u_pll/inst/clk_in1_clk_wiz_0
  MMCME2_ADV_X1Y2                      r
u_pll/inst/mmcm_adv_inst/CLKIN1
  MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
                              -7.324      -4.751 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
  net (fo=1, routed)          1.639      -3.112
u_pll/inst/clk_out1_clk_wiz_0
  BUFGCTRL_X0Y16                      r
u_pll/inst/clkout1_buf/I
  BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)      0.091      -3.021 r
u_pll/inst/clkout1_buf/O
  net (fo=1039, routed)      1.601      -1.419
u_FPGA_Core/CLK
  SLICE_X7Y90      FDCE                      r
u_FPGA_Core/Unit_Delay_out1_reg[17]/C

Slack:          inf
Source:          In1[8]
                  (input port)
Destination:     u_FPGA_Core/Unit_Delay_out1_reg[16]/D
                  (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group:      (none)
Path Type:       Setup (Max at Slow Process Corner)
Data Path Delay: 16.644ns (logic 5.891ns (35.395%) route 10.753ns
(64.605%))
Logic Levels:    14 (CARRY4=6 IBUF=1 LUT3=4 LUT4=2 LUT5=1)
Clock Path Skew: -1.419ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD): -1.419ns
  Source Clock Delay (SCD): 0.000ns
  Clock Pessimism Removal (CPR): 0.000ns
Clock Uncertainty: 0.179ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
  Total System Jitter (TSJ): 0.050ns
  Discrete Jitter (DJ): 0.154ns
  Phase Error (PE): 0.099ns

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

T8 (IN)		0.000	0.000 r	In1[8]
	net (fo=0)	0.000	0.000	In1[8]
T8 In1_IBUF[8]_inst/I			r	

T8	IBUF (Prop_ibuf_I_0)	1.497	1.497	r
In1_IBUF[8]_inst/0				
	net (fo=3, routed)	5.467	6.964	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8]				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2				
SLICE_X1Y98	LUT3 (Prop_lut3_I2_0)	0.152	7.116	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/0				
	net (fo=2, routed)	0.469	7.585	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3				
SLICE_X1Y98	LUT4 (Prop_lut4_I3_0)	0.326	7.911	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/0				
	net (fo=1, routed)	0.000	7.911	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]				
SLICE_X1Y98	CARRY4 (Prop_carry4_S[1]_O[2])	0.580	8.491	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/O[2]				
	net (fo=2, routed)	0.582	9.073	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_5				
SLICE_X4Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86/I1				
SLICE_X4Y98	LUT3 (Prop_lut3_I1_0)	0.331	9.404	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86/0				
	net (fo=2, routed)	0.690	10.094	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86_n_0				
SLICE_X4Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90/I3				
SLICE_X4Y98	LUT4 (Prop_lut4_I3_0)	0.327	10.421	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90/0				
	net (fo=1, routed)	0.000	10.421	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90_n_0				
SLICE_X4Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53/S[3]				
SLICE_X4Y98	CARRY4 (Prop_carry4_S[3]_O[3])	0.248	10.669	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53/O[3]				
	net (fo=2, routed)	0.833	11.502	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53_n_4				
SLICE_X5Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26/I1				
SLICE_X5Y98	LUT3 (Prop_lut3_I1_0)	0.335	11.837	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26/0				
	net (fo=2, routed)	0.783	12.620	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26_n_0				
SLICE_X5Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/DI[3]				

SLICE_X5Y98	CARRY4 (Prop_carry4_DI[3]_CO[3])	0.588	13.208	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/CO[3]	net (fo=1, routed)	0.000	13.208	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14_n_0	SLICE_X5Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/CI	SLICE_X5Y99	CARRY4 (Prop_carry4_CI_0[1])	0.334	13.542
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/O[1]	net (fo=3, routed)	0.783	14.325	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14_n_6	SLICE_X6Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11/I2	SLICE_X6Y92	LUT3 (Prop_lut3_I2_0)	0.303	14.628
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11/O	net (fo=2, routed)	0.515	15.142	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11_n_0	SLICE_X7Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3/I1	SLICE_X7Y92	LUT5 (Prop_lut5_I1_0)	0.124	15.266
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3/O	net (fo=2, routed)	0.631	15.898	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3_n_0	SLICE_X7Y89			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/DI[2]	SLICE_X7Y89	CARRY4 (Prop_carry4_DI[2]_CO[3])	0.398	16.296
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/CO[3]	net (fo=1, routed)	0.000	16.296	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1_n_0	SLICE_X7Y90			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/CI	SLICE_X7Y90	CARRY4 (Prop_carry4_CI_0[1])	0.348	16.644
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/O[1]	net (fo=3, routed)	0.000	16.644	
u_FPGA_Core/Discrete_FIR_Filter_out1[32]	SLICE_X7Y90	FDCE		r
u_FPGA_Core/Unit_Delay_out1_reg[16]/D	-----			

(clock clk_out1_clk_wiz_0_1 rise edge)		0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1	E3			r

```

u_pll/inst/clkin1_ibufg/I
  E3          IBUF (Prop_ibuf_I_0)          1.411      1.411 r
u_pll/inst/clkin1_ibufg/O
  net (fo=1, routed)          1.162      2.573
u_pll/inst/clk_in1_clk_wiz_0
  MMCME2_ADV_X1Y2                                r
u_pll/inst/mmcm_adv_inst/CLKIN1
  MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
                                -7.324      -4.751 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
  net (fo=1, routed)          1.639      -3.112
u_pll/inst/clk_out1_clk_wiz_0
  BUFGCTRL_X0Y16                                r
u_pll/inst/clkout1_buf/I
  BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)          0.091      -3.021 r
u_pll/inst/clkout1_buf/O
  net (fo=1039, routed)        1.601      -1.419
u_FPGA_Core/CLK
  SLICE_X7Y90          FDCE                                r
u_FPGA_Core/Unit_Delay_out1_reg[16]/C

Slack:          inf
Source:          In1[8]
                  (input port)
Destination:      u_FPGA_Core/Unit_Delay_out1_reg[15]/D
                  (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group:      (none)
Path Type:        Setup (Max at Slow Process Corner)
Data Path Delay:  16.531ns (logic 5.778ns (34.953%) route 10.753ns
(65.047%))
Logic Levels:     14 (CARRY4=6 IBUF=1 LUT3=4 LUT4=2 LUT5=1)
Clock Path Skew:  -1.419ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD):  -1.419ns
  Source Clock Delay (SCD):        0.000ns
  Clock Pessimism Removal (CPR):   0.000ns
Clock Uncertainty: 0.179ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
  Total System Jitter (TSJ):       0.050ns
  Discrete Jitter (DJ):            0.154ns
  Phase Error (PE):                0.099ns

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

T8 (IN)		0.000	0.000 r	In1[8]
	net (fo=0)	0.000	0.000	In1[8]
T8 In1_IBUF[8]_inst/I			r	

T8	IBUF (Prop_ibuf_I_0)	1.497	1.497	r
In1_IBUF[8]_inst/0				
	net (fo=3, routed)	5.467	6.964	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8]				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2				
SLICE_X1Y98	LUT3 (Prop_lut3_I2_0)	0.152	7.116	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/0				
	net (fo=2, routed)	0.469	7.585	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3				
SLICE_X1Y98	LUT4 (Prop_lut4_I3_0)	0.326	7.911	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/0				
	net (fo=1, routed)	0.000	7.911	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]				
SLICE_X1Y98	CARRY4 (Prop_carry4_S[1]_O[2])	0.580	8.491	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/O[2]				
	net (fo=2, routed)	0.582	9.073	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_5				
SLICE_X4Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86/I1				
SLICE_X4Y98	LUT3 (Prop_lut3_I1_0)	0.331	9.404	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86/0				
	net (fo=2, routed)	0.690	10.094	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86_n_0				
SLICE_X4Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90/I3				
SLICE_X4Y98	LUT4 (Prop_lut4_I3_0)	0.327	10.421	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90/0				
	net (fo=1, routed)	0.000	10.421	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90_n_0				
SLICE_X4Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53/S[3]				
SLICE_X4Y98	CARRY4 (Prop_carry4_S[3]_O[3])	0.248	10.669	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53/O[3]				
	net (fo=2, routed)	0.833	11.502	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53_n_4				
SLICE_X5Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26/I1				
SLICE_X5Y98	LUT3 (Prop_lut3_I1_0)	0.335	11.837	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26/0				
	net (fo=2, routed)	0.783	12.620	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26_n_0				
SLICE_X5Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/DI[3]				

SLICE_X5Y98	CARRY4 (Prop_carry4_DI[3]_CO[3])	0.588	13.208	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/CO[3]	net (fo=1, routed)	0.000	13.208	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14_n_0	SLICE_X5Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/CI	SLICE_X5Y99	CARRY4 (Prop_carry4_CI_0[1])	0.334	13.542
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/O[1]	net (fo=3, routed)	0.783	14.325	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14_n_6	SLICE_X6Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11/I2	SLICE_X6Y92	LUT3 (Prop_lut3_I2_0)	0.303	14.628
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11/O	net (fo=2, routed)	0.515	15.142	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11_n_0	SLICE_X7Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3/I1	SLICE_X7Y92	LUT5 (Prop_lut5_I1_0)	0.124	15.266
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3/O	net (fo=2, routed)	0.631	15.898	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3_n_0	SLICE_X7Y89			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/DI[2]	SLICE_X7Y89	CARRY4 (Prop_carry4_DI[2]_CO[3])	0.398	16.296
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/CO[3]	net (fo=1, routed)	0.000	16.296	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1_n_0	SLICE_X7Y90			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/CI	SLICE_X7Y90	CARRY4 (Prop_carry4_CI_0[0])	0.235	16.531
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_1/O[0]	net (fo=3, routed)	0.000	16.531	
u_FPGA_Core/Discrete_FIR_Filter_out1[31]	SLICE_X7Y90	FDCE		r
u_FPGA_Core/Unit_Delay_out1_reg[15]/D	-----			

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1	E3			r

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u_pll/inst/clkin1_ibufg/I
  E3          IBUF (Prop_ibuf_I_0)          1.411      1.411 r
u_pll/inst/clkin1_ibufg/O
  net (fo=1, routed)          1.162      2.573
u_pll/inst/clk_in1_clk_wiz_0
  MMCME2_ADV_X1Y2                                r
u_pll/inst/mmcm_adv_inst/CLKIN1
  MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
                                -7.324      -4.751 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
  net (fo=1, routed)          1.639      -3.112
u_pll/inst/clk_out1_clk_wiz_0
  BUFGCTRL_X0Y16                                r
u_pll/inst/clkout1_buf/I
  BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)          0.091      -3.021 r
u_pll/inst/clkout1_buf/O
  net (fo=1039, routed)        1.601      -1.419
u_FPGA_Core/CLK
  SLICE_X7Y90          FDCE                                r
u_FPGA_Core/Unit_Delay_out1_reg[15]/C

Slack:          inf
Source:          In1[8]
                 (input port)
Destination:     u_FPGA_Core/Unit_Delay_out1_reg[14]/D
                 (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group:      (none)
Path Type:       Setup (Max at Slow Process Corner)
Data Path Delay:  16.353ns (logic 5.600ns (34.245%) route 10.753ns
(65.755%))
Logic Levels:    13 (CARRY4=5 IBUF=1 LUT3=4 LUT4=2 LUT5=1)
Clock Path Skew: -1.419ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD):  -1.419ns
  Source Clock Delay (SCD):        0.000ns
  Clock Pessimism Removal (CPR):   0.000ns
Clock Uncertainty: 0.179ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
  Total System Jitter (TSJ):       0.050ns
  Discrete Jitter (DJ):            0.154ns
  Phase Error (PE):                0.099ns

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Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

T8		0.000	0.000 r	In1[8]
(IN)				
	net (fo=0)	0.000	0.000	In1[8]
T8			r	
In1_IBUF[8]_inst/I				

T8	IBUF (Prop_ibuf_I_0)	1.497	1.497	r
In1_IBUF[8]_inst/0				
	net (fo=3, routed)	5.467	6.964	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8]				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2				
SLICE_X1Y98	LUT3 (Prop_lut3_I2_0)	0.152	7.116	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/0				
	net (fo=2, routed)	0.469	7.585	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3				
SLICE_X1Y98	LUT4 (Prop_lut4_I3_0)	0.326	7.911	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/0				
	net (fo=1, routed)	0.000	7.911	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0				
SLICE_X1Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]				
SLICE_X1Y98	CARRY4 (Prop_carry4_S[1]_O[2])	0.580	8.491	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/O[2]				
	net (fo=2, routed)	0.582	9.073	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_5				
SLICE_X4Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86/I1				
SLICE_X4Y98	LUT3 (Prop_lut3_I1_0)	0.331	9.404	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86/0				
	net (fo=2, routed)	0.690	10.094	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86_n_0				
SLICE_X4Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90/I3				
SLICE_X4Y98	LUT4 (Prop_lut4_I3_0)	0.327	10.421	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90/0				
	net (fo=1, routed)	0.000	10.421	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90_n_0				
SLICE_X4Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53/S[3]				
SLICE_X4Y98	CARRY4 (Prop_carry4_S[3]_O[3])	0.248	10.669	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53/O[3]				
	net (fo=2, routed)	0.833	11.502	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53_n_4				
SLICE_X5Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26/I1				
SLICE_X5Y98	LUT3 (Prop_lut3_I1_0)	0.335	11.837	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26/0				
	net (fo=2, routed)	0.783	12.620	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26_n_0				
SLICE_X5Y98				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/DI[3]				

SLICE_X5Y98	CARRY4 (Prop_carry4_DI[3]_CO[3])	0.588	13.208	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/CO[3]	net (fo=1, routed)	0.000	13.208	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14_n_0	SLICE_X5Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/CI	SLICE_X5Y99	CARRY4 (Prop_carry4_CI_0[1])	0.334	13.542
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/O[1]	net (fo=3, routed)	0.783	14.325	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14_n_6	SLICE_X6Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11/I2	SLICE_X6Y92	LUT3 (Prop_lut3_I2_0)	0.303	14.628
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11/O	net (fo=2, routed)	0.515	15.142	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_11_n_0	SLICE_X7Y92			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3/I1	SLICE_X7Y92	LUT5 (Prop_lut5_I1_0)	0.124	15.266
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3/O	net (fo=2, routed)	0.631	15.898	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_3_n_0	SLICE_X7Y89			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/DI[2]	SLICE_X7Y89	CARRY4 (Prop_carry4_DI[2]_O[3])	0.455	16.353
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/O[3]	net (fo=3, routed)	0.000	16.353	
u_FPGA_Core/Discrete_FIR_Filter_out1[30]	SLICE_X7Y89	FDCE		r
u_FPGA_Core/Unit_Delay_out1_reg[14]/D				

(clock clk_out1_clk_wiz_0_1 rise edge)				
E3		0.000	0.000	r
(IN)		0.000	0.000	r sys_clk
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I	E3	IBUF (Prop_ibuf_I_0)	1.411	1.411
u_pll/inst/clkin1_ibufg/O	net (fo=1, routed)	1.162	2.573	
u_pll/inst/clk_in1_clk_wiz_0	MMCME2_ADV_X1Y2			r
u_pll/inst/mmcm_adv_inst/CLKIN1				

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MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
                        -7.324    -4.751 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
                        net (fo=1, routed)      1.639    -3.112
u_pll/inst/clk_out1_clk_wiz_0
  BUFGCTRL_X0Y16                                           r
u_pll/inst/clkout1_buf/I
  BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)      0.091    -3.021 r
u_pll/inst/clkout1_buf/O
                        net (fo=1039, routed)    1.601    -1.419
u_FPGA_Core/CLK
  SLICE_X7Y89          FDCE                                           r
u_FPGA_Core/Unit_Delay_out1_reg[14]/C

Slack:                inf
Source:                In1[8]
                      (input port)
Destination:          u_FPGA_Core/Unit_Delay_out1_reg[13]/D
                      (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group:            (none)
Path Type:             Setup (Max at Slow Process Corner)
Data Path Delay:       16.282ns (logic 5.584ns (34.296%) route 10.698ns
(65.704%))
Logic Levels:         13 (CARRY4=5 IBUF=1 LUT3=4 LUT4=2 LUT5=1)
Clock Path Skew:       -1.419ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD):    -1.419ns
  Source Clock Delay (SCD):         0.000ns
  Clock Pessimism Removal (CPR):    0.000ns
Clock Uncertainty:     0.179ns ((TSJ^2 + DJ^2)^1/2) / 2 + PE
  Total System Jitter (TSJ):        0.050ns
  Discrete Jitter (DJ):             0.154ns
  Phase Error (PE):                 0.099ns

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
T8 (IN)		0.000	0.000 r	In1[8]
	net (fo=0)	0.000	0.000	In1[8]
T8 In1_IBUF[8]_inst/I			r	
T8 In1_IBUF[8]_inst/O	IBUF (Prop_ibuf_I_0)	1.497	1.497 r	
	net (fo=3, routed)	5.467	6.964	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[8] SLICE_X1Y98			r	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/I2 SLICE_X1Y98	LUT3 (Prop_lut3_I2_0)	0.152	7.116 r	

u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182/0	net (fo=2, routed)	0.469	7.585	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_182_n_0	SLICE_X1Y98			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/I3	SLICE_X1Y98 LUT4 (Prop_lut4_I3_0)	0.326	7.911	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186/0	net (fo=1, routed)	0.000	7.911	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[18]_i_186_n_0	SLICE_X1Y98			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/S[1]	SLICE_X1Y98 CARRY4 (Prop_carry4_S[1]_O[2])	0.580	8.491	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159/O[2]	net (fo=2, routed)	0.582	9.073	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_159_n_5	SLICE_X4Y98			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86/I1	SLICE_X4Y98 LUT3 (Prop_lut3_I1_0)	0.331	9.404	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86/0	net (fo=2, routed)	0.690	10.094	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_86_n_0	SLICE_X4Y98			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90/I3	SLICE_X4Y98 LUT4 (Prop_lut4_I3_0)	0.327	10.421	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90/0	net (fo=1, routed)	0.000	10.421	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_90_n_0	SLICE_X4Y98			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53/S[3]	SLICE_X4Y98 CARRY4 (Prop_carry4_S[3]_O[3])	0.248	10.669	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53/O[3]	net (fo=2, routed)	0.833	11.502	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_53_n_4	SLICE_X5Y98			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26/I1	SLICE_X5Y98 LUT3 (Prop_lut3_I1_0)	0.335	11.837	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26/0	net (fo=2, routed)	0.783	12.620	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_26_n_0	SLICE_X5Y98			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/DI[3]	SLICE_X5Y98 CARRY4 (Prop_carry4_DI[3]_CO[3])	0.588	13.208	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14/CO[3]	net (fo=1, routed)	0.000	13.208	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_14_n_0	SLICE_X5Y99			r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/CI				

SLICE_X5Y99	CARRY4 (Prop_carry4_CI_0[0])	0.223	13.431	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14/0[0]				
	net (fo=3, routed)	0.659	14.090	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[18]_i_14_n_7				
SLICE_X7Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_12/I2				
SLICE_X7Y92	LUT3 (Prop_lut3_I2_0)	0.299	14.389	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_12/0				
	net (fo=2, routed)	0.532	14.921	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_12_n_0				
SLICE_X6Y92				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_4/I1				
SLICE_X6Y92	LUT5 (Prop_lut5_I1_0)	0.124	15.045	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_4/0				
	net (fo=2, routed)	0.684	15.728	
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1[14]_i_4_n_0				
SLICE_X7Y89				r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/DI[1]				
SLICE_X7Y89	CARRY4 (Prop_carry4_DI[1]_0[2])	0.554	16.282	r
u_FPGA_Core/u_Discrete_FIR_Filter/Unit_Delay_out1_reg[14]_i_1/0[2]				
	net (fo=3, routed)	0.000	16.282	
u_FPGA_Core/Discrete_FIR_Filter_out1[29]				
SLICE_X7Y89	FDCE			r
u_FPGA_Core/Unit_Delay_out1_reg[13]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	1.411	1.411	r
u_pll/inst/clkin1_ibufg/0				
	net (fo=1, routed)	1.162	2.573	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-7.324	-4.751	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	1.639	-3.112	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				

```

    BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)      0.091    -3.021 r
u_pll/inst/clkout1_buf/0
                        net (fo=1039, routed)      1.601    -1.419
u_FPGA_Core/CLK
    SLICE_X7Y89         FDCE                      r
u_FPGA_Core/Unit_Delay_out1_reg[13]/C

```

Min Delay Paths

```

---
Slack:                inf
Source:               In1[1]
                    (input port)
Destination:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][1]/D
                    (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group:          (none)
Path Type:           Hold (Min at Fast Process Corner)
Data Path Delay:     0.617ns (logic 0.247ns (40.088%) route 0.370ns
(59.912%))
Logic Levels:       1 (IBUF=1)
Clock Path Skew:    -0.796ns (DCD - SCD - CPR)
Destination Clock Delay (DCD): -0.796ns
Source Clock Delay (SCD): 0.000ns
Clock Pessimism Removal (CPR): -0.000ns

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
L16 (IN)		0.000	0.000 r	In1[1]
L16	net (fo=0)	0.000	0.000 r	In1[1]
In1_IBUF[1]_inst/I				
L16	IBUF (Prop_ibuf_I_0)	0.247	0.247 r	
In1_IBUF[1]_inst/O				
	net (fo=3, routed)	0.370	0.617	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[1]				
SLICE_X0Y95	FDCE		r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][1]/D				

(clock clk_out1_clk_wiz_0_1 rise edge)

		0.000	0.000	r	
E3		0.000	0.000	r	sys_clk
(IN)					
	net (fo=0)	0.000	0.000		
u_pll/inst/clk_in1					
E3				r	
u_pll/inst/clkin1_ibufg/I					
E3	IBUF (Prop_ibuf_I_0)	0.438	0.438	r	
u_pll/inst/clkin1_ibufg/O					
	net (fo=1, routed)	0.480	0.918		
u_pll/inst/clk_in1_clk_wiz_0					
MMCME2_ADV_X1Y2				r	
u_pll/inst/mmcm_adv_inst/CLKIN1					
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245	r	
u_pll/inst/mmcm_adv_inst/CLKOUT0					
	net (fo=1, routed)	0.544	-1.702		
u_pll/inst/clk_out1_clk_wiz_0					
BUFGCTRL_X0Y16				r	
u_pll/inst/clkout1_buf/I					
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673	r	
u_pll/inst/clkout1_buf/O					
	net (fo=1039, routed)	0.877	-0.796		
u_FPGA_Core/u_Discrete_FIR_Filter/CLK					
SLICE_X0Y95	FDCE			r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][1]/C					

Slack: inf

Source: In1[2]
(input port)

Destination:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][2]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: (none)

Path Type: Hold (Min at Fast Process Corner)

Data Path Delay: 0.714ns (logic 0.253ns (35.432%) route 0.461ns
(64.568%))

Logic Levels: 1 (IBUF=1)

Clock Path Skew: -0.796ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): -0.796ns

Source Clock Delay (SCD): 0.000ns

Clock Pessimism Removal (CPR): -0.000ns

Location	Delay type	Incr(ns)	Path(ns)	Netlist
Resource(s)				

M13		0.000	0.000	r In1[2]
(IN)				

	net (fo=0)	0.000	0.000	In1[2]
M13				r
In1_IBUF[2]_inst/I				
M13	IBUF (Prop_ibuf_I_0)	0.253	0.253	r
In1_IBUF[2]_inst/O				
	net (fo=3, routed)	0.461	0.714	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[2]				
SLICE_X3Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][2]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)	0.000	0.000	r
E3		0.000	0.000	r sys_clk
(IN)				
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3				r
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.438	0.438	r
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.480	0.918	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2				r
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245	r
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.544	-1.702	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16				r
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673	r
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.877	-0.796	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				
SLICE_X3Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][2]/C				

Slack: inf

Source: In1[0]
(input port)

Destination:

u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][0]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})

Path Group: (none)

Path Type: Hold (Min at Fast Process Corner)

Data Path Delay: 0.723ns (logic 0.245ns (33.947%) route 0.478ns)

(66.053%))

Logic Levels: 1 (IBUF=1)
Clock Path Skew: -0.796ns (DCD - SCD - CPR)
Destination Clock Delay (DCD): -0.796ns
Source Clock Delay (SCD): 0.000ns
Clock Pessimism Removal (CPR): -0.000ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
-------------------------	------------	----------	----------	---------

J15		0.000	0.000 r	In1[0]
(IN)				
J15	net (fo=0)	0.000	0.000 r	In1[0]
In1_IBUF[0]_inst/I				
J15	IBUF (Prop_ibuf_I_0)	0.245	0.245 r	
In1_IBUF[0]_inst/O				
	net (fo=3, routed)	0.478	0.723	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[0]				
SLICE_X3Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][0]/D				

(clock clk_out1_clk_wiz_0_1 rise edge)

E3		0.000	0.000 r	
(IN)		0.000	0.000 r	sys_clk
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1				
E3			r	
u_pll/inst/clkin1_ibufg/I				
E3	IBUF (Prop_ibuf_I_0)	0.438	0.438 r	
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.480	0.918	
u_pll/inst/clk_in1_clk_wiz_0				
MMCME2_ADV_X1Y2			r	
u_pll/inst/mmcm_adv_inst/CLKIN1				
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245 r	
u_pll/inst/mmcm_adv_inst/CLKOUT0				
	net (fo=1, routed)	0.544	-1.702	
u_pll/inst/clk_out1_clk_wiz_0				
BUFGCTRL_X0Y16			r	
u_pll/inst/clkout1_buf/I				
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673 r	
u_pll/inst/clkout1_buf/O				
	net (fo=1039, routed)	0.877	-0.796	
u_FPGA_Core/u_Discrete_FIR_Filter/CLK				

SLICE_X3Y95 FDCE r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][0]/C

Slack: inf
Source: In1[7]
(input port)
Destination:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][7]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group: (none)
Path Type: Hold (Min at Fast Process Corner)
Data Path Delay: 0.751ns (logic 0.275ns (36.672%) route 0.476ns
(63.328%))
Logic Levels: 1 (IBUF=1)
Clock Path Skew: -0.796ns (DCD - SCD - CPR)
Destination Clock Delay (DCD): -0.796ns
Source Clock Delay (SCD): 0.000ns
Clock Pessimism Removal (CPR): -0.000ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

R13 (IN)		0.000	0.000 r	In1[7]
	net (fo=0)	0.000	0.000	In1[7]
R13 In1_IBUF[7]_inst/I			r	
R13 In1_IBUF[7]_inst/O	IBUF (Prop_ibuf_I_0)	0.275	0.275 r	
	net (fo=3, routed)	0.476	0.751	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[7] SLICE_X3Y95 FDCE			r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][7]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)			
E3 (IN)		0.000	0.000 r	
		0.000	0.000 r	sys_clk
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1 E3			r	
u_pll/inst/clkin1_ibufg/I E3	IBUF (Prop_ibuf_I_0)	0.438	0.438 r	
u_pll/inst/clkin1_ibufg/O				
	net (fo=1, routed)	0.480	0.918	
u_pll/inst/clk_in1_clk_wiz_0				

```

    MMCME2_ADV_X1Y2
u_pll/inst/mmcm_adv_inst/CLKIN1
    MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
                                -3.163    -2.245 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
                                net (fo=1, routed)      0.544    -1.702
u_pll/inst/clk_out1_clk_wiz_0
    BUFGCTRL_X0Y16
u_pll/inst/clkout1_buf/I
    BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)      0.029    -1.673 r
u_pll/inst/clkout1_buf/O
                                net (fo=1039, routed)    0.877    -0.796
u_FPGA_Core/u_Discrete_FIR_Filter/CLK
    SLICE_X3Y95      FDCE
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][7]/C

```

```

Slack:          inf
Source:         In1[5]
                (input port)

Destination:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][5]/D
                (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group:     (none)
Path Type:      Hold (Min at Fast Process Corner)
Data Path Delay: 0.852ns (logic 0.265ns (31.120%) route 0.587ns
(68.880%))
Logic Levels:   1 (IBUF=1)
Clock Path Skew: -0.796ns (DCD - SCD - CPR)
  Destination Clock Delay (DCD):  -0.796ns
  Source Clock Delay (SCD):      0.000ns
  Clock Pessimism Removal (CPR): -0.000ns

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
T18 (IN)		0.000	0.000 r	In1[5]
T18	net (fo=0)	0.000	0.000 r	In1[5]
In1_IBUF[5]_inst/I				
T18	IBUF (Prop_ibuf_I_0)	0.265	0.265 r	
In1_IBUF[5]_inst/O				
	net (fo=3, routed)	0.587	0.852	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[5]				
SLICE_X0Y96	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][5]/D				

```

(clock clk_out1_clk_wiz_0_1 rise edge)
0.000 0.000 r
E3 0.000 0.000 r sys_clk
(IN)
net (fo=0) 0.000 0.000
u_pll/inst/clk_in1
E3 r
u_pll/inst/clkin1_ibufg/I
E3 IBUF (Prop_ibuf_I_0) 0.438 0.438 r
u_pll/inst/clkin1_ibufg/O
net (fo=1, routed) 0.480 0.918
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2 r
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
-3.163 -2.245 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
net (fo=1, routed) 0.544 -1.702
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16 r
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16 BUFG (Prop_bufg_I_0) 0.029 -1.673 r
u_pll/inst/clkout1_buf/O
net (fo=1039, routed) 0.877 -0.796
u_FPGA_Core/u_Discrete_FIR_Filter/CLK
SLICE_X0Y96 FDCE r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][5]/C

```

```

Slack: inf
Source: In1[3]
(input port)
Destination:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][3]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group: (none)
Path Type: Hold (Min at Fast Process Corner)
Data Path Delay: 0.915ns (logic 0.245ns (26.772%) route 0.670ns
(73.228%))
Logic Levels: 1 (IBUF=1)
Clock Path Skew: -0.796ns (DCD - SCD - CPR)
Destination Clock Delay (DCD): -0.796ns
Source Clock Delay (SCD): 0.000ns
Clock Pessimism Removal (CPR): -0.000ns

```

Location	Delay type	Incr(ns)	Path(ns)	Netlist
Resource(s)				

R15		0.000	0.000	r	In1[3]
(IN)					
	net (fo=0)	0.000	0.000		In1[3]
R15				r	
In1_IBUF[3]_inst/I					
R15	IBUF (Prop_ibuf_I_0)	0.245	0.245	r	
In1_IBUF[3]_inst/O					
	net (fo=3, routed)	0.670	0.915		
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[3]					
SLICE_X3Y95	FDCE			r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][3]/D					

	(clock clk_out1_clk_wiz_0_1 rise edge)				
		0.000	0.000	r	
E3		0.000	0.000	r	sys_clk
(IN)					
	net (fo=0)	0.000	0.000		
u_pll/inst/clk_in1					
E3				r	
u_pll/inst/clkin1_ibufg/I					
E3	IBUF (Prop_ibuf_I_0)	0.438	0.438	r	
u_pll/inst/clkin1_ibufg/O					
	net (fo=1, routed)	0.480	0.918		
u_pll/inst/clk_in1_clk_wiz_0					
MMCME2_ADV_X1Y2				r	
u_pll/inst/mmcm_adv_inst/CLKIN1					
MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245	r	
u_pll/inst/mmcm_adv_inst/CLKOUT0					
	net (fo=1, routed)	0.544	-1.702		
u_pll/inst/clk_out1_clk_wiz_0					
BUFGCTRL_X0Y16				r	
u_pll/inst/clkout1_buf/I					
BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673	r	
u_pll/inst/clkout1_buf/O					
	net (fo=1039, routed)	0.877	-0.796		
u_FPGA_Core/u_Discrete_FIR_Filter/CLK					
SLICE_X3Y95	FDCE			r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][3]/C					
Slack:	inf				
Source:	In1[6]				
	(input port)				
Destination:					
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][6]/D					
	(rising edge-triggered cell FDCE clocked by				
clk_out1_clk_wiz_0_1	{rise@0.000ns fall@12.500ns period=25.000ns})				
Path Group:	(none)				

Path Type: Hold (Min at Fast Process Corner)
 Data Path Delay: 0.921ns (logic 0.262ns (28.431%) route 0.659ns (71.569%))
 Logic Levels: 1 (IBUF=1)
 Clock Path Skew: -0.796ns (DCD - SCD - CPR)
 Destination Clock Delay (DCD): -0.796ns
 Source Clock Delay (SCD): 0.000ns
 Clock Pessimism Removal (CPR): -0.000ns

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

U18 (IN)		0.000	0.000 r	In1[6]
	net (fo=0)	0.000	0.000	In1[6]
U18 In1_IBUF[6]_inst/I			r	
U18 In1_IBUF[6]_inst/O	IBUF (Prop_ibuf_I_0)	0.262	0.262 r	
	net (fo=3, routed)	0.659	0.921	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[6] SLICE_X3Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][6]/D				

	(clock clk_out1_clk_wiz_0_1 rise edge)			
		0.000	0.000 r	
E3 (IN)		0.000	0.000 r	sys_clk
	net (fo=0)	0.000	0.000	
u_pll/inst/clk_in1 E3			r	
u_pll/inst/clkin1_ibufg/I E3	IBUF (Prop_ibuf_I_0)	0.438	0.438 r	
u_pll/inst/clkin1_ibufg/O	net (fo=1, routed)	0.480	0.918	
u_pll/inst/clk_in1_clk_wiz_0 MMCME2_ADV_X1Y2			r	
u_pll/inst/mmcm_adv_inst/CLKIN1 MMCME2_ADV_X1Y2	MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)	-3.163	-2.245 r	
u_pll/inst/mmcm_adv_inst/CLKOUT0	net (fo=1, routed)	0.544	-1.702	
u_pll/inst/clk_out1_clk_wiz_0 BUFGCTRL_X0Y16			r	
u_pll/inst/clkout1_buf/I BUFGCTRL_X0Y16	BUFG (Prop_bufg_I_0)	0.029	-1.673 r	
u_pll/inst/clkout1_buf/O				

```

net (fo=1039, routed)          0.877    -0.796
u_FPGA_Core/u_Discrete_FIR_Filter/CLK
  SLICE_X3Y95      FDCE
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][6]/C

Slack:          inf
Source:         In1[4]
                (input port)
Destination:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][4]/D
                (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group:     (none)
Path Type:      Hold (Min at Fast Process Corner)
Data Path Delay: 1.012ns (logic 0.260ns (25.718%) route 0.752ns
(74.282%))
Logic Levels:   1 (IBUF=1)
Clock Path Skew: -0.796ns (DCD - SCD - CPR)
  Destination Clock Delay (DCD): -0.796ns
  Source Clock Delay (SCD): 0.000ns
  Clock Pessimism Removal (CPR): -0.000ns

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

R17 (IN)		0.000	0.000 r	In1[4]
	net (fo=0)	0.000	0.000	In1[4]
R17			r	
In1_IBUF[4]_inst/I				
R17	IBUF (Prop_ibuf_I_0)	0.260	0.260 r	
In1_IBUF[4]_inst/O				
	net (fo=3, routed)	0.752	1.012	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[4]				
SLICE_X2Y95	FDCE		r	
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][4]/D				

```

(clock clk_out1_clk_wiz_0_1 rise edge)
0.000    0.000 r
E3        0.000    0.000 r sys_clk
(IN)
net (fo=0) 0.000    0.000
u_pll/inst/clk_in1
E3
u_pll/inst/clkin1_ibufg/I
E3 IBUF (Prop_ibuf_I_0) 0.438    0.438 r
u_pll/inst/clkin1_ibufg/O

```

```

net (fo=1, routed) 0.480 0.918
u_pll/inst/clk_in1_clk_wiz_0
  MMCME2_ADV_X1Y2 r
u_pll/inst/mmcm_adv_inst/CLKIN1
  MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
  -3.163 -2.245 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
  net (fo=1, routed) 0.544 -1.702
u_pll/inst/clk_out1_clk_wiz_0
  BUFGCTRL_X0Y16 r
u_pll/inst/clkout1_buf/I
  BUFGCTRL_X0Y16 BUFG (Prop_bufg_I_0) 0.029 -1.673 r
u_pll/inst/clkout1_buf/O
  net (fo=1039, routed) 0.877 -0.796
u_FPGA_Core/u_Discrete_FIR_Filter/CLK
  SLICE_X2Y95 FDCE r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][4]/C

```

```

Slack: inf
Source: In1[11]
        (input port)

Destination:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][11]/D
        (rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group: (none)
Path Type: Hold (Min at Fast Process Corner)
Data Path Delay: 1.087ns (logic 0.270ns (24.804%) route 0.818ns
(75.196%))
Logic Levels: 1 (IBUF=1)
Clock Path Skew: -0.796ns (DCD - SCD - CPR)
  Destination Clock Delay (DCD): -0.796ns
  Source Clock Delay (SCD): 0.000ns
  Clock Pessimism Removal (CPR): -0.000ns

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist

T13 (IN)		0.000	0.000 r	In1[11]
	net (fo=0)	0.000	0.000	In1[11]
T13			r	
In1_IBUF[11]_inst/I				
T13	IBUF (Prop_ibuf_I_0)	0.270	0.270 r	
In1_IBUF[11]_inst/O				
	net (fo=3, routed)	0.818	1.087	
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[11]				
SLICE_X3Y95	FDCE			r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][11]/D				

```

-----
(clock clk_out1_clk_wiz_0_1 rise edge)
E3 0.000 0.000 r sys_clk
(IN) 0.000 0.000 r
net (fo=0) 0.000 0.000
u_pll/inst/clk_in1
E3 r
u_pll/inst/clkin1_ibufg/I
E3 IBUF (Prop_ibuf_I_0) 0.438 0.438 r
u_pll/inst/clkin1_ibufg/O
net (fo=1, routed) 0.480 0.918
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2 r
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2 MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
-3.163 -2.245 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
net (fo=1, routed) 0.544 -1.702
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16 r
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16 BUFG (Prop_bufg_I_0) 0.029 -1.673 r
u_pll/inst/clkout1_buf/O
net (fo=1039, routed) 0.877 -0.796
u_FPGA_Core/u_Discrete_FIR_Filter/CLK
SLICE_X3Y95 FDCE r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][11]/C

Slack: inf
Source: In1[10]
(input port)

Destination:
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][10]/D
(rising edge-triggered cell FDCE clocked by
clk_out1_clk_wiz_0_1 {rise@0.000ns fall@12.500ns period=25.000ns})
Path Group: (none)
Path Type: Hold (Min at Fast Process Corner)
Data Path Delay: 1.346ns (logic 0.250ns (18.564%) route 1.096ns
(81.436%))
Logic Levels: 1 (IBUF=1)
Clock Path Skew: -0.796ns (DCD - SCD - CPR)
Destination Clock Delay (DCD): -0.796ns
Source Clock Delay (SCD): 0.000ns
Clock Pessimism Removal (CPR): -0.000ns

```

Location Resource(s)	Delay type	Incr(ns)	Path(ns)	Netlist
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```

-----
R16                                0.000    0.000 r  In1[10]
(IN)
      net (fo=0)                    0.000    0.000    In1[10]
R16                                r
In1_IBUF[10]_inst/I
R16      IBUF (Prop_ibuf_I_0)        0.250    0.250 r
In1_IBUF[10]_inst/O
      net (fo=3, routed)            1.096    1.346
u_FPGA_Core/u_Discrete_FIR_Filter/In1_IBUF[10]
SLICE_X3Y95      FDCE                                r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][10]/D
-----

      (clock clk_out1_clk_wiz_0_1 rise edge)
      0.000    0.000 r
E3                                0.000    0.000 r  sys_clk
(IN)
      net (fo=0)                    0.000    0.000
u_pll/inst/clk_in1
E3                                r
u_pll/inst/clkin1_ibufg/I
E3      IBUF (Prop_ibuf_I_0)        0.438    0.438 r
u_pll/inst/clkin1_ibufg/O
      net (fo=1, routed)            0.480    0.918
u_pll/inst/clk_in1_clk_wiz_0
MMCME2_ADV_X1Y2                                r
u_pll/inst/mmcm_adv_inst/CLKIN1
MMCME2_ADV_X1Y2      MMCME2_ADV (Prop_mmcme2_adv_CLKIN1_CLKOUT0)
      -3.163    -2.245 r
u_pll/inst/mmcm_adv_inst/CLKOUT0
      net (fo=1, routed)            0.544    -1.702
u_pll/inst/clk_out1_clk_wiz_0
BUFGCTRL_X0Y16                                r
u_pll/inst/clkout1_buf/I
BUFGCTRL_X0Y16      BUFG (Prop_bufg_I_0)        0.029    -1.673 r
u_pll/inst/clkout1_buf/O
      net (fo=1039, routed)        0.877    -0.796
u_FPGA_Core/u_Discrete_FIR_Filter/CLK
SLICE_X3Y95      FDCE                                r
u_FPGA_Core/u_Discrete_FIR_Filter/delay_pipeline_1_reg[0][10]/C

```